

# Traceability Matrix

REMEMBER GORDON! – Rulebook  $\rightleftharpoons$  Implementation Mapping

Generated from `docs/traceability.toml`

## Overview

| Implemented | Descriptive | Implicit | Out-of-scope |
|-------------|-------------|----------|--------------|
| 86          | 33          | 2        | 3            |

Total mappings: 124 · Total impl sites: 261

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 1/3 implemented

# §1 – Introduction

## §1 – Introduction

**descriptive**  
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### Introduction

## §1.1 – General Comments

**implemented**  
manual page 1

### General Comments

“REMEMBER GORDON!” — THE BATTLE OF OMDURMAN is a simulation of the final battle in Great Britain’s two-year campaign to reassert her presence in the Sudan (1896–1898). Fought September 2nd, 1898, Omdurman finally broke the back of the fanatical Dervish rebellion and gained Britain a million square miles of desolate territory and two million impoverished subjects. With two players, one assumes the role of Herbert Kitchener, Sirdar (CIC) of the Anglo-Egyptian army; the other player becomes the Khalifa, Abdullah the Taiasha, absolute ruler of the Dervishes. The game is also suited for multi-player participation, with each player assuming command of one or more Dervish tribes or Anglo-Egyptian brigades.

While “REMEMBER GORDON!” — THE BATTLE OF OMDURMAN is not, strictly speaking, a beginner’s game, the mechanics of play should be familiar to players of modest experience. It is suggested that the bonus game, FALL OF KHARTOUM, and the historical scenario be played first to familiarize players with the game system prior (see manual for full text)

| File                                                                                                  | Symbol                   | Code Snippet                                                                                                                                                                                                                                                                                                                                                                   |
|-------------------------------------------------------------------------------------------------------|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/unit_profiles.rs:194</a><br>GH:omdurman-rules/src/unit_profiles.rs:194 | <b>command_owns_unit</b> | <pre>192   /// treat *nobody-claims-it* units as the faction's communal pool, so `false` 193   /// here means "any member of the faction may act on this unit". 194   pub fn command_owns_unit(scope: &amp;omdurman_types::CommandScope, identity: &amp;UnitIdentity) -&gt; bool { 195       match scope { 196           omdurman_types::CommandScope::Army =&gt; false,</pre> |

Covered by tests: [omdurman-rules::src::unit\\_profiles::command\\_owns\\_unit\\_follows\\_tribes\\_and\\_brigades](#)  
[omdurman-net::src::lib::start\\_game\\_commands\\_round\\_trip](#)  
[omdurman-net::src::lib::legacy\\_start\\_game\\_without\\_commands\\_still\\_loads](#)  
[omdurman-net::src::lib::brigade\\_scope\\_display\\_and\\_membership](#)  
[omdurman-app::src::tests::replayed\\_start\\_game\\_stages\\_commands\\_and\\_resets\\_ready](#)

## §1.2 – Game Scale

**descriptive**  
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### Game Scale

Each hexagon of the mapsheet represents approximately 400–440 yards of real terrain and each day turn is the equivalent of two hours of real time. Each counter of infantry and cavalry represents between 400 and 700 men, and each of the gunboats present at the battle has its own counter. The upper echelon of command is represented by individual leader counters for the Anglo-Egyptian force; and leaders plus their retinues for the Dervish army.

## §2 – Game Components

### §2 – Game Components

descriptive

manual page unknown

#### Game Components

### §2.1 – The Game Maps

descriptive

manual page unknown

#### The Game Maps

The Omdurman battle map represents approximately 100 square miles of real territory and includes the area north of Omdurman in which the historical battle took place as well as the dominant terrain features that influenced the course of the battle. Note that the mapsheet also contains the Turn Record Track, Turn Sequence, and Terrain Effects Chart at the top; and the Combat Tables and Howitzer Fire Scattergram in the lower right corner. The large letters “A”, “D”, “Y”, etc. are set-up hexes for the historical scenario only (9.2) and should be ignored in the campaign game. Similarly, the hexsides of the Zariba exist only in the historical scenario and should be considered clear terrain in the campaign game. Note, however, that the Anglo-Egyptian player may “construct” the Zariba in the campaign game if desired (see 5.3). All full hexes of the Omdurman game map are playable, including the seven hexes of the Howitzer Fire Scattergram.

The mini-map for the bonus game, FALL OF KHARTOUM, represents that city as it appeared in 1885. The portion (see manual for full text)

See also: §5.3, §9.2

### §2.2 – Play Aids

descriptive

manual page unknown

#### Play Aids

Certain charts and tables are needed to play the game. The Terrain Effects Chart lists all terrain found on the mapsheet and the effect of each type on movement and combat. The Combat Tables describe the range effects on various weapon types and includes the Combat Results Table. Also note the Line of Sight Table on the back of this rulebook. It tells players when certain terrain types block line of sight, thus preventing direct fire attacks on enemy units. Players should become familiar with these various charts and tables prior to the beginning of play.

(see manual for full text)

### §2.3 – The Units

implemented

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#### The Units

| File | Symbol | Code Snippet |
|------|--------|--------------|
|------|--------|--------------|

|                                                                                     |                         |                                                                                                                                                                                                                                                                             |
|-------------------------------------------------------------------------------------|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-types/src/lib.rs:1052</a><br>GH:omdurman-types/src/lib.rs:1052 | <b>BrigadeId</b>        | <pre> 1050   /// same field for uniform handling. 1051   #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, PartialOrd, Ord, Hash, Debug)] 1052   pub struct BrigadeId { 1053       pub number: u8, 1054       pub nationality: BrigadeNationality,</pre>         |
| <a href="#">omdurman-rules/src/unit.rs:491</a><br>GH:omdurman-rules/src/unit.rs:491 | <b>UnitProfile</b>      | <pre> 489   /// print no melee value). 490   #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 491   pub struct UnitProfile { 492       pub kind: UnitKind, 493       pub identity: UnitIdentity,</pre>                                                  |
| <a href="#">omdurman-rules/src/lib.rs:14</a><br>GH:omdurman-rules/src/lib.rs:14     | <b>BrigadeId</b>        | <pre> 12   13   use omdurman_types::{ 14       BrigadeId, BrigadeNationality, DayNight, DervishTribe, HexCoord, Player, UnitKind, 15   }; 16  </pre>                                                                                                                        |
| <a href="#">omdurman-types/src/lib.rs:686</a><br>GH:omdurman-types/src/lib.rs:686   | <b>SpriteAnnotation</b> | <pre> 684   /// as an optional overlay over the compiled `sprite_data` fallback. 685   #[derive(Serialize, Deserialize, Clone, Debug, PartialEq)] 686   pub struct SpriteAnnotation { 687       pub color: SpriteColor, 688       pub faction: Option&lt;Faction&gt;,</pre> |

Covered by tests: [omdurman-rules::src::unit\\_profiles::british\\_army\\_row\\_zero\\_specials\\_classify\\_by\\_counter](#)  
[omdurman-rules::src::unit\\_profiles::egyptian\\_army\\_row\\_zero\\_specials\\_classify\\_by\\_counter](#)  
[omdurman-rules::src::unit\\_profiles::tribe\\_stats\\_come\\_from\\_annotation](#)  
[omdurman-rules::src::unit\\_profiles::section\\_owner\\_dervish\\_sections](#)  
[omdurman-rules::src::unit\\_profiles::section\\_owner\\_green\\_sections\\_are\\_dervish](#)

## §2.4 – Game Parts Inventory

**out-of-scope**

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### Game Parts Inventory

Your complete copy of “REMEMBER GORDON!” — THE BATTLE OF OMDURMAN includes:

- One 22 × 28 Battle of Omdurman mapsheet
- One 8½ × 11 bonus game: FALL OF KHARTOUM mapsheet
- One Rules Booklet
- One die-cut Unit Counter Sheet
- One Campaign Game Order of Appearance Card
- One ten-sided die
- One game box

## §2.31 – Dervish weapon types

**implemented**

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Dervish artillery, gunboats, and forts fire on the “artillery” line of the Dervish Range Effects Table; Jehadia and Danagla units fire on the “rifles” line as does the Isa Zachneih unit. All other Dervish units (including leaders) are armed with spears and swords.

| File | Symbol | Code Snippet |
|------|--------|--------------|
|------|--------|--------------|

|                                                                                       |                         |                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|---------------------------------------------------------------------------------------|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| omdurman-rules/src/unit.rs:167<br>GH:omdurman-rules/src/unit.rs:167                   | <b>WeaponClass</b>      | <pre> 165   Serialize, Deserialize, Clone, Copy, PartialEq, Eq, PartialOrd, Ord, Hash, Debug, strum::Display, 166   )] 167   pub enum WeaponClass { 168       /// Dervish spears and swords -- no ranged fire at all. 169       Melee,</pre>                                                                                                                                                                                                |
| omdurman-rules/src/unit_profiles.rs:578<br>GH:omdurman-rules/src/unit_profiles.rs:578 | <b>dervish_tribe</b>    | <pre> 576   /// Resolve a Dervish tribal foot counter (§2.31): Jehadia, Danagla and 577   /// Isa Zachneih fire on the rifles line; every other tribe is spear-armed. 578   pub fn dervish_tribe(tribe: DervishTribe) -&gt; Option&lt;Classification&gt; { 579       // §2.31: "Jehadia and Danagla units fire on the 'rifles' line as does the 580       // Isa Zachneih unit. All other Dervish units (including leaders) are armed</pre> |
| omdurman-rules/src/unit_profiles.rs:344<br>GH:omdurman-rules/src/unit_profiles.rs:344 | <b>khalifa_abdullah</b> | <pre> 342   /// battle (§9.322). All three are interchangeable, so they share the 343   /// `DervishArtillery` identity. 344   pub fn khalifa_abdullah(col: u32, row: u32) -&gt; Option&lt;Classification&gt; { 345       let artillery =    { 346           Some(Classification {</pre>                                                                                                                                                    |

Covered by tests: [omdurman-rules::src::unit\\_profiles::dervish\\_weapon\\_class\\_follows\\_the\\_rifles\\_line](#)

## §2.32 – Anglo-Egyptian weapon types

**implemented**

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All Anglo-Egyptian units (except gunboats, Maxims, artillery, and leaders) are armed with rifles. Maxims fire on the “Maxims” line of the Anglo-Egyptian Range Effects Table, and artillery and old gunboats fire on the “Artillery” line. New type (named) gunboats may fire on the “Howitzer”, “Artillery”, and “Maxims” lines of the Range Effects Table. (See 6.52 for the fire capabilities of the “Friendlies”).

**\*\*Sample Dervish Units\*\*** (printed on counters): combat unit (Combat / Melee / Movement values, plus Tribe identifier); Leader (e.g. OSMAN DIGNA); Camel unit (e.g. Danagla, 4-6-12); Fort.

**\*\*Sample Anglo-Egyptian Units\*\*** (printed on counters): Cavalry (e.g. 21 Lancers); Artillery (e.g. 32 Battery); Old Gunboat (e.g. LORD KITCHENER, 0-0-15); New Gunboat — named (e.g. Sultan, with artillery and howitzer factor, plus movement downstream / movement upstream values); Maxim Guns (fire twice per turn); Infantry (Fire Combat Factor / Melee / Movement, plus Battalion ID and Brigade ID — e.g. “2B” = 2nd British Brigade, “3E” = 3rd Egyptian Brigade).

(see manual for full text)

See also: §6.52

| File                                                                | Symbol              | Code Snippet                                                                                                                                                                                                                                                                                                   |
|---------------------------------------------------------------------|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| omdurman-rules/src/unit.rs:119<br>GH:omdurman-rules/src/unit.rs:119 | <b>GunboatId</b>    | <pre> 117   /// fire; "old" gunboats do not (rulebook §2.32). 118   #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)] 119   pub enum GunboatId { 120       /// One of the five new-type named gunboats with howitzer capability. 121       Named(NamedGunboat),</pre> |
| omdurman-rules/src/unit.rs:138<br>GH:omdurman-rules/src/unit.rs:138 | <b>NamedGunboat</b> | <pre> 136   /// The five named gunboats with howitzer capability (rulebook §6.64, §2.32). 137   #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)] 138   pub enum NamedGunboat { 139       Sultan, 140       Melik,</pre>                                              |

|                                                                                     |                                   |                                                                                                                                                                                                                                                      |
|-------------------------------------------------------------------------------------|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/unit.rs:151</a><br>GH:omdurman-rules/src/unit.rs:151 | <b>OldGunboat</b>                 | <pre>149   /// in the Maxim Second Fire and Howitzer subphase (\$6.42). 150   #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)] 151   pub enum OldGunboat { 152       LordKitchener, 153       Tamai,</pre> |
| <a href="#">omdurman-rules/src/unit.rs:151</a><br>GH:omdurman-rules/src/unit.rs:151 | <b>GunboatId::Old</b>             | <pre>149   /// in the Maxim Second Fire and Howitzer subphase (\$6.42). 150   #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)] 151   pub enum OldGunboat { 152       LordKitchener, 153       Tamai,</pre> |
| <a href="#">omdurman-rules/src/unit.rs:125</a><br>GH:omdurman-rules/src/unit.rs:125 | <b>GunboatId::Dervish-Gunboat</b> | <pre>123       Old(OldGunboat), 124       /// A Dervish gunboat (\$9.111, \$10.14). 125       DervishGunboat(u8), 126   } 127  </pre>                                                                                                                |

Covered

by

tests:

omdurman-rules::src::effects::tests::old\_gunboat\_lacks\_howitzer

omdurman-rules::src::effects::tests::old\_gunboat\_rejected\_from\_howitzer\_subphase

0/1 implemented

## | §3 – Getting Started

### §3 – Getting Started

out-of-scope

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#### Getting Started

Spread out the mapsheet on a table. It should lie flat if you backfold it against the scored lines. The Dervish player should sit next to the west edge of the map and the Anglo-Egyptian player opposite him on the east edge. Read through the rules once, looking over the various charts as they are referred to in the various sections. Next, select a scenario and punch out only those unit counters needed to play. Later on, the rest of the unit counters should be punched out, sorted and stored by unit type.

(see manual for full text)

■ 1/1 implemented

## §4 – Turn Sequence

### §4 – Turn Sequence

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#### Turn Sequence

“REMEMBER GORDON!” — THE BATTLE OF OMDURMAN is played in “Game Turns”, each of which has two “Player Turns”. The player moving first will vary according to the scenario being played. In the Campaign Game, for example, the Anglo-Egyptian player moves first.

### A) Anglo-Egyptian Player Turn:

1. Anglo-Egyptian Movement Phase
2. Fire Combat Phase a. Dervish Defensive Fire b. Anglo-Egyptian Offensive Fire
  1. Direct Fire Subphase
  2. Maxim Second Fire and Howitzer Fire Subphase

(see manual for full text)

| File                                                                                                  | Symbol                | Code Snippet                                                                                                                                                                                                                                                                                                                                                           |
|-------------------------------------------------------------------------------------------------------|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/scalars.rs:184</a><br>GH:omdurman-rules/src/scalars.rs:184             | <b>GameTurnIndex</b>  | <pre>182   /// One-based Game Turn index (1, 2, ... up to the 183   scenario length) (rulebook §4). 184   #[derive(Serialize, Deserialize, Clone, Copy, 185   PartialEq, Eq, PartialOrd, Ord, Hash, Debug)] 186   pub struct GameTurnIndex(pub(crate) u8); 187   188   impl GameTurnIndex {</pre>                                                                      |
| <a href="#">omdurman-rules/src/turn.rs:16</a><br>GH:omdurman-rules/src/turn.rs:16                     | <b>Phase</b>          | <pre>14   /// etc. 15   #[derive(Serialize, Deserialize, Clone, Copy, 16   PartialEq, Eq, Hash, Debug, Default)] 17   pub enum Phase { 18       /// Pre-game deployment (§9.2/§9.3/§10): fixed 19       /// units are placed, each side 20       /// deploys its order of battle within its legal 21       /// zone, and river</pre>                                   |
| <a href="#">omdurman-rules/src/effects/state.rs:21</a><br>GH:omdurman-rules/src/effects/state.rs:21   | <b>GameState</b>      | <pre>19   /// All mutable state of a game in progress (rulebook 20   §4). 21   #[derive(Serialize, Deserialize, Clone, Debug)] 22   pub struct GameState { 23       pub scenario: Scenario, 24       pub current_turn: GameTurnIndex,</pre>                                                                                                                            |
| <a href="#">omdurman-rules/src/effects/state.rs:178</a><br>GH:omdurman-rules/src/effects/state.rs:178 | <b>GameState::new</b> | <pre>176   impl GameState { 177       /// Create a fresh game state for a given scenario 178       /// (rulebook §4). 179       pub fn new(scenario: Scenario) -&gt; Self { 180           let first = scenario_turn(scenario, 181   GameTurnIndex::new(1)); 182           // First player to *move* per scenario: 183           Campaign -- Anglo-Egyptian moves</pre> |
| <a href="#">omdurman-rules/src/effects/effect.rs:17</a><br>GH:omdurman-rules/src/effects/effect.rs:17 | <b>AdvancePhase</b>   | <pre>15       /// At end-of-turn, disrupted units recover and 16       /// per-turn tracking is 17       /// cleared. 18       AdvancePhase, 19       // -- Movement 20       -----</pre>                                                                                                                                                                              |
| <a href="#">omdurman-rules/src/effects/dispatch.rs:192</a><br>GH:omdurman-                            | <b>advance_phase</b>  | <pre>190   191   /// Advance the game state to the next phase (rulebook 192   §4).</pre>                                                                                                                                                                                                                                                                               |

|                                                                                                             |                           |                                                                                                                                                                                                                                                                                                 |
|-------------------------------------------------------------------------------------------------------------|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| rules/src/effects/<br>dispatch.rs:192                                                                       |                           | <pre>192   pub fn advance_phase(state: &amp;mut GameState) -&gt;<br/>Result&lt;(), RuleError&gt; {<br/>193       let old_phase = state.phase;<br/>194       #[cfg(not(feature = "kani"))]</pre>                                                                                                 |
| omdurman-<br>rules/src/effects/<br>dispatch.rs:306<br>GH:omdurman-<br>rules/src/effects/<br>dispatch.rs:306 | end_player_turn           | <pre>304  <br/>305   /// End the current player's turn: recover disrupted<br/>units, switch active player, advance turn index (rulebook<br/>§4).<br/>306   pub fn end_player_turn(state: &amp;mut GameState) {<br/>307       #[cfg(not(feature = "kani"))]<br/>308       debug!({</pre>         |
| omdurman-<br>rules/src/<br>scalars.rs:190<br>GH:omdurman-<br>rules/src/<br>scalars.rs:190                   | GameTurnIn-<br>dex::value | <pre>188           Self(value)<br/>189       }<br/>190       pub fn value(self) -&gt; u8 {<br/>191           self.0<br/>192       }</pre>                                                                                                                                                       |
| omdurman-<br>rules/src/effects/<br>state.rs:170<br>GH:omdurman-<br>rules/src/effects/<br>state.rs:170       | PendingMelee              | <pre>168   /// resolution after the reaction window is<br/>deterministic and host-ordered (rulebook §7.5).<br/>169   #[derive(Serialize, Deserialize, Clone, Debug)]<br/>170   pub struct PendingMelee {<br/>171       pub attack: MeleeAttack,<br/>172       pub attacker_roll: DieRoll,</pre> |

Covered by tests: omdurman-rules::src::effects::tests::both\_ready\_auto\_advances\_out\_of\_setup  
omdurman-rules::src::effects::tests::fire\_combat\_wrong\_phase\_rejected omdurman-rules::src::effects::tests::new\_game\_starts\_in\_setup  
omdurman-rules::src::turn\_track::scenario\_turn\_dispatches\_correctly  
omdurman-rules::src::effects::tests::turn\_advances\_through\_phases  
omdurman-rules::src::unit\_profiles::game\_turn\_marker\_cell\_returns\_none  
omdurman-rules::src::effects::tests::declared\_melee\_blocks\_phase\_advance

18/23 implemented

## §5 – Movement Phase

### §5 – Movement Phase (general)

descriptive

manual page unknown

#### Movement Phase

### §5.1 – General Rules

descriptive

manual page unknown

#### General Rules

### §5.2 – Movement Restrictions

descriptive

manual page unknown

#### Movement Restrictions

### §5.3 – Constructing the Zariba

implemented

manual page unknown

#### Constructing the Zariba

The Zariba trench and thorn hedge hexsides are built and in place in the historical scenario only. These hexsides are considered clear terrain in the campaign game. The Anglo-Egyptian player may, however, find it useful to construct this defensive position during the campaign game. The Zariba hexsides may only be built in their position as displayed on the mapsheet. Construction procedure is as follows: any Anglo-Egyptian infantry unit that begins and ends the Anglo-Egyptian player turn adjacent to (and on the Nile side of) Zariba hexsides has constructed all Zariba hexsides to which he is adjacent. The constructing unit may neither fire offensively nor melee attack during the turn of construction. Use a blank counter to denote units constructing Zariba hexsides. See 9.23 for defensive benefits and movement restrictions of Zariba hexsides.

(see manual for full text)

See also: §9.23

| File                                                                                | Symbol                     | Code Snippet                                                                                                                                                                                                                                                                                                                                                                                |
|-------------------------------------------------------------------------------------|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/unit.rs:525</a><br>GH:omdurman-rules/src/unit.rs:525 | <b>constructing_zariba</b> | <pre>523       /// Set while the unit is building Zariba hexsides --         -- neither offensive 524       /// fire nor melee allowed that turn (§5.3). 525       pub constructing_zariba: bool, 526       /// Set when the Royal Engineers are committed to a           demolition this turn 527       /// (§6.53) -- neither offensive fire nor melee           allowed that turn.</pre> |
| <a href="#">omdurman-rules/src/effects/effect.rs:170</a><br>GH:omdurman-            | <b>ConstructZariba</b>     | <pre>168   169       /// Begin constructing a Zariba hexside (rulebook           §5.3). 170       ConstructZariba {</pre>                                                                                                                                                                                                                                                                   |

|                                                                                                           |                               |                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------------------------------------------------------------------------------------|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| rules/src/effects/<br>effect.rs:170                                                                       |                               | <pre> 171           unit_ids: Vec&lt;UnitId&gt;, 172           hexside: HexsideRef,</pre>                                                                                                                                                                                                                                                                                                                                              |
| <a href="#">omdurman-rules/src/effects/dispatch.rs:49</a><br>GH:omdurman-rules/src/effects/dispatch.rs:49 | <b>apply_construct_zariba</b> | <pre> 47           GameEffect::RecoverUnit { unit_id } =&gt;               apply_recover_unit(state, *unit_id), 48           GameEffect::ConstructZariba { unit_ids,               hexside } =&gt; { 49               apply_construct_zariba(state, unit_ids,                   *hexside) 50           } 51           GameEffect::Demolition { unit_id, target } =&gt;               apply_demolition(state, *unit_id, *target),</pre> |

Covered by tests: [omdurman-rules::src::effects::tests::construct\\_zariba\\_marks\\_builders\\_and\\_records\\_hexside](#)  
[omdurman-rules::src::effects::tests::zariba\\_builders\\_may\\_not\\_fire\\_offensively\\_or\\_melee\\_until\\_turn\\_end](#)  
[omdurman-rules::src::effects::tests::zariba\\_construction\\_is\\_a\\_campaign\\_movement\\_phase\\_action](#)  
[omdurman-rules::src::effects::tests::world\\_lens\\_composes\\_breaches\\_and\\_constructed\\_zariba](#)

## §5.4 – Zones of Control

[descriptive](#)

manual page unknown

### Zones of Control

## §5.5 – Stacking

[descriptive](#)

manual page unknown

### Stacking

## §5.11 – Movement allowances printed on units

[implemented](#)

manual page unknown

The movement allowances of the various unit types are printed directly on the units (see 2.3). A unit may move up to this printed movement allowance, paying varying costs for different terrain types (see the Terrain Effects Chart).

See also: §2.3

| File                                                                                    | Symbol                   | Code Snippet                                                                                                                                                                                                                                                                                                       |
|-----------------------------------------------------------------------------------------|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/scalars.rs:63</a><br>GH:omdurman-rules/src/scalars.rs:63 | <b>MovementAllowance</b> | <pre> 61           /// is a named variant. 62           #[derive(Serialize, Deserialize, Clone, Copy,               PartialEq, Eq, Hash, Debug)] 63           pub enum MovementAllowance { 64               /// Immobile (forts, wrecked gunboats). 65               Immobile = 0,</pre>                           |
| <a href="#">omdurman-rules/src/unit.rs:502</a><br>GH:omdurman-rules/src/unit.rs:502     | <b>UnitMovement</b>      | <pre> 500   /// Movement allowance -- uniform for land units, split       for gunboats (rulebook §5.11, §5.24, §5.25). 501   #[derive(Serialize, Deserialize, Clone, Copy,       PartialEq, Eq, Debug)] 502   pub enum UnitMovement { 503       Land(MovementAllowance), 504       Gunboat(GunboatMovement),</pre> |
| <a href="#">omdurman-rules/src/scalars.rs:105</a><br>GH:omdurman-                       | <b>MovementPoints</b>    | <pre> 103           Serialize, Deserialize, Clone, Copy, PartialEq, Eq,               PartialOrd, Ord, Hash, Debug, Default, 104       )] 105       pub struct MovementPoints(pub(crate) i16);</pre>                                                                                                               |

|                                                                                                 |                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-------------------------------------------------------------------------------------------------|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| rules/src/<br>scalars.rs:105                                                                    |                                  | 106  <br>107   <code>impl MovementPoints {</code>                                                                                                                                                                                                                                                                                                                                                                                                     |
| omdurman-<br>rules/src/terrain_chart.rs:21<br>GH:omdurman-<br>rules/src/<br>terrain_chart.rs:21 | <b>terrain_effects_chart</b>     | 19   <code>///</code><br>20   <code>/// Source: printed Terrain Effects Chart on the</code><br>mapsheet.<br>21   <code>pub fn terrain_effects_chart(terrain: Terrain) -&gt;</code><br>TerrainEntry {<br>22   <code>    match terrain {</code><br>23   <code>        Terrain::Clear { .. } =&gt; TerrainEntry {</code>                                                                                                                                 |
| omdurman-<br>rules/src/terrain_chart.rs:77<br>GH:omdurman-<br>rules/src/<br>terrain_chart.rs:77 | <b>movement_cost</b>             | 75   <code>/// Convenience: get the movement cost for a terrain</code><br>type (rulebook §5.11, Terrain Effects Chart).<br>76   <code>/// Returns `None` for impassable terrain (Nile).</code><br>77   <code>pub fn movement_cost(terrain: Terrain) -&gt;</code><br>Option<MovementAllowance> {<br>78   <code>    terrain_effects_chart(terrain).movement_cost</code><br>79   <code>}</code>                                                          |
| omdurman-<br>rules/src/terrain_chart.rs:87<br>GH:omdurman-<br>rules/src/<br>terrain_chart.rs:87 | <b>movement_cost_with_ro</b>     | 85   <code>/// movement overlay only -- combat/LOS use the</code><br>underlying terrain ("Per<br>86   <code>/// other terrain in hex"). The Nile stays impassable</code><br>to land units.<br>87   <code>pub fn movement_cost_with_road(terrain: Terrain,</code><br>along_road: bool) -> Option<MovementAllowance> {<br>88   <code>    if along_road &amp;&amp; !terrain.is_nile() {</code><br>89   <code>        Some(MovementAllowance::One)</code> |
| omdurman-<br>types/src/<br>lib.rs:505<br>GH:omdurman-<br>types/src/lib.rs:505                   | <b>Terrain::has_road</b>         | 503  <br>504   <code>/// Whether this hex has any road touching it.</code><br>505   <code>pub fn has_road(self) -&gt; bool {</code><br>506   <code>    !matches!(self.road(), Road::None)</code><br>507   <code>}</code>                                                                                                                                                                                                                              |
| omdurman-<br>types/src/<br>lib.rs:442<br>GH:omdurman-<br>types/src/lib.rs:442                   | <b>Terrain::passable_by_land</b> | 440  <br>441   <code>/// Whether this terrain may be entered by land</code><br>units (rulebook §5.11).<br>442   <code>pub fn passable_by_land(self) -&gt; bool {</code><br>443   <code>    !self.is_nile()</code><br>444   <code>}</code>                                                                                                                                                                                                             |
| omdurman-<br>types/src/<br>lib.rs:510<br>GH:omdurman-<br>types/src/lib.rs:510                   | <b>Terrain::is_crossroad</b>     | 508  <br>509   <code>/// Whether roads converge at this hex's centre.</code><br>510   <code>pub fn is_crossroad(self) -&gt; bool {</code><br>511   <code>    matches!(self.road(), Road::Crossroad)</code><br>512   <code>}</code>                                                                                                                                                                                                                    |
| omdurman-<br>rules/src/<br>board.rs:341<br>GH:omdurman-<br>rules/src/<br>board.rs:341           | <b>BoardInfo::roads</b>          | 339   <code>    tiles: tiles.into_iter().collect(),</code><br>340   <code>    hexsides: Vec::new(),</code><br>341   <code>    roads: Vec::new(),</code><br>342   <code>    excluded: BTreeSet::new(),</code><br>343   <code>    overlay: default_overlay(),</code>                                                                                                                                                                                    |

Proven by: `omdurman-rules::src::terrain_chart::movement_column_matches_the_printed_chart`  
`omdurman-types::src::lib::terrain_road_state_round_trips`

Covered by tests: `omdurman-rules::src::effects::tests::road_rate_applies_only_along_the_road`  
`omdurman-rules::src::effects::tests::crossing_a_hexside_costs_the_chart_surcharge`  
`omdurman-rules::src::terrain_chart::step_cost_adds_the_crossed_hexside`  
`omdurman-rules::src::terrain_chart::engine_matches_the_transcribed_terrain_effects_chart`  
`omdurman-rules::src::effects::tests::extreme_coordinates_are_rejected_without_panicking`  
`omdurman-rules::src::effects::tests::move_cost_is_engine_computed_and_caller_cost_ignored`  
`omdurman-rules::src::effects::tests::move_path_must_be_contiguous_and_end_at_destination`  
`omdurman-rules::src::terrain_chart::clear_terrain_no_bonus`  
`omdurman-rules::src::terrain_chart::nile_is_impassable` `omdurman-rules::src::terrain_chart::rough_movement_and_defense`  
`omdurman-rules::src::terrain_chart::swamp_movement_and_defense`  
`omdurman-rules::src::terrain_chart::hilltop_movement_and_defense`  
`omdurman-rules::src::terrain_chart::huts_movement_and_defense`  
`omdurman-rules::src::terrain_chart::movement_cost_convenience_matches_chart`

omdurman-rules::src::terrain\_chart::movement\_cost\_with\_road\_always\_one  
 omdurman-rules::src::effects::tests::land\_unit\_may\_not\_enter\_nile  
 omdurman-rules::src::terrain\_chart::movement\_cost\_without\_road\_matches\_terrain  
 omdurman-rules::src::effects::tests::movement\_cost\_for\_uses\_terrain  
 omdurman-rules::src::effects::tests::movement\_cost\_for\_road\_costs\_one  
 omdurman-rules::src::terrain\_chart::road\_gives\_crossroad      omdurman-rules::src::terrain\_chart::terrain\_movement\_costs\_in\_bounds  
 omdurman-rules::src::terrain\_chart::terrain\_chart\_road\_always\_costs\_one  
 omdurman-rules::src::terrain\_chart::trees\_cost\_one\_and\_give\_no\_defense

## §5.12 – Move up to allowance, hex by hex (cumulative MP cap)

implemented

manual page unknown

A player may move as many or as few of his units as desired during each movement phase, limited only by the units' movement allowance, the terrain costs paid in moving from hex to hex, and enemy zones of control (see 5.4).

See also: §5.4

| File                                                                                                                  | Symbol               | Code Snippet                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-----------------------------------------------------------------------------------------------------------------------|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/effects/state/movement.rs:25</a><br>GH:omdurman-rules/src/effects/state/movement.rs:25 | <b>can_move_unit</b> | <pre> 23       /// Returns the same `RuleError` the `MoveUnit`           effect would on rejection. 24       /// Lets the UI gate input without mutating or           duplicating the rules. 25       pub fn can_move_unit(&amp;self, unit_id: UnitId, cost:           MovementPoints) -&gt; Result&lt;(), RuleError&gt; { 26           let unit = self.unit_or_err(unit_id)?; 27           self.movement_preconditions(unit)?;           </pre> |
| <a href="#">omdurman-rules/src/effects/state.rs:389</a><br>GH:omdurman-rules/src/effects/state.rs:389                 | <b>mp_spent</b>      | <pre> 387   388       /// Movement points `unit_id` has already spent           this turn (§5.11/§5.12). 389       pub fn mp_spent(&amp;self, unit_id: UnitId) -&gt; i16 { 390   391           self.mp_spent_this_turn.get(&amp;unit_id).copied().unwrap_or(0)           </pre>                                                                                                                                                                  |

Covered by tests: omdurman-rules::src::effects::tests::cumulative\_move\_cost\_may\_not\_exceed\_allowance  
 omdurman-rules::src::effects::tests::wrong\_faction\_move\_is\_rejected  
 omdurman-rules::src::effects::tests::wrong\_faction\_gunboat\_move\_is\_rejected

## §5.13 – No MP accumulation between turns

implemented

manual page unknown

A unit may never accumulate movement points from turn to turn, nor may a unit transfer unused movement points to other units. A unit's unused movement points in any given turn are considered lost.

| File                                                                                                        | Symbol                 | Code Snippet                                                                                                                                                                                                                                                                                                    |
|-------------------------------------------------------------------------------------------------------------|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/effects/dispatch.rs:306</a><br>GH:omdurman-rules/src/effects/dispatch.rs:306 | <b>end_player_turn</b> | <pre> 304   305       /// End the current player's turn: recover disrupted           units, switch active player, advance turn index (rulebook           §4). 306       pub fn end_player_turn(state: &amp;mut GameState) { 307           #[cfg(not(feature = "kani"))] 308           debug! (           </pre> |

Covered by tests: omdurman-rules::src::effects::tests::recover\_unit\_is\_the\_owners\_end\_of\_turn\_only  
 omdurman-rules::src::effects::tests::unused\_movement\_points\_do\_not\_carry\_over

## §5.21 – Friendlies transport via gunboat

implemented

manual page unknown

In general, naval transport missions are not allowed, i.e. gunboats may not carry any land units. The sole exception is that the Anglo-Egyptian player may transport the surviving units of the “Friendlies” brigade from the east bank of the Nile to the west bank after, and only after, the Dervish east bank unit (Isa Zachneih) has been eliminated. The transport is accomplished in the following sequence: a) on any turn that a “Friendlies” unit and any Anglo-Egyptian gunboat start their turn adjacent, that unit may load onto (i.e. stack with) the gunboat; b) during the Anglo-Egyptian player’s next turn the gunboat may move to any Nile hex adjacent to a west bank hex (up to the gunboat’s movement allowance); c) on the Anglo-Egyptian player’s third turn the “Friendlies” unit may disembark and move normally, paying the normal terrain cost for the first hex entered. The gunboat may also move normally that turn.

(see manual for full text)

| File                                                                                                                    | Symbol                            | Code Snippet                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------------------------------------------------------------------------------------------------------------------------|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/unit.rs:317</a><br>GH:omdurman-rules/src/unit.rs:317                                     | <b>is_friendlylies</b>            | <pre>315       /// "Friendlies" units obey several special rules           (\$5.21, \$5.23, \$6.52, 316       /// \$9.14 victory conditions). 317       pub fn is_friendlylies(&amp;self) -&gt; bool { 318           matches!( 319               self,</pre>                                                                                                                                                                      |
| <a href="#">omdurman-rules/src/effects/state/movement.rs:492</a><br>GH:omdurman-rules/src/effects/state/movement.rs:492 | <b>friendlylies_transport_off</b> | <pre>490       /// regardless of selection. Pairs with           [`GameEffect::FriendlyliesTransport`] 491       /// so the UI can offer exactly the action the           engine would accept. 492       pub fn friendlylies_transport_offer(&amp;self, selected:           Option&lt;UnitId&gt;) -&gt; Option&lt;FriendlyliesAction&gt; { 493           match self.friendlylies_transport { 494               None =&gt; {</pre> |
| <a href="#">omdurman-rules/src/unit.rs:522</a><br>GH:omdurman-rules/src/unit.rs:522                                     | <b>loaded_on</b>                  | <pre>520       pub disrupted: bool, 521       /// `Some(gunboat)` after a "Friendlies" unit loads           onto a gunboat (§5.21). 522       pub loaded_on: Option&lt;UnitId&gt;, 523       /// Set while the unit is building Zariba hexsides           -- neither offensive 524       /// fire nor melee allowed that turn (§5.3).</pre>                                                                                       |
| <a href="#">omdurman-rules/src/effects/effect.rs:197</a><br>GH:omdurman-rules/src/effects/effect.rs:197                 | <b>FriendlyliesTransport</b>      | <pre>195   196       /// Load/disembark the "Friendlies" brigade via           gunboat (rulebook §5.21). 197       FriendlyliesTransport(crate::FriendlyliesAction), 198   199       // -- Optional rules           -----</pre>                                                                                                                                                                                                   |
| <a href="#">omdurman-rules/src/effects/river.rs:12</a><br>GH:omdurman-rules/src/effects/river.rs:12                     | <b>apply_friendlylies_transpo</b> | <pre>10       /// unit is freed (a disembarking `MoveUnit` should           follow, costed by 11       /// terrain). 12       pub fn apply_friendlylies_transport( 13           state: &amp;mut GameState, 14           action: FriendlyliesAction,</pre>                                                                                                                                                                         |
| <a href="#">omdurman-rules/src/transport.rs:20</a><br>GH:omdurman-rules/src/transport.rs:20                             | <b>FriendlyliesAction</b>         | <pre>18       /// tracks each unit-gunboat pair independently. 19       #[derive(Serialize, Deserialize, Clone, Copy,           PartialEq, Eq, Debug)] 20       pub enum FriendlyliesAction { 21           /// Turn N (the load turn): unit and gunboat               started adjacent; unit 22           /// loads onto (stacks with) the gunboat.</pre>                                                                         |
| <a href="#">omdurman-rules/src/</a>                                                                                     | <b>TransportState</b>             | <pre>38       /// third turn. 39       #[derive(Serialize, Deserialize, Clone, Copy,</pre>                                                                                                                                                                                                                                                                                                                                        |

|                                                          |  |                                                                                                                                                                                                |
|----------------------------------------------------------|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| transport.rs:40<br>GH:omdurman-rules/src/transport.rs:40 |  | PartialEq, Eq, Debug))<br>40   pub enum TransportState {<br>41       /// Turn N (the load turn): unit and gunboat started adjacent; unit<br>42       /// loads onto (stacks with) the gunboat. |
|----------------------------------------------------------|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Covered by tests: [omdurman-rules::src::effects::tests::a\\_mined\\_gunboat\\_scores\\_and\\_takes\\_its\\_friendlies\\_down](#)  
[omdurman-rules::src::effects::tests::friendlies\\_transport\\_offer\\_load\\_requires\\_prerequisites](#)  
[omdurman-rules::src::effects::tests::friendlies\\_transport\\_offer\\_follows\\_state\\_machine](#)

## §5.22 – Land units may never enter a Nile River hex

implemented

manual page unknown

With the exception of 5.21, land units may never enter a Nile River hex. Only gunboats may enter and move along Nile River hexes.

See also: §5.21

| File                                                                                                                  | Symbol                    | Code Snippet                            |
|-----------------------------------------------------------------------------------------------------------------------|---------------------------|-----------------------------------------|
| <a href="#">omdurman-rules/src/effects/state/movement.rs:43</a><br>GH:omdurman-rules/src/effects/state/movement.rs:43 | <b>can_move_unit_to</b>   | 41  <br>42  <br>43  <br>44  <br>45      |
| <a href="#">omdurman-rules/src/effects/state/setup.rs:215</a><br>GH:omdurman-rules/src/effects/state/setup.rs:215     | <b>in_deployment_zone</b> | 213  <br>214  <br>215  <br>216  <br>217 |

Covered by tests: [omdurman-rules::src::effects::tests::campaign\\_deployment\\_is\\_boat\\_land\\_exclusive](#)  
[omdurman-rules::src::effects::tests::fok\\_ae\\_gunboat\\_deploys\\_only\\_on\\_nile](#)  
[omdurman-rules::src::effects::tests::fok\\_ae\\_land\\_unit\\_rejected\\_on\\_nile](#)  
[omdurman-rules::src::effects::tests::deploy\\_via\\_real\\_sprite\\_resolution\\_matches\\_engine](#)  
[omdurman-rules::src::effects::tests::fok\\_dervish\\_land\\_unit\\_rejected\\_on\\_nile](#)  
[omdurman-rules::src::effects::tests::retreat\\_before\\_melee\\_may\\_not\\_land\\_on\\_nile](#)

## §5.23 – Walled city entry restrictions

implemented

manual page unknown

Only certain units may enter the walled portion of Omdurman. For the Dervish player these are the Khalifa unit, the three Dervish artillery units, and the Taiasha units (the Khalifa's bodyguard). Any Anglo-Egyptian units that can get to the walled city may enter it (except gunboats and "Friendlies"). Units entering and/or exiting the walled city may only do so through a gate or breach hexside.

| File                                          | Symbol            | Code Snippet          |
|-----------------------------------------------|-------------------|-----------------------|
| <a href="#">omdurman-types/src/lib.rs:152</a> | <b>HexsideRef</b> | 150  <br>151  <br>152 |

|                                                                                       |                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------------------------------------------------------------------------------------|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GH:omdurman-types/src/lib.rs:152                                                      |                              | <pre> 153   pub a: HexCoord, 154   pub b: HexCoord,</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| omdurman-types/src/lib.rs:187<br>GH:omdurman-types/src/lib.rs:187                     | <b>HexsideKind</b>           | <pre> 185   strum::EnumIter, 186   }} 187   pub enum HexsideKind { 188       /// City wall (Khartoum, walled city of Omdurman). 189       /// movement except across gates/breaches (§5.23), 190       blocks ZOC into the city</pre>                                                                                                                                                                                                                                                                           |
| omdurman-types/src/lib.rs:265<br>GH:omdurman-types/src/lib.rs:265                     | <b>blocks_movement</b>       | <pre> 263       /// `omdurman-rules`. The trench *end* variants 264       are therefore intentionally 265       not blocking. 266       pub fn blocks_movement(self) -&gt; bool { 267           matches!( 268               self,</pre>                                                                                                                                                                                                                                                                         |
| omdurman-rules/src/unit.rs:343<br>GH:omdurman-rules/src/unit.rs:343                   | <b>may_enter_walled_city</b> | <pre> 341       /// Taiasha bodyguard may enter. Anglo-Egyptian: 342       any unit that can reach the 343       /// walled city *except* gunboats and "Friendlies". 344       pub fn may_enter_walled_city(&amp;self) -&gt; bool { 345           match self { 346               // §5.23 Dervish: Khalifa, artillery, 347               Taiasha.</pre>                                                                                                                                                         |
| omdurman-rules/src/board.rs:249<br>GH:omdurman-rules/src/board.rs:249                 | <b>is_walled_city</b>        | <pre> 247       /// are its seeds). The set is derived once from 248       the board data, replacing 249       /// the older "at least two of six hexsides are 250       Wall/Gate/Breach" heuristic. 251       pub fn is_walled_city(&amp;self, hex: HexCoord) -&gt; bool { 252           { 253               // Membership in the precomputed enclosed area 254               (see `walled_city`). 255               // Palace/Tomb hexes are always part of it 256               (they are the seeds).</pre> |
| omdurman-rules/src/effects/error.rs:101<br>GH:omdurman-rules/src/effects/error.rs:101 | <b>WalledCityEntry</b>       | <pre> 99   100       #[error("unit {0:?} is not eligible to enter the 101       walled city of Omdurman at {1:?} (§5.23)")] 102       WalledCityEntry(UnitId, HexCoord), 103   104       #[error("movement cost {cost:?} exceeds allowance 105       {allowance:?}")]</pre>                                                                                                                                                                                                                                     |

Proven

by:

[omdurman-types::src::lib::hexside\\_ref\\_is\\_order\\_independent](#) [omdurman-types::src::lib::hexside\\_blocking\\_classifiers\\_are\\_exact](#)

Covered

by

tests:

[omdurman-rules::src::effects::tests::a\\_wall\\_blocks\\_every\\_step\\_not\\_just\\_the\\_endpoints](#)  
[omdurman-rules::src::effects::tests::can\\_move\\_rejects\\_wall\\_hexside](#) [omdurman-rules::src::effects::tests::can\\_move\\_allows\\_gate\\_hexside](#)  
[omdurman-rules::src::effects::tests::walled\\_city\\_entry\\_allows\\_khalifa](#)  
[omdurman-rules::src::effects::tests::walled\\_city\\_entry\\_rejects\\_unauthorized\\_dervish](#)  
[omdurman-rules::src::effects::tests::walled\\_city\\_entry\\_rejects\\_ae\\_gunboat](#)  
[omdurman-rules::src::effects::tests::walled\\_city\\_entry\\_not\\_enforced\\_for\\_fok](#)  
[omdurman-rules::src::board\\_data::campaign\\_walled\\_city\\_is\\_enclosed\\_by\\_walls](#)  
[omdurman-rules::src::effects::tests::breach\\_is\\_an\\_opening\\_for\\_los\\_and\\_effective\\_kinds](#)

## §5.24 – Gunboat upstream/downstream movement

**implemented**

manual page unknown

Note that gunboats have two movement allowances separated by a slash, e.g. 10/16. The smaller number is the movement allowance when moving upstream, i.e. against the current (the direction of the current is indicated by arrows in the Nile). The larger number is the movement allowance when moving downstream, i.e. with the current. Gunboats may combine movement in both directions, but if they move even one hex upstream, their upstream movement allowance is their maximum movement allowance for that turn, and may not be exceeded.

(see manual for full text)

| File                                                                                | Symbol                 | Code Snippet                                                                                                                                                                                                         |
|-------------------------------------------------------------------------------------|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/unit.rs:243</a><br>GH:omdurman-rules/src/unit.rs:243 | <b>GunboatMovement</b> | <pre>241   /// the turn. 242   #[derive(Serialize, Deserialize, Clone, Copy, 243   pub struct GunboatMovement { 244       pub upstream: MovementAllowance, 245       pub downstream: MovementAllowance,</pre>        |
| <a href="#">omdurman-types/src/lib.rs:976</a><br>GH:omdurman-types/src/lib.rs:976   | <b>is_boat</b>         | <pre>974   975       /// Gunboats use the split upstream/downstream movement allowance (§5.24). 976       pub fn is_boat(self) -&gt; bool { 977           matches!(self, UnitKind::Gunboat { .. }) 978       }</pre> |

Proven by: [omdurman-types::src::lib::unit\\_kind\\_fire\\_and\\_movement\\_capability\\_law\\_is\\_exact](#)  
[omdurman-types::src::lib::hex\\_direction\\_from\\_index\\_is\\_total\\_mod\\_six](#)  
[omdurman-types::src::lib::nile\\_flow\\_rotation\\_composes\\_and\\_never\\_leaves\\_the\\_nile](#)

Covered by tests: [omdurman-rules::src::unit\\_profiles::boat\\_annotation\\_yields\\_split\\_gunboat\\_movement](#)  
[omdurman-rules::src::effects::tests::gunboat\\_upstream\\_cap\\_is\\_sticky\\_across\\_moves](#)  
[omdurman-app::src::picker::movement::gunboat\\_cap\\_gate\\_mirrors\\_the\\_engine](#)  
[omdurman-app::src::picker::movement::remaining\\_mp\\_respects\\_the\\_sticky\\_upstream\\_cap](#)

## §5.25 – Dervish forts may not move

[implemented](#)

manual page unknown

Dervish forts may not move in any way once placed.

| File                                                                                | Symbol                        | Code Snippet                                                                                                                                                                                                                                                                                              |
|-------------------------------------------------------------------------------------|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/unit.rs:506</a><br>GH:omdurman-rules/src/unit.rs:506 | <b>Immobile</b>               | <pre>504   Gunboat(GunboatMovement), 505       /// Forts may not move once placed (§5.25). 506       Immobile, 507   } 508  </pre>                                                                                                                                                                        |
| <a href="#">omdurman-types/src/lib.rs:933</a><br>GH:omdurman-types/src/lib.rs:933   | <b>UnitKind::Fort</b>         | <pre>931       /// Permanent emplacement (§6.54): fire (artillery) / melee (defensive). 932       /// May not move once placed (§5.25). 933       Fort { fire: i32, melee: i32 }, 934       /// Dervish leader (§6.51): fire / melee / movement. May melee attack (§7.4). 935       DervishLeader {</pre> |
| <a href="#">omdurman-rules/src/unit.rs:506</a><br>GH:omdurman-rules/src/unit.rs:506 | <b>UnitMovement::Immobile</b> | <pre>504   Gunboat(GunboatMovement), 505       /// Forts may not move once placed (§5.25). 506       Immobile, 507   } 508  </pre>                                                                                                                                                                        |

Covered by tests: [omdurman-rules::src::effects::tests::forts\\_are\\_never\\_advance\\_eligible](#)  
[omdurman-rules::src::effects::tests::immobile\\_fort\\_rejects\\_move\\_unit](#)

## §5.26 – Units stop on entering enemy ZOC

implemented

manual page unknown

Units must stop their movement immediately upon entering an enemy zone of control (see 5.4).

See also: §5.4

| File                                                                                                                    | Symbol                  | Code Snippet                                                                                                                                                                                                                                       |
|-------------------------------------------------------------------------------------------------------------------------|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/effects/state/movement.rs:43</a><br>GH:omdurman-rules/src/effects/state/movement.rs:43   | <b>can_move_unit_to</b> | <pre>41       /// 42       /// Stacking (§5.51) is checked when the move is          applied, not here. 43       pub fn can_move_unit_to( 44           &amp;self, 45           unit_id: UnitId,</pre>                                              |
| <a href="#">omdurman-rules/src/effects/state/stacking.rs:218</a><br>GH:omdurman-rules/src/effects/state/stacking.rs:218 | <b>hex_in_enemy_zoc</b> | <pre>216       /// does not extend into or out of a Nile hex. With          no board loaded these 217       /// reduce to the plain adjacency rule. 218       pub fn hex_in_enemy_zoc( 219           &amp;self, 220           hex: HexCoord,</pre> |

Covered by tests: [omdurman-rules::src::effects::tests::unit\\_entering\\_enemy\\_zoc\\_may\\_move\\_no\\_further\\_that\\_turn](#)  
[omdurman-rules::src::effects::tests::zoc\\_transit\\_check\\_uses\\_the\\_actual\\_path](#)

## §5.41 – All units except AE leaders exert ZOC

implemented

manual page unknown

All units except Anglo-Egyptian leaders exert a zone of control (hereafter called a ZOC) into their six adjacent hexes (exception: Gunboats exert a ZOC only against enemy gunboats). Disrupted units have no ZOC.

| File                                                                                                                    | Symbol                        | Code Snippet                                                                                                                                                                                                                                                                                                                                                                                                              |
|-------------------------------------------------------------------------------------------------------------------------|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/effects/state/stacking.rs:110</a><br>GH:omdurman-rules/src/effects/state/stacking.rs:110 | <b>unit_projects_zoc_rule</b> | <pre>108       /// * A fort projects ZOC out of its hex even when          unoccupied (§5.44), 109       /// modelled by the fort unit projecting normally. 110       pub fn unit_projects_zoc_rule( 111           unit: &amp;UnitPlacement, 112           mover_player: Player,</pre>                                                                                                                                    |
| <a href="#">omdurman-rules/src/unit.rs:561</a><br>GH:omdurman-rules/src/unit.rs:561                                     | <b>ZocReason</b>              | <pre>559       /// Used by the engine when answering "is this hex in          an enemy ZOC?". 560       #[derive(Serialize, Deserialize, Clone, Copy,          PartialEq, Eq, Debug)] 561       pub enum ZocReason { 562           /// Normal ZOC: any non-disrupted unit other than              an Anglo-Egyptian 563           /// leader (§5.41) projects ZOC into each of its              six adjacent hexes.</pre> |
| <a href="#">omdurman-rules/src/effects/state/stacking.rs:200</a><br>GH:omdurman-rules/src/effects/state/stacking.rs:200 | <b>unit_projects_zoc</b>      | <pre>198       /// §5.44) need the game map, which the engine does          not hold; the app layers 199       /// those on top. This is the position/kind/          disruption core of the rule. 200       pub fn unit_projects_zoc(</pre>                                                                                                                                                                               |

|                                                                                                                           |                  |                                                                                                                                                                                                                       |
|---------------------------------------------------------------------------------------------------------------------------|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| fects/state/<br>stacking.rs:200                                                                                           |                  | 201         &self,<br>202         unit: &UnitPlacement,                                                                                                                                                               |
| <a href="#">omdurman-rules/src/ef-fects/state/stacking.rs:289</a><br>GH:omdurman-rules/src/ef-fects/state/stacking.rs:289 | <b>zoc_hexes</b> | 287         /// ZOC covers a given hex; this function returns<br>*which* hexes a<br>288         /// specific unit covers.<br>289         pub fn zoc_hexes(<br>290         &self,<br>291         unit: &UnitPlacement, |

Proven by: [omdurman-rules::src::effects::unit\\_projects\\_zoc\\_matches\\_manual\\_clauses](#)  
[omdurman-types::src::lib::player\\_opponent\\_is\\_a\\_fixed\\_point\\_free\\_involution](#)

Covered by tests: [omdurman-rules::src::effects::tests::zoc\\_hexes\\_empty\\_for\\_anglo\\_egyptian\\_leader](#)  
[omdurman-rules::src::effects::tests::zoc\\_hexes\\_normal\\_unit\\_projects\\_six\\_adjacent\\_minus\\_exclusions](#)  
[omdurman-rules::src::effects::tests::zoc\\_hexes\\_empty\\_for\\_disrupted\\_unit](#)  
[omdurman-rules::src::effects::tests::zoc\\_hexes\\_matches\\_hex\\_in\\_enemy\\_zoc](#) [omdurman-rules::src::effects::tests::zoc\\_hexes\\_excludes\\_nile](#)  
[omdurman-rules::src::effects::tests::zoc\\_hexes\\_excludes\\_khor](#)

## §5.42 – No MP cost to enter/leave enemy ZOC

**implemented**

manual page unknown

There is no movement point cost to enter or leave an enemy ZOC.

| File                                                                                                                      | Symbol                   | Code Snippet                                                                                                                                                                                       |
|---------------------------------------------------------------------------------------------------------------------------|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/ef-fects/state/movement.rs:437</a><br>GH:omdurman-rules/src/ef-fects/state/movement.rs:437 | <b>movement_cost_for</b> | 435         ///<br>436         /// §5.42: entering or leaving an enemy ZOC adds no<br>MP cost.<br>437         pub fn movement_cost_for(<br>438         &self,<br>439         unit: &UnitPlacement, |

Covered by tests: [omdurman-rules::src::effects::tests::entering\\_enemy\\_zoc\\_costs\\_no\\_extra\\_mp](#)

## §5.43 – Units stop when entering enemy ZOC

**implemented**

manual page unknown

All units must stop when they enter an enemy ZOC and may move no further that turn. In their next movement phase they may withdraw or, if desired, move directly into another enemy ZOC.

| File                                                                                                                    | Symbol                  | Code Snippet                                                                                                                                                                                   |
|-------------------------------------------------------------------------------------------------------------------------|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/ef-fects/state/movement.rs:43</a><br>GH:omdurman-rules/src/ef-fects/state/movement.rs:43 | <b>can_move_unit_to</b> | 41         ///<br>42         /// Stacking (§5.51) is checked when the move is<br>applied, not here.<br>43         pub fn can_move_unit_to(<br>44         &self,<br>45         unit_id: UnitId, |
| <a href="#">omdurman-rules/src/ef-fects/state/</a>                                                                      | <b>hex_in_enemy_zoc</b> | 216         /// does not extend into or out of a Nile hex. With<br>no board loaded these<br>217         /// reduce to the plain adjacency rule.                                                |

|                                                                                     |  |                                                                                            |
|-------------------------------------------------------------------------------------|--|--------------------------------------------------------------------------------------------|
| stacking.rs:218<br>GH:omdurman-<br>rules/src/ef-<br>fects/state/<br>stacking.rs:218 |  | 218       pub fn hex_in_enemy_zoc(<br>219           &self,<br>220           hex: HexCoord, |
|-------------------------------------------------------------------------------------|--|--------------------------------------------------------------------------------------------|

Covered by tests: `omdurman-rules::src::effects::tests::unit_entering_enemy_zoc_may_move_no_further_that_turn`

## §5.44 – ZOC limitations (walls, khor, fort, Nile, Zariba)

**implemented**

manual page unknown

ZOCs do not extend into or out of a Nile River hex (exception: Gunboats, see 5.41). ZOCs do not extend across a khor, into a fort, or into a hex inside the walled city across a wall hexside. ZOCs do extend out of a fort (even if unoccupied), and from a walled city hex into an adjacent non-walled-city hex across a wall hexside. ZOCs also extend out of (but not into) a walled city hex across a gate hexside. ZOCs extend both ways across a breach hexside. ZOCs also extend out of, but not into, a hut or building hex. In the historical scenario ZOCs extend out of, but not into, the Zariba across a Zariba hexside (also in the campaign game if the Zariba is constructed).

(see manual for full text)

See also: §5.41

| File                                                                                | Symbol                   | Code Snippet                                                                                                                                                                                                                                                                                                                                                                               |
|-------------------------------------------------------------------------------------|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| omdurman-<br>rules/src/<br>unit.rs:561<br>GH:omdurman-<br>rules/src/<br>unit.rs:561 | <b>ZocReason</b>         | 559       /// Used by the engine when answering "is this hex in<br>an enemy ZOC?".<br>560       #[derive(Serialize, Deserialize, Clone, Copy,<br>PartialEq, Eq, Debug)]<br>561       pub enum ZocReason {<br>562           /// Normal ZOC: any non-disrupted unit other than<br>an Anglo-Egyptian<br>563           /// leader (§5.41) projects ZOC into each of its<br>six adjacent hexes. |
| omdurman-<br>types/src/<br>lib.rs:192<br>GH:omdurman-<br>types/src/lib.rs:192       | <b>Wall</b>              | 190       /// (§5.44), blocks melee (§7.2), blocks advance-<br>after-combat (§6.82).<br>191       #[default]<br>192       Wall,<br>193       /// Gate hexside in a wall. ZOC extends *out of*<br>the walled city through<br>194       /// gates but not into it (§5.44). Melee may be<br>made through a gate (§7.2).                                                                       |
| omdurman-<br>types/src/<br>lib.rs:201<br>GH:omdurman-<br>types/src/lib.rs:201       | <b>Khor</b>              | 199       /// Khor -- gully/wadi. ZOCs do not extend across<br>(§5.44); advance after<br>200       /// combat may not cross (§6.82).<br>201       Khor,<br>202       /// Crest line. Blocks LOS unless the firer is on<br>the higher side<br>203       /// (§6.3 note 7).                                                                                                                  |
| omdurman-<br>rules/src/<br>unit.rs:574<br>GH:omdurman-<br>rules/src/<br>unit.rs:574 | <b>ZocReason::Zariba</b> | 572       /// Zariba hexside ZOC behaviour in the historical<br>scenario / when the<br>573       /// Zariba is constructed (§5.44).<br>574       Zariba,<br>575       }<br>576                                                                                                                                                                                                             |
| omdurman-<br>rules/src/ef-<br>fects/state/                                          | <b>unit_projects_zoc</b> | 198       /// §5.44) need the game map, which the engine does<br>not hold; the app layers<br>199       /// those on top. This is the position/kind/                                                                                                                                                                                                                                        |

|                                                                                                         |                         |                                                                                                                                                                                                                                                                                                                       |
|---------------------------------------------------------------------------------------------------------|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| stacking.rs:200<br>GH:omdurman-rules/src/effects/state/stacking.rs:200                                  |                         | disruption core of the rule.<br>200       pub fn unit_projects_zoc(<br>201           &self,<br>202           unit: &UnitPlacement,                                                                                                                                                                                    |
| omdurman-types/src/lib.rs:282<br>GH:omdurman-types/src/lib.rs:282                                       | HexsideKind::blocks_zoc | 280       /// cannot express; those are left to the caller.<br>This predicate captures the<br>281       /// symmetric "does not extend across" cases.<br>282       pub fn blocks_zoc(self) -> bool {<br>283           matches!(<br>284               self,                                                            |
| omdurman-rules/src/effects/state/stacking.rs:218<br>GH:omdurman-rules/src/effects/state/stacking.rs:218 | hex_in_enemy_zoc        | 216       /// does not extend into or out of a Nile hex. With<br>no board loaded these<br>217       /// reduce to the plain adjacency rule.<br>218       pub fn hex_in_enemy_zoc(<br>219           &self,<br>220           hex: HexCoord,                                                                             |
| omdurman-rules/src/effects/state/stacking.rs:242<br>GH:omdurman-rules/src/effects/state/stacking.rs:242 | zoc_extends             | 240       ///     only outward ("out of, but not into, a walled<br>city hex"), across a<br>241       ///     breach both ways.<br>242       pub fn zoc_extends(&self, unit: &UnitPlacement,<br>into: HexCoord) -> bool {<br>243           use omdurman_types::HexsideKind;<br>244           let from = unit.position; |

Proven by: omdurman-rules::src::effects::unit\_projects\_zoc\_matches\_manual\_clauses  
omdurman-types::src::lib::hexside\_blocking\_classifiers\_are\_exact

Covered by tests: omdurman-rules::src::effects::tests::zoc\_extends\_out\_of\_but\_not\_into\_buildings\_and\_forts  
omdurman-rules::src::effects::tests::zoc\_crosses\_walls\_and\_gates\_only\_outward  
omdurman-rules::src::effects::tests::zoc\_hexes\_excludes\_nile omdurman-rules::src::effects::tests::zoc\_hexes\_excludes\_khor  
omdurman-rules::src::effects::tests::zoc\_hexes\_normal\_unit\_projects\_six\_adjacent\_minus\_exclusions

## §5.51 – Stacking limit (4 units + leaders, gunboats isolated)

implemented

manual page unknown

No more than four units may occupy a hex, with the exception of leaders and gunboats. All leader units are free stacking, i.e. they may stack in addition to the four-unit-per-hex stacking limitation. Gunboats may not stack with any other unit (Exception: 5.21). Players may move through friendly units at no additional cost in movement points. The stacking limitation applies only at the end of the movement phase and during combat.

See also: §5.21

| File                                                                | Symbol       | Code Snippet                                                                                                                                                                                                                                                                                                                                              |
|---------------------------------------------------------------------|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| omdurman-rules/src/unit.rs:586<br>GH:omdurman-rules/src/unit.rs:586 | OverLimit    | 584       /// (§6.51: a Dervish unit occupying his hex<br>eliminates him).<br>585       #[error("hex stack exceeds the four-unit limit<br>[§5.51]")]<br>586       OverLimit,<br>587       /// "Gunboats may not stack with any other unit"<br>(§5.51, exception §5.21).<br>588       #[error("gunboats may not stack with non-gunboat<br>units [§5.51]")] |
| omdurman-rules/src/unit.rs:589<br>GH:omdurman-                      | GunboatStack | 587       /// "Gunboats may not stack with any other unit"<br>(§5.51, exception §5.21).<br>588       #[error("gunboats may not stack with non-gunboat<br>units [§5.51]")]                                                                                                                                                                                 |

|                                                                                                                                   |                          |                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------------------------------------------------------------------------------------------------------------------------|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| rules/src/<br>unit.rs:589                                                                                                         |                          | <pre> 589   GunboatStack, 590       /// "Units of different Dervish tribes may not       stack together" (§5.52). 591       /// The Dervish artillery (§9.322's three guns) is       not a tribe but is not </pre>                                                                                                                                               |
| omdurman-<br>rules/src/<br>unit.rs:599<br>GH:omdurman-<br>rules/src/<br>unit.rs:599                                               | <b>EnemyCohabitation</b> | <pre> 597       /// lone Anglo-Egyptian leader is exempt (§6.51). 598       #[error("enemy units may not share a hex; melee,       not movement, engages them [§5.51, §7.1]")] 599       EnemyCohabitation, 600       /// "If Dervish leaders elect to stack, they may       only stack with units of 601       /// their command (i.e. colour)" (§5.53). </pre> |
| omdurman-<br>rules/src/ef-<br>fects/state/<br>stacking.rs:151<br>GH:omdurman-<br>rules/src/ef-<br>fects/state/<br>stacking.rs:151 | <b>check_stacking</b>    | <pre> 149       /// Dervish artillery is its own group) may not       stack together. 150       /// * §5.53 -- a Dervish leader may stack only with       units of its command. 151       pub fn check_stacking( 152           &amp;self, 153           mover: &amp;UnitPlacement, </pre>                                                                        |
| omdurman-<br>rules/src/ef-<br>fects/state/<br>stacking.rs:21<br>GH:omdurman-<br>rules/src/effects/<br>state/stacking.rs:21        | <b>stacking_rule</b>     | <pre> 19       /// artillery as its own group) may not share a hex. 20       /// * §5.53: a Dervish leader stacks only with units of       its command. 21       pub fn stacking_rule(occupants: &amp;[&amp;UnitPlacement]) -&gt;       Result&lt;(), crate::StackingError&gt; { 22           use crate::StackingError; 23   </pre>                              |

Proven by:

- [omdurman-rules::src::effects::stacking\\_rule\\_cohabitation\\_is\\_exact](#)
- [omdurman-rules::src::effects::stacking\\_rule\\_is\\_symmetric](#)
- [omdurman-rules::src::effects::stacking\\_rule\\_leaders\\_are\\_free\\_stacking](#)
- [omdurman-rules::src::effects::stacking\\_rule\\_limit\\_is\\_four\\_counted\\_units](#)
- [omdurman-rules::src::effects::stacking\\_rule\\_gunboat\\_never\\_shares](#)

Covered by tests:

- [omdurman-rules::src::effects::tests::stacking\\_over\\_limit\\_rejected](#)
- [omdurman-rules::src::effects::tests::mid\\_move\\_stacking\\_allows\\_pass\\_through](#)
- [omdurman-rules::src::effects::tests::mid\\_move\\_stacking\\_rejects\\_over\\_limit\\_destination](#)
- [omdurman-rules::src::effects::tests::validate\\_stacking\\_invariants\\_clean\\_state](#)
- [omdurman-rules::src::effects::tests::validate\\_stacking\\_invariants\\_catches\\_stacking\\_violation](#)
- [omdurman-rules::src::effects::tests::validate\\_stacking\\_invariants\\_allows\\_leaders\\_stacking](#)
- [omdurman-rules::src::effects::tests::deploy\\_rejects\\_enemy\\_cohabitation\\_during\\_setup](#)
- [omdurman-rules::src::effects::tests::validate\\_stacking\\_invariants\\_catches\\_enemy\\_cohabitation](#)
- [omdurman-rules::src::effects::tests::ae\\_garrison\\_stacks\\_freely\\_under\\_gordon](#)
- [omdurman-rules::src::effects::tests::melee\\_mandatory\\_advance\\_leaders\\_are\\_free\\_stacking](#)

## §5.52 – Different Dervish tribes may not stack together

implemented

manual page unknown

The units of different Dervish tribes may not stack together, even if they are the same color (e.g. although both are green, Mulazmin and Jehadia units may not stack with each other).

| File                                                                                | Symbol                 | Code Snippet                                                                                                                                                                                                                                                                                                                                           |
|-------------------------------------------------------------------------------------|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| omdurman-<br>rules/src/<br>unit.rs:595<br>GH:omdurman-<br>rules/src/<br>unit.rs:595 | <b>DervishTribeMix</b> | <pre> 593       /// [<code>UnitIdentity::dervish_stacking_group`</code>]). 594       #[error("Dervish units of different tribes may not       stack [§5.52]")] 595       DervishTribeMix, 596       /// A unit may never share a hex with enemy units       (§5.51, §7.1); only the 597       /// lone Anglo-Egyptian leader is exempt (§6.51). </pre> |

|                                                                     |                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---------------------------------------------------------------------|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| omdurman-rules/src/unit.rs:395<br>GH:omdurman-rules/src/unit.rs:395 | <b>dervish_stacking_group</b> | <pre> 393       /// `None` for every unit the §5.52 law does not           constrain: Dervish 394       /// leaders, forts and gunboats, and all Anglo-           Egyptian units. 395       pub fn dervish_stacking_group(&amp;self) -&gt;           Option&lt;DervishStackingGroup&gt; { 396           match self { 397               UnitIdentity::DervishTribal { tribe } =&gt;                   Some(DervishStackingGroup::Tribe(*tribe)), </pre> |
|---------------------------------------------------------------------|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Proven by: [omdurman-rules::src::effects::stacking\\_rule\\_group\\_purity\\_is\\_exact](#)

Covered by tests:

- [omdurman-rules::src::unit\\_profiles::green\\_sections\\_are\\_mulazmin\\_tribal\\_units](#)
- [omdurman-rules::src::effects::tests::deploy\\_rejects\\_dervish\\_tribe\\_mix](#)
- [omdurman-rules::src::effects::tests::deploy\\_rejects\\_hadendowa\\_on\\_dervish\\_gun](#)
- [omdurman-rules::src::effects::tests::dervish\\_guns\\_stack\\_with\\_guns\\_and\\_their\\_leader](#)
- [omdurman-rules::src::effects::tests::validate\\_stacking\\_invariants\\_clean\\_state](#)

## §5.53 – Leader stacking with command colour only

**implemented**

manual page unknown

Leader units are not required to stack. If Dervish leaders elect to stack, however, they may only stack with units of their command (i.e. color). For example, Sheik El Din may only stack with Mulazmins or Jehadias.

**\*\*5.54) Anglo-Egyptian Brigade Integrity:\*\*** All British, Sudanese, and Egyptian infantry units have their brigade designation printed in the upper right corner (e.g. “2B” = 2nd British Brigade; “3E” = 3rd Egyptian Brigade, etc.). In any combat phase in which all four infantry battalions belonging to any Anglo-Egyptian infantry brigade are stacked in the same hex they are said to have brigade integrity. Stacks having brigade integrity receive a +1 modifier to their fire combat die roll provided they all fire at the same enemy occupied hex. This modifier is in addition to the normal +1 bonus given to all Anglo-Egyptian direct fire attacks (see 6.24).

(see manual for full text)

See also: §5.54, §6.24

| File                                                                | Symbol                              | Code Snippet                                                                                                                                                                                                                 |
|---------------------------------------------------------------------|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| omdurman-rules/src/unit.rs:603<br>GH:omdurman-rules/src/unit.rs:603 | <b>DervishLeaderCommandMismatch</b> | <pre> 601       /// their command (i.e. colour)" (§5.53). 602       #[error("Dervish leader may only stack with units           of their own command [§5.53]")] 603       DervishLeaderCommandMismatch, 604   } 605   </pre> |

Covered by tests:

- [omdurman-rules::src::effects::tests::dervish\\_leader\\_stacks\\_only\\_with\\_command\\_colour](#)
- [omdurman-rules::src::unit\\_profiles::command\\_owns\\_unit\\_follows\\_tribes\\_and\\_brigades](#)

## §5.54 – Anglo-Egyptian Brigade Integrity

**implemented**

manual page unknown

| File                                                                | Symbol                  | Code Snippet                                                                                                                                                                                                                               |
|---------------------------------------------------------------------|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| omdurman-rules/src/unit.rs:609<br>GH:omdurman-rules/src/unit.rs:609 | <b>BrigadeIntegrity</b> | <pre> 607       /// stack contains all four battalions of a single           Anglo-Egyptian brigade. 608       #[derive(Serialize, Deserialize, Clone, Copy,           PartialEq, Eq, Debug)] 609       pub enum BrigadeIntegrity { </pre> |

|                                                                                     |                                       |                                                                                                                                                                                                                                                                                                                                                                                |
|-------------------------------------------------------------------------------------|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                     |                                       | <pre> 610       None, 611       Integrated(BrigadeId), </pre>                                                                                                                                                                                                                                                                                                                  |
| <a href="#">omdurman-rules/src/unit.rs:459</a><br>GH:omdurman-rules/src/unit.rs:459 | <b>brigade_integrity</b>              | <pre> 457   /// Only a full stack of four battalions qualifies. 458   /// stack and fire, but they receive no brigade- 459   pub fn brigade_integrity(identities: &amp;[UnitIdentity]) - 460       &gt; BrigadeIntegrity { 461       let Some(brigade) = identities.first().and_then( i  462           i.brigade()) else { 463           return BrigadeIntegrity::None; </pre> |
| <a href="#">omdurman-rules/src/unit.rs:609</a><br>GH:omdurman-rules/src/unit.rs:609 | <b>FireModifier::BrigadeIntegrity</b> | <pre> 607   /// stack contains all four battalions of a single 608   Anglo-Egyptian brigade. 609   #[derive(Serialize, Deserialize, Clone, Copy, 610   PartialEq, Eq, Debug)] 611   pub enum BrigadeIntegrity { 612       None, 613       Integrated(BrigadeId), </pre>                                                                                                        |
| <a href="#">omdurman-rules/src/unit.rs:20</a><br>GH:omdurman-rules/src/unit.rs:20   | <b>BattalionOrdinal</b>               | <pre> 18       /// brigade integrity requires all four stacked in 19       one hex (\$5.54). 20       #[derive(Serialize, Deserialize, Clone, Copy, 21   PartialEq, Eq, Hash, Debug, strum::Display)] 22       pub enum BattalionOrdinal { 23           First = 1, 24           Second = 2, </pre>                                                                             |
| <a href="#">omdurman-types/src/lib.rs:1052</a><br>GH:omdurman-types/src/lib.rs:1052 | <b>BrigadeId</b>                      | <pre> 1050   /// same field for uniform handling. 1051   #[derive(Serialize, Deserialize, Clone, Copy, 1052   PartialEq, Eq, PartialOrd, Ord, Hash, Debug)] 1053   pub struct BrigadeId { 1054       pub number: u8, 1055       pub nationality: BrigadeNationality, </pre>                                                                                                    |

Proven by: [omdurman-rules::src::verification::brigade\\_integrity\\_requires\\_all\\_four\\_distinct\\_battalions\\_of\\_one\\_brigade](#)

Covered by tests:

- [omdurman-rules::src::unit\\_profiles::brigade\\_designation\\_ignored\\_for\\_non\\_infantry](#)
- [omdurman-rules::src::unit\\_profiles::printed\\_brigade\\_designation\\_overrides\\_column](#)
- [omdurman-rules::src::unit\\_profiles::section\\_owner\\_anglo\\_egyptian\\_sections](#)
- [omdurman-rules::src::tests::brigade\\_integrity\\_four\\_battalions\\_returns\\_integrated](#)
- [omdurman-rules::src::unit\\_profiles::ae\\_infantry\\_fourth\\_battalion\\_from\\_col\\_3](#)
- [omdurman-rules::src::tests::brigade\\_integrity\\_empty\\_slice](#)
- [omdurman-rules::src::tests::brigade\\_integrity\\_friendly\\_returns\\_none](#)
- [omdurman-rules::src::tests::brigade\\_integrity\\_three\\_battalions\\_returns\\_none](#)
- [omdurman-rules::src::tests::unit\\_identity\\_brigade\\_and\\_battalion\\_accessors](#)
- [omdurman-rules::src::unit\\_profiles::ae\\_infantry\\_brigade\\_number\\_three\\_from\\_col\\_7](#)
- [omdurman-rules::src::tests::brigade\\_integrity\\_non\\_infantry\\_returns\\_none](#)
- [omdurman-rules::src::tests::brigade\\_integrity\\_mixed\\_brigades\\_returns\\_none](#)
- [omdurman-rules::src::unit\\_profiles::ae\\_infantry\\_third\\_battalion\\_from\\_col\\_2](#)
- [omdurman-rules::src::effects::tests::brigade\\_integrity\\_modifier\\_is\\_engine\\_derived](#)

23/31 implemented

## §6 – Fire Combat Phase

### §6 – Fire Combat Phase

descriptive

manual page unknown

Fire Combat Phase

### §6.1 – General Rules

descriptive

manual page unknown

General Rules

### §6.2 – How To Have Fire Combat

descriptive

manual page unknown

How To Have Fire Combat

### §6.3 – Line of Sight Table

implemented

manual page unknown

Line of Sight Table

This table is located on the back of this rulebook and should be self-explanatory. Locate the terrain type the firing unit is in and cross-index it with the terrain type the target unit is in. Terrain types in the intersecting box block line of sight, with exceptions as footnoted. Also study the “Special LOS Notes” given and remember that (with the exception of howitzer fire — see 6.64) you can’t fire at anything you can’t see!

See also: §6.64

| File                                                                                          | Symbol              | Code Snippet                                                                                                                                                                                                                                                                   |
|-----------------------------------------------------------------------------------------------|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/los_table.rs:57</a><br>GH:omdurman-rules/src/los_table.rs:57   | <b>LosLevel</b>     | <pre>55       serde::Serialize, serde::Deserialize, Clone, Copy, 56       PartialEq, Eq, PartialOrd, Ord, Hash, Debug, 57   )] 58   pub enum LosLevel { 59       Ground,     Rough,</pre>                                                                                      |
| <a href="#">omdurman-rules/src/los_table.rs:81</a><br>GH:omdurman-rules/src/los_table.rs:81   | <b>LosFeature</b>   | <pre>79   /// authored RON spellings. 80   #[derive(serde::Serialize, serde::Deserialize, Clone, 81   Copy, PartialEq, Eq, Hash, Debug)] 82   pub enum LosFeature { 83       /// A hex containing units (gunboats/forts excluded     per note a).     Units,</pre>             |
| <a href="#">omdurman-rules/src/los_table.rs:103</a><br>GH:omdurman-rules/src/los_table.rs:103 | <b>LosCondition</b> | <pre>101   /// A positional condition from the LOS table Detail     footnotes. 102   #[derive(serde::Serialize, serde::Deserialize, Clone, 103   Copy, PartialEq, Eq, Hash, Debug)] 104   pub enum LosCondition {     /// (1) Blocks only if the ray passes through more</pre> |

|                                                                                               |                                |                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------------------------------------------------------------------------------------|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                               |                                | than two such features.<br><b>105</b>       MoreThanTwo,                                                                                                                                                                                                                                                                                                         |
| <a href="#">omdurman-rules/src/los_table.rs:183</a><br>GH:omdurman-rules/src/los_table.rs:183 | <b>los_level</b>               | <b>181</b>       ///<br><b>182</b>       /// For all other units, the level is derived from the terrain at `hex`.<br><b>183</b>       pub fn los_level_for_unit(<br><b>184</b>           kind: UnitKind,<br><b>185</b>           hex: HexCoord,                                                                                                                  |
| <a href="#">omdurman-rules/src/los_table.rs:183</a><br>GH:omdurman-rules/src/los_table.rs:183 | <b>los_level_for_unit</b>      | <b>181</b>       ///<br><b>182</b>       /// For all other units, the level is derived from the terrain at `hex`.<br><b>183</b>       pub fn los_level_for_unit(<br><b>184</b>           kind: UnitKind,<br><b>185</b>           hex: HexCoord,                                                                                                                  |
| <a href="#">omdurman-rules/src/los_table.rs:223</a><br>GH:omdurman-rules/src/los_table.rs:223 | <b>blocking_rules</b>          | <b>221</b>       /// always blocks. Indexing is in-bounds by construction (both enums have<br><b>222</b>       /// exactly three variants).<br><b>223</b>       pub fn blocking_rules(firer: LosLevel, target: LosLevel) -> &'static [BlockingRule] {<br><b>224</b>           crate::tables_data::LOS_CELLS[firer.index()][target.index()]<br><b>225</b>       } |
| <a href="#">omdurman-rules/src/los_table.rs:558</a><br>GH:omdurman-rules/src/los_table.rs:558 | <b>has_los</b>                 | <b>556</b>       /// board-only tools).<br><b>557</b>       #[allow(clippy::too_many_arguments)]<br><b>558</b>       pub fn has_los(<br><b>559</b>           board: &crate::board::BoardInfo,<br><b>560</b>           from: HexCoord,                                                                                                                            |
| <a href="#">omdurman-types/src/lib.rs:230</a><br>GH:omdurman-types/src/lib.rs:230             | <b>HexsideKind::blocks_los</b> | <b>228</b>       /// (LOS table conditions 2-4, 7) and note (e) are handled by the engine<br><b>229</b>       /// in `omdurman_rules::los_table`, not by this predicate.<br><b>230</b>       pub fn blocks_los(self) -> bool {<br><b>231</b>           matches!(self, HexsideKind::Wall   HexsideKind::Crest)<br><b>232</b>       }                              |
| <a href="#">omdurman-types/src/lib.rs:230</a><br>GH:omdurman-types/src/lib.rs:230             | <b>Terrain::blocks_los</b>     | <b>228</b>       /// (LOS table conditions 2-4, 7) and note (e) are handled by the engine<br><b>229</b>       /// in `omdurman_rules::los_table`, not by this predicate.<br><b>230</b>       pub fn blocks_los(self) -> bool {<br><b>231</b>           matches!(self, HexsideKind::Wall   HexsideKind::Crest)<br><b>232</b>       }                              |
| <a href="#">omdurman-types/src/lib.rs:459</a><br>GH:omdurman-types/src/lib.rs:459             | <b>Terrain::is_los_trees</b>   | <b>457</b>       /// (§6.3 note 1). Retained for compatibility; the full LOS engine<br><b>458</b>       /// checks `Terrain::Trees` directly.<br><b>459</b>       pub fn is_los_trees(self) -> bool {<br><b>460</b>           matches!(self, Terrain::Trees { .. })<br><b>461</b>       }                                                                        |

Proven

by: [omdurman-types::src::lib::line\\_between\\_forms\\_a\\_connected\\_ray](#) [omdurman-types::src::lib::hexside\\_blocking\\_classifiers\\_are\\_exact](#)  
[omdurman-rules::src::los\\_table::los\\_level\\_depends\\_only\\_on\\_the\\_ground](#)  
[omdurman-rules::src::los\\_table::los\\_level\\_overrides\\_hold\\_for\\_gunboats\\_and\\_forts](#)  
[omdurman-rules::src::los\\_table::blocking\\_grid\\_is\\_total\\_and\\_walls\\_block\\_ground\\_firers](#)

Covered by tests: [omdurman-rules::src::los\\_table::los\\_level\\_mapping](#) [omdurman-rules::src::los\\_table::blocking\\_rules\\_all\\_cells\\_covered](#)  
[omdurman-rules::src::los\\_table::has\\_los\\_empty\\_board\\_is\\_clear](#) [omdurman-rules::src::los\\_table::has\\_los\\_adjacent\\_clear](#)  
[omdurman-rules::src::los\\_table::has\\_los\\_howitzzer\\_bypasses](#)  
[omdurman-rules::src::los\\_table::has\\_los\\_wall\\_hexside\\_blocks\\_ground\\_to\\_ground](#)  
[omdurman-rules::src::los\\_table::has\\_los\\_gate\\_hexside\\_passes](#) [omdurman-rules::src::los\\_table::has\\_los\\_breach\\_hexside\\_passes](#)  
[omdurman-rules::src::effects::tests::breach\\_is\\_an\\_opening\\_for\\_los\\_and\\_effective\\_kinds](#)  
[omdurman-rules::src::los\\_table::has\\_los\\_rough\\_intervening\\_blocks\\_ground\\_to\\_ground](#)

```

omdurman-rules::src::los_table::has_los_two_tree_hexes_pass_ground_to_ground
omdurman-rules::src::los_table::has_los_three_tree_hexes_block_ground_to_ground
omdurman-rules::src::los_table::has_los_two_hut_hexes_pass_ground_to_ground
omdurman-rules::src::los_table::has_los_three_hut_hexes_block_ground_to_ground
omdurman-rules::src::los_table::has_los_hilltop_to_hilltop_clear_no_units
omdurman-rules::src::los_table::has_los_hilltop_to_hilltop_blocked_by_hilltop_unit
omdurman-rules::src::los_table::has_los_hilltop_to_hilltop_not_blocked_by_ground_unit
omdurman-rules::src::los_table::has_los_rough_to_rough_unit_at_lower_level_passes
omdurman-rules::src::los_table::has_los_rough_to_rough_unit_at_same_level_blocks
omdurman-rules::src::los_table::has_los_rough_to_rough_hilltop_blocks
omdurman-rules::src::los_table::has_los_ground_to_hilltop_intervening_hilltop_blocks
omdurman-rules::src::los_table::has_los_building_blocks_like_huts_ground_to_ground
omdurman-rules::src::los_table::has_los_two_building_hexes_pass_ground_to_ground
omdurman-rules::src::los_table::los_level_for_unit_gunboat_is_rough
omdurman-rules::src::los_table::los_level_for_unit_fort_is_ground
omdurman-rules::src::los_table::los_level_for_unit_walled_city_adj_wall_is_rough
omdurman-rules::src::los_table::gunboat_firer_uses_rough_row_not_ground
omdurman-rules::src::los_table::los_reflexive_all_terrains
omdurman-rules::src::los_table::los_reflexive_hilltop
omdurman-rules::src::los_table::los_symmetric_ground_to_ground_no_units
omdurman-rules::src::los_table::los_howitzer_always_has_los
omdurman-rules::src::los_table::los_howitzer_same_hex
omdurman-rules::src::los_table::los_blocking_rules_match_reference_table
omdurman-rules::src::los_table::los_ground_to_ground_features_block_as_expected

```

## §6.4 – Fire Combat Sequence

implemented

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### Fire Combat Sequence

The sequence of fire combat resolution is the same for both defensive and offensive fire combat. During the Dervish player turn the Anglo-Egyptian player executes 6.41 AND 6.42 as defensive fire, after which the Dervish player executes 6.41 as offensive fire. During the Anglo-Egyptian player turn the Dervish player executes 6.41 as defensive fire, after which the Anglo-Egyptian player executes 6.41 AND 6.42 as offensive fire.

**\*\*6.41) Direct Fire Subphase (Dervish and Anglo-Egyptian players):\*\*** The firing player must first allocate all of his fire attacks, combining his units' direct fire combat factors in any manner he wishes. After all fire has been allocated, the firing player then resolves his attacks in any order he wishes.

**\*\*6.42) Maxim Second Fire and Howitzer Fire Subphase (Anglo-Egyptian player only):\*\*** Anglo-Egyptian named gunboats may now fire their artillery factor as howitzer fire (see 6.64) and all Maxim guns may fire a second time. Once

(see manual for full text)

See also: §6.41, §6.42, §6.64

| File                                                                                                  | Symbol              | Code Snippet                                                                                                                                                                                                                                                                                     |
|-------------------------------------------------------------------------------------------------------|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/turn.rs:44</a><br>GH:omdurman-rules/src/turn.rs:44                     | <b>FireSubPhase</b> | <pre> 42   43   #[derive(Serialize, Deserialize, Clone, Copy, 44           PartialEq, Eq, Hash, Debug)] 45   pub enum FireSubPhase { 46       /// Direct fire (§6.41). Both sides participate in     this sub-phase. 47       DirectFire, </pre>                                                 |
| <a href="#">omdurman-rules/src/effects/effect.rs:73</a><br>GH:omdurman-rules/src/effects/effect.rs:73 | <b>FireCombat</b>   | <pre> 71       /// - Firers marked as fired; target hex marked as     fired-at. 72       /// - Victory points awarded for eliminations. 73       FireCombat { attack: FireAttack, roll: DieRoll }, 74   75       /// Resolve a howitzer bombardment (two rolls: CRT     + impact scatter) </pre> |

Covered by tests: [omdurman-rules::src::effects::tests::turn\\_advances\\_through\\_phases](#)

§6.5 – Special Unit Capabilities

descriptive  
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Special Unit Capabilities

**\*\*6.51) Leader Units:\*\*** Dervish leader units have a fire factor, a melee factor, and a movement factor. They may thus attack, melee, and be eliminated like any other combat unit. Their special benefit is that they stack free.

Anglo-Egyptian leaders have a movement factor only. They are eliminated if a) they are alone in a hex when a Dervish unit occupies or passes through that hex, or b) if all of the combat units a leader is stacked with are eliminated in fire combat or melee. The special function of Anglo-Egyptian leaders is that at least one must survive to occupy the Mahdi’s tomb hex if it is to be taken from the Dervish player (see 9.14).

**\*\*6.52) Anglo-Egyptian “Friendlies” Brigade:\*\*** These units represent native volunteers in the Anglo-Egyptian army. They fire rifles on the Dervish Range Effects Table and melee with the Dervish melee modifier. They may not enter the walled city of Omdurman (see 5.23). They may be transferred to the west bank (see 5.21).

(see manual for full text)

See also: §5.21, §5.23, §5.44, §6.51, §6.52, §6.53, §6.54, §6.62, §6.63, §7.6, §9.14

§6.6 – Special Artillery Capabilities

descriptive  
manual page unknown

Special Artillery Capabilities

§6.7 – Defensive Fire

implemented  
manual page unknown

Defensive Fire

In Defensive Fire phase, all of the non-moving player’s units may fire at any of the moving player’s units in range, within the limitations imposed by the rules of combat (see 6.1 to 6.6). There is no advance after combat as a result of defensive fires.

See also: §6.1, §6.6

| File                                                                                              | Symbol                 | Code Snippet                                                                                                                                                                                                                                                                                                                                                                                                            |
|---------------------------------------------------------------------------------------------------|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| omdurman-rules/src/effects/state/melee.rs:141<br>GH:omdurman-rules/src/effects/state/melee.rs:141 | can_advance_after_comb | 139   // player's unit, not artillery, adjacent to `to`,<br>and `to` now empty.<br>140   // Wall/khor hexside restrictions are not enforced<br>(no hexside map data).<br>141   pub fn can_advance_after_combat(&self, unit_id:<br>UnitId, to: HexCoord) -> Result<(), RuleError> {<br>142   let unit = self.unit_or_err(unit_id)?;<br>143   // §6.7: there is no advance after combat as a<br>result of defensive fire. |

Covered by tests: omdurman-rules::src::effects::tests::no\_advance\_after\_defensive\_fire  
omdurman-rules::src::effects::tests::defensive\_fire\_opens\_no\_advance\_window

§6.8 – Offensive Fire

descriptive  
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Offensive Fire

## §6.11 – Fire combat factor printed on units

implemented

manual page unknown

The fire combat factor of the various unit types is printed directly on the units and is a numerical expression of the unit's fire strength.

| File                                                                                    | Symbol            | Code Snippet                                                                                                                                                                                                                                                |
|-----------------------------------------------------------------------------------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/scalars.rs:18</a><br>GH:omdurman-rules/src/scalars.rs:18 | <b>FireFactor</b> | <pre>16   // Every possible value from the annotated counter 17   set is a named variant. 18   #[derive(Serialize, Deserialize, Clone, Copy, 19   PartialEq, Eq, Hash, Debug, strum::Display)] 20   pub enum FireFactor {     One = 1,     Three = 3,</pre> |

Covered by tests: [omdurman-rules::src::tests::fire\\_factor\\_sum\\_to\\_row](#)

## §6.12 – Fire combat is always voluntary

implicit

manual page unknown

Fire combat is always voluntary. A unit is never required to fire at enemy units merely because they are in range or adjacent.

## §6.13 – Fire factor is unitary (may not be divided)

implemented

manual page unknown

If a unit elects to fire, its fire combat factor at an enemy unit, that fire combat factor is unitary. A unit's fire combat factor may not be divided up to fire at enemy units on different hexes.

| File                                                                                                | Symbol                        | Code Snippet                                                                                                                                                                                                                                     |
|-----------------------------------------------------------------------------------------------------|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/effects/fire.rs:561</a><br>GH:omdurman-rules/src/effects/fire.rs:561 | <b>build_fire_attack_from</b> | <pre>559   // derived exactly as in [`build_fire_attack`]. 560   Returns `None` for an empty, 561   duplicated, or not-fully-legal list. 562   pub fn build_fire_attack_from( 563       gs: &amp;GameState, 564       firer_hex: HexCoord,</pre> |

Covered by tests: [omdurman-rules::src::effects::tests::fire\\_rejects\\_duplicate\\_firers\\_and\\_a\\_foreign\\_firing\\_player](#)  
[omdurman-rules::src::effects::tests::single\\_unit\\_fires\\_alone\\_from\\_a\\_stack](#)

## §6.14 – Players may combine fire factors into one attack

implemented

manual page unknown

Players may combine fire during fire combat phase, i.e. they may fire at an enemy-occupied hex with as many friendly units as may legally do so, combining all of their fire combat factors into one attack. Note that in any given fire combat phase, however, a combat unit may only fire once and may only be fired at once (exceptions: Maxim guns and gunboats — see 6.4).

See also: §6.4

| File                                                                                                          | Symbol                              | Code Snippet                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|---------------------------------------------------------------------------------------------------------------|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/scalars.rs:32</a><br>GH:omdurman-rules/src/scalars.rs:32                       | <b>sum_to_row</b>                   | <pre> 30   impl FireFactor { 31       /// Sum multiple fire factors and return the     corresponding Combat Results Table row (rulebook §6.11). 32       pub fn sum_to_row&lt;'a&gt;(factors: impl     IntoIterator&lt;Item = &amp;'a FireFactor&gt;) -&gt; FireFactorRow { 33           let total: u16 = factors.into_iter().map( f      f.value()).sum(); 34           crate::combat_results_table::FireFactorRow::from_total(total) </pre> |
| <a href="#">omdurman-rules/src/effects/state/fire.rs:49</a><br>GH:omdurman-rules/src/effects/state/fire.rs:49 | <b>GameS-<br/>tate::can_fire_at</b> | <pre> 47       /// modifier in the [<code>FireAttack</code>] and is     responsible for the LOS gate. 48       /// (Howitzer fire ignores LOS entirely -- §6.64.) 49       pub fn can_fire_at( 50           &amp;self, 51           firer: UnitId, </pre>                                                                                                                                                                                     |

Covered by tests: [omdurman-rules::src::effects::tests::dead\\_units\\_are\\_pruned\\_from\\_every\\_tracker](#)  
[omdurman-rules::src::effects::tests::unit\\_may\\_only\\_be\\_fired\\_at\\_once\\_per\\_phase](#)  
[omdurman-rules::src::effects::tests::gunboat\\_and\\_maxim\\_may\\_be\\_fired\\_at\\_repeatedly](#)  
[omdurman-rules::src::effects::tests::rejected\\_fire\\_attack\\_does\\_not\\_mark\\_firers\\_as\\_fired](#)  
[omdurman-rules::src::effects::tests::single\\_unit\\_fires\\_alone\\_from\\_a\\_stack](#)  
[omdurman-rules::src::effects::tests::infantry\\_may\\_fire\\_only\\_once\\_per\\_subphase](#)  
[omdurman-rules::src::effects::tests::maxim\\_may\\_fire\\_twice\\_per\\_turn](#)  
[omdurman-rules::src::effects::tests::fire\\_from\\_two\\_hexes\\_combines\\_into\\_one\\_attack](#)

## §6.15 – May divide a stack to fire at different hexes

**implemented**

manual page unknown

Players may also divide a stack of units in order to fire at different enemy-occupied hexes. Anglo-Egyptian infantry units having brigade integrity, however, do not receive their +1 direct fire modifier unless they all fire at the same enemy-occupied hex (see 5.54).

See also: §5.54

| File                                                                                                | Symbol                        | Code Snippet                                                                                                                                                                                                                                                                                    |
|-----------------------------------------------------------------------------------------------------|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/effects/fire.rs:561</a><br>GH:omdurman-rules/src/effects/fire.rs:561 | <b>build_fire_attack_from</b> | <pre> 559       /// derived exactly as in [<code>build_fire_attack</code>].     Returns <code>None</code> for an empty, 560       /// duplicated, or not-fully-legal list. 561       pub fn build_fire_attack_from( 562           gs: &amp;GameState, 563           firer_hex: HexCoord, </pre> |

Covered by tests: [omdurman-rules::src::effects::tests::stack\\_may\\_split\\_its\\_fire\\_across\\_hexes](#)

## §6.16 – Halving fire strength rounds down, minimum 1

**implemented**

manual page unknown

When halving fire combat strength, always round down each individual unit. For example, an Egyptian brigade of four battalions, each having a printed strength of 9 fire factors, will fire a total of 16 factors when halved. However, a unit's firing strength is never reduced below one by halving.

| File                                | Symbol           | Code Snippet                                                                                                               |
|-------------------------------------|------------------|----------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/</a> | <b>RangeBand</b> | <pre> 208       /// multiplied at a given distance (§6.22). 209       #[derive(Serialize, Deserialize, Clone, Copy, </pre> |

|                                                          |  |                                                                                                  |
|----------------------------------------------------------|--|--------------------------------------------------------------------------------------------------|
| unit.rs:210<br>GH:omdurman-<br>rules/src/<br>unit.rs:210 |  | PartialEq, Eq, Debug))<br>210   pub enum RangeBand {<br>211       Tripled,<br>212       Doubled, |
|----------------------------------------------------------|--|--------------------------------------------------------------------------------------------------|

Proven by: [omdurman-rules::src::verification::range\\_band\\_halved\\_is\\_max\\_of\\_one\\_and\\_floor\\_half](#)  
[omdurman-rules::src::verification::range\\_band\\_apply\\_is\\_monotone\\_in\\_raw](#)  
[omdurman-rules::src::verification::disrupt\\_half\\_is\\_rounded\\_up](#)

Covered by tests: [omdurman-rules::src::tests::halving\\_rounds\\_down\\_and\\_never\\_below\\_one](#)

## §6.21 – First check LOS before firing

[implemented](#)

manual page unknown

When combat units wish to fire at enemy units, first check the Line of Sight Table to be sure the firing unit can see the target hex (exception: howitzer fire, see 6.64).

See also: §6.64

| File                                                                                          | Symbol                         | Code Snippet                                                                                                                                                                                                                                                                                                        |
|-----------------------------------------------------------------------------------------------|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/los_table.rs:223</a><br>GH:omdurman-rules/src/los_table.rs:223 | <a href="#">blocking_rules</a> | 221   /// always blocks. Indexing is in-bounds by<br>construction (both enums have<br>222   /// exactly three variants).<br>223   pub fn blocking_rules(firer: LosLevel, target:<br>LosLevel) -> &'static [BlockingRule] {<br>224       crate::tables_data::LOS_CELLS[firer.index()<br>[target.index()]]<br>225   } |
| <a href="#">omdurman-rules/src/los_table.rs:558</a><br>GH:omdurman-rules/src/los_table.rs:558 | <a href="#">has_los</a>        | 556   /// board-only tools).<br>557   #[allow(clippy::too_many_arguments)]<br>558   pub fn has_los(<br>559       board: &crate::board::BoardInfo,<br>560       from: HexCoord,                                                                                                                                      |

Covered by tests: [omdurman-rules::src::effects::tests::can\\_fire\\_at\\_rejects\\_blocked\\_los](#) [omdurman-rules::src::effects::tests::can\\_fire\\_at\\_allows\\_clear\\_los](#)

## §6.22 – Consult Range Effects Table

[implemented](#)

manual page unknown

Next consult the Range Effects Table to see if the firing unit's fire combat factor is tripled, doubled, normal, halved, or if the target hex is out of range. Add up the total number of fire combat factors firing at the enemy-occupied hex.

| File                                                                                                | Symbol                                | Code Snippet                                                                                                                                                                                                                                                                                                     |
|-----------------------------------------------------------------------------------------------------|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/range_effects.rs:34</a><br>GH:omdurman-rules/src/range_effects.rs:34 | <a href="#">ae_range_effects</a>      | 32   /// Look up the range band for an Anglo-Egyptian weapon<br>(\$6.22, \$6.24).<br>33   /// Distances > 10 are out of range for all weapons.<br>34   pub fn ae_range_effects(weapon: WeaponClass, distance:<br>HexDistance) -> RangeBand {<br>35       band_at(faction_rows(true), weapon, distance)<br>36   } |
| <a href="#">omdurman-rules/src/range_effects.rs:40</a><br>GH:omdurman-                              | <a href="#">dervish_range_effects</a> | 38   /// Look up the range band for a Dervish weapon<br>(\$6.22).<br>39   /// Distances > 10 are out of range for all weapons.<br>40   pub fn dervish_range_effects(weapon: WeaponClass,                                                                                                                         |

|                                                                                           |                    |                                                                                                                                                                                                                                 |
|-------------------------------------------------------------------------------------------|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| rules/src/<br>range_effects.rs:40                                                         |                    | distance: HexDistance) -> RangeBand {<br>41       band_at(faction_rows(false), weapon, distance)<br>42   }                                                                                                                      |
| omdurman-<br>rules/src/<br>unit.rs:210<br>GH:omdurman-<br>rules/src/<br>unit.rs:210       | <b>RangeBand</b>   | 208   /// multiplied at a given distance (§6.22).<br>209   #[derive(Serialize, Deserialize, Clone, Copy,<br>PartialEq, Eq, Debug)]<br>210   pub enum RangeBand {<br>211       Tripled,<br>212       Doubled,                    |
| omdurman-<br>rules/src/<br>scalars.rs:119<br>GH:omdurman-<br>rules/src/<br>scalars.rs:119 | <b>HexDistance</b> | 117   /// (rulebook §6.22, §7.5).<br>118   #[derive(Serialize, Deserialize, Clone, Copy,<br>PartialEq, Eq, PartialOrd, Ord, Hash, Debug)]<br>119   pub struct HexDistance(pub(crate) u16);<br>120  <br>121   impl HexDistance { |

Proven by: [omdurman-types::src::lib::adjacency\\_iff\\_distance\\_one](#)  
[omdurman-rules::src::verification::fire\\_factor\\_row\\_from\\_total\\_matches\\_printed\\_bands](#)  
[omdurman-rules::src::verification::range\\_band\\_multiplier\\_arithmetic\\_is\\_exact](#)  
[omdurman-rules::src::range\\_effects::range\\_effects\\_are\\_out\\_of\\_range\\_outside\\_the\\_printed\\_ten\\_hex\\_distance\\_window](#)  
[omdurman-rules::src::range\\_effects::max\\_day\\_range\\_is\\_the\\_last\\_in\\_range\\_hex](#)  
[omdurman-rules::src::effects::fire::range\\_band\\_for\\_routes\\_to\\_the\\_right\\_faction\\_table](#)

Covered by tests: [omdurman-rules::src::effects::tests::numbered\\_result\\_eliminate\\_without\\_disrupting\\_survivors](#)  
[omdurman-rules::src::range\\_effects::ae\\_rifles\\_doubled\\_at\\_range\\_1](#) [omdurman-rules::src::range\\_effects::ae\\_rifles\\_halved\\_at\\_range\\_4](#)  
[omdurman-rules::src::range\\_effects::ae\\_howitzer\\_range](#) [omdurman-rules::src::range\\_effects::dervish\\_rifles\\_shorter\\_range](#)  
[omdurman-rules::src::range\\_effects::melee\\_only\\_range\\_1](#) [omdurman-rules::src::range\\_effects::ae\\_range\\_effects\\_artillery\\_full](#)  
[omdurman-rules::src::range\\_effects::ae\\_range\\_effects\\_maxims\\_match\\_rifles](#)  
[omdurman-rules::src::range\\_effects::ae\\_range\\_effects\\_distance\\_over\\_10](#)  
[omdurman-rules::src::range\\_effects::ae\\_range\\_effects\\_howitzer\\_halved\\_4\\_to\\_10](#)  
[omdurman-rules::src::range\\_effects::dervish\\_range\\_effects\\_rifles](#) [omdurman-rules::src::range\\_effects::dervish\\_range\\_effects\\_artillery](#)  
[omdurman-rules::src::range\\_effects::dervish\\_range\\_effects\\_maxims\\_and\\_howitzer](#)  
[omdurman-rules::src::range\\_effects::dervish\\_range\\_effects\\_melee](#)  
[omdurman-rules::src::range\\_effects::dervish\\_range\\_effects\\_distance\\_over\\_10](#)  
[omdurman-rules::src::effects::tests::fire\\_combat\\_eliminate\\_target](#) [omdurman-rules::src::range\\_effects::max\\_day\\_range\\_all\\_combos](#)  
[omdurman-rules::src::range\\_effects::range\\_effects\\_every\\_cell\\_ae\\_artillery](#)  
[omdurman-rules::src::range\\_effects::range\\_effects\\_every\\_cell\\_ae\\_howitzer](#)  
[omdurman-rules::src::range\\_effects::range\\_effects\\_every\\_cell\\_ae\\_maxims](#)  
[omdurman-rules::src::range\\_effects::range\\_effects\\_every\\_cell\\_ae\\_rifles](#)  
[omdurman-rules::src::range\\_effects::range\\_effects\\_every\\_cell\\_dervish\\_artillery](#)  
[omdurman-rules::src::range\\_effects::range\\_effects\\_every\\_cell\\_dervish\\_maxims\\_howitzer](#)  
[omdurman-rules::src::range\\_effects::range\\_effects\\_every\\_cell\\_dervish\\_rifles](#)  
[omdurman-rules::src::range\\_effects::range\\_effects\\_every\\_cell\\_dervish\\_spears](#)  
[omdurman-rules::src::effects::tests::can\\_fire\\_at\\_gates\\_phase\\_range\\_and\\_player](#)  
[omdurman-rules::src::effects::tests::mixed\\_attack\\_bands\\_per\\_firer](#)  
[omdurman-rules::src::range\\_effects::ae\\_range\\_effects\\_monotone\\_non\\_increasing](#)  
[omdurman-rules::src::range\\_effects::ae\\_howitzer\\_has\\_minimum\\_range](#)  
[omdurman-rules::src::range\\_effects::dervish\\_range\\_effects\\_monotone\\_non\\_increasing](#)  
[omdurman-rules::src::range\\_effects::range\\_effects\\_first\\_range\\_max\\_effect\\_last\\_range\\_oor](#)

## §6.23 – Terrain defensive modifier

**implemented**

manual page unknown

Next check the Terrain Effects Chart to see if the enemy-occupied hex fired upon contains any terrain which gives the enemy units in that hex a defensive benefit. If so, apply this negative modifier to the roll of the ten-sided die and cross-index your net die roll on the Combat Results Table with the number of combat factors firing.

| File | Symbol | Code Snippet |
|------|--------|--------------|
|------|--------|--------------|

|                                                                                     |                              |                                                                                                                                                                                                                                                               |
|-------------------------------------------------------------------------------------|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| omdurman-rules/src/terrain_chart.rs:59<br>GH:omdurman-rules/src/terrain_chart.rs:59 | <b>defense_modifier</b>      | <pre> 57   58   /// Convenience: get the defense modifier for a terrain       type (rulebook §6.23, Terrain Effects Chart). 59   pub fn defense_modifier(terrain: Terrain) -&gt; i16 { 60       terrain_effects_chart(terrain).defense_modifier 61   } </pre> |
| omdurman-rules/src/combat.rs:26<br>GH:omdurman-rules/src/combat.rs:26               | <b>FireModifier::Terrain</b> | <pre> 24       /// Negative modifier from the Terrain Effects       Chart applied to the 25       /// defender's hex (§6.23). 26       Terrain(i16), 27       /// -2 thorn-hedge defensive modifier (§9.231). 28       ZaribaThornHedge, </pre>               |

Proven by: [omdurman-rules::src::verification::fire\\_modifier\\_keeps\\_roll\\_legal](#)  
[omdurman-rules::src::terrain\\_chart::defence\\_column\\_never\\_helps\\_the\\_attacker](#)

Covered by tests:

- [omdurman-rules::src::terrain\\_chart::engine\\_matches\\_the\\_transcribed\\_terrain\\_effects\\_chart](#)
- [omdurman-rules::src::effects::tests::fire\\_across\\_a\\_crest\\_or\\_wall\\_is\\_penalised](#)
- [omdurman-rules::src::terrain\\_chart::clear\\_terrain\\_no\\_bonus](#)
- [omdurman-rules::src::terrain\\_chart::building\\_gives\\_minus\\_3](#)
- [omdurman-rules::src::terrain\\_chart::trees\\_cost\\_one\\_and\\_give\\_no\\_defense](#)
- [omdurman-rules::src::terrain\\_chart::rough\\_movement\\_and\\_defense](#)
- [omdurman-rules::src::terrain\\_chart::swamp\\_movement\\_and\\_defense](#)
- [omdurman-rules::src::terrain\\_chart::hilltop\\_movement\\_and\\_defense](#)
- [omdurman-rules::src::terrain\\_chart::huts\\_movement\\_and\\_defense](#)
- [omdurman-rules::src::terrain\\_chart::defense\\_modifier\\_convenience\\_matches\\_chart](#)
- [omdurman-rules::src::terrain\\_chart::terrain\\_movement\\_costs\\_in\\_bounds](#)
- [omdurman-rules::src::terrain\\_chart::terrain\\_defense\\_modifier\\_non\\_positive](#)

## §6.24 – Anglo-Egyptian direct fire accuracy bonus and brigade integrity

**implemented**

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All Anglo-Egyptian direct fire attacks receive a +1 modifier to their die roll as an accuracy bonus. In addition, any stack of Anglo-Egyptian infantry having brigade integrity (see 5.54) receives a +1 modifier to their die roll if all four fire at the same enemy-occupied hex. These modifiers are cumulative.

See also: §5.54

| File                                                                  | Symbol                            | Code Snippet                                                                                                                                                                                                                                                                                                                                         |
|-----------------------------------------------------------------------|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| omdurman-rules/src/combat.rs:20<br>GH:omdurman-rules/src/combat.rs:20 | <b>AngloEgyptianDirectFire</b>    | <pre> 18   pub enum FireModifier { 19       /// +1 to all Anglo-Egyptian *direct* fire (§6.24). 20       AngloEgyptianDirectFire, 21       /// +1 brigade integrity, applied only if all four       battalions fire at 22       /// the same enemy-occupied hex (§5.54, §6.24). </pre>                                                               |
| omdurman-rules/src/unit.rs:609<br>GH:omdurman-rules/src/unit.rs:609   | <b>BrigadeIntegrity</b>           | <pre> 607   /// stack contains all four battalions of a single       Anglo-Egyptian brigade. 608   #[derive(Serialize, Deserialize, Clone, Copy,       PartialEq, Eq, Debug)] 609   pub enum BrigadeIntegrity { 610       None, 611       Integrated(BrigadeId), </pre>                                                                              |
| omdurman-rules/src/combat.rs:36<br>GH:omdurman-rules/src/combat.rs:36 | <b>FireModifier::die_modifier</b> | <pre> 34   impl FireModifier { 35       /// Return the numeric die-roll modifier for this       bonus/penalty (rulebook §6.24, §5.54, §6.23, §9.231, §9.232). 36       pub fn die_modifier(self) -&gt; i16 { 37           match self { 38               FireModifier::AngloEgyptianDirectFire         FireModifier::BrigadeIntegrity =&gt; 1, </pre> |

Proven by: [omdurman-rules::src::verification::die\\_roll\\_apply\\_modifier\\_is\\_total](#)

Covered by tests: [omdurman-rules::src::effects::tests::brigade\\_integrity\\_modifier\\_is\\_engine\\_derived](#)  
[omdurman-rules::src::effects::tests::fire\\_modifiers\\_are\\_engine\\_derived\\_and\\_mismatches\\_rejected](#)

## §6.41 – Direct Fire Subphase

implemented

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| File                                                                              | Symbol            | Code Snippet                                                                                                                                                                                                                                                                              |
|-----------------------------------------------------------------------------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/turn.rs:46</a><br>GH:omdurman-rules/src/turn.rs:46 | <b>DirectFire</b> | <pre>44   pub enum FireSubPhase { 45       /// Direct fire (§6.41). Both sides participate in           this sub-phase. 46       DirectFire, 47       /// Anglo-Egyptian only: Maxim second fire + named-           gunboat howitzer fire (§6.42). 48       MaximSecondAndHowitzer,</pre> |

Covered by tests: [omdurman-rules::src::effects::tests::fire\\_rejects\\_duplicate\\_firers\\_and\\_a\\_foreign\\_firing\\_player](#)  
[omdurman-rules::src::effects::tests::disrupted\\_unit\\_cannot\\_fire](#) [omdurman-rules::src::effects::tests::fire\\_combat\\_elimimates\\_target](#)

## §6.42 – Maxim Second Fire and Howitzer Fire Subphase

implemented

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| File                                                                              | Symbol                        | Code Snippet                                                                                                                                                                                                                                                                                                 |
|-----------------------------------------------------------------------------------|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/turn.rs:48</a><br>GH:omdurman-rules/src/turn.rs:48 | <b>MaximSecondAndHowitzer</b> | <pre>46       DirectFire, 47       /// Anglo-Egyptian only: Maxim second fire + named-           gunboat howitzer fire (§6.42). 48       MaximSecondAndHowitzer, 49   } 50  </pre>                                                                                                                           |
| <a href="#">omdurman-types/src/lib.rs:994</a><br>GH:omdurman-types/src/lib.rs:994 | <b>fires_twice</b>            | <pre>992       /// Maxim guns fire twice per turn -- once in the           Direct Fire Subphase and 993       /// again in the Maxim Second Fire Subphase           (rulebook §6.42). 994       pub fn fires_twice(self) -&gt; bool { 995           matches!(self, UnitKind::Maxim { .. }) 996       }</pre> |

Proven by: [omdurman-types::src::lib::unit\\_kind\\_fire\\_and\\_movement\\_capability\\_law\\_is\\_exact](#)

Covered by tests: [omdurman-rules::src::effects::tests::fall\\_of\\_khartoum\\_has\\_no\\_maxim\\_howitzer\\_subphase](#)  
[omdurman-rules::src::howitzer\\_scatter::howitzer\\_on\\_target\\_7\\_to\\_10](#)  
[omdurman-rules::src::howitzer\\_scatter::howitzer\\_scatters\\_below\\_7](#)  
[omdurman-rules::src::effects::tests::advance\\_window\\_bridges\\_fire\\_subphase\\_and\\_closes\\_at\\_melee](#)  
[omdurman-rules::src::effects::tests::fired\\_at\\_tracker\\_resets\\_at\\_maxim\\_subphase](#)  
[omdurman-rules::src::effects::tests::maxim\\_may\\_fire\\_twice\\_per\\_turn](#)  
[omdurman-rules::src::effects::tests::maxim\\_that\\_skipped\\_direct\\_may\\_fire\\_once\\_in\\_second\\_subphase](#)

## §6.51 – Leader Units

implemented

manual page unknown

| File                                                           | Symbol               | Code Snippet                                                                                                                                                                                                  |
|----------------------------------------------------------------|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/unit.rs:108</a><br>GH:omdurman- | <b>BritishLeader</b> | <pre>106       /// to claim the Mahdi's Tomb (§9.14). 107       #[derive(Serialize, Deserialize, Clone, Copy,           PartialEq, Eq, Hash, Debug, strum::Display)] 108       pub enum BritishLeader {</pre> |

|                                                                               |                           |                                                                                                                                                                                                                                                                                |
|-------------------------------------------------------------------------------|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| rules/src/<br>unit.rs:108                                                     |                           | 109       Kitchener,<br>110       Gatacre,                                                                                                                                                                                                                                     |
| omdurman-<br>types/src/<br>lib.rs:941<br>GH:omdurman-<br>types/src/lib.rs:941 | <b>BritishLeader</b>      | 939       },<br>940       /// Anglo-Egyptian leader (\$6.51): movement only.<br>941       BritishLeader { movement: i32 },<br>942       /// Wall-breach marker placed by artillery fire<br>(\$6.63). Not a combat unit.<br>943       Breech,                                   |
| omdurman-<br>types/src/<br>lib.rs:982<br>GH:omdurman-<br>types/src/lib.rs:982 | <b>has_combat_factors</b> | 980       /// British leaders print a movement factor only<br>(\$6.51); other kinds carry<br>981       /// fire and/or melee factors. Markers carry no<br>stats.<br>982       pub fn has_combat_factors(self) -> bool {<br>983           !matches!(<br>984               self, |

Proven by: [omdurman-types::src::lib::unit\\_kind\\_fire\\_and\\_movement\\_capability Law is exact](#)

Covered by tests: [omdurman-rules::src::unit\\_profiles::zero\\_factor\\_is\\_none\\_not\\_zero](#)  
[omdurman-rules::src::unit\\_profiles::kitchener\\_block\\_resolves\\_leaders\\_friendlys\\_camel\\_and\\_sudanese](#)  
[omdurman-rules::src::unit\\_profiles::dervish\\_leader\\_sections\\_resolve\\_leader\\_and\\_retinue\\_per\\_cell](#)  
[omdurman-rules::src::effects::tests::deploy\\_rejects\\_enemy\\_cohabitation\\_during\\_setup](#)  
[omdurman-rules::src::effects::tests::dervish\\_move\\_through\\_lone\\_ae\\_leader\\_hex\\_eliminate\\_the\\_leader](#)  
[omdurman-rules::src::effects::tests::dervish\\_move\\_onto\\_lone\\_ae\\_leader\\_hex\\_eliminate\\_the\\_leader](#)  
[omdurman-rules::src::effects::tests::ae\\_leader\\_with\\_combat\\_unit\\_is\\_not\\_overrun\\_by\\_dervish\\_move](#)  
[omdurman-rules::src::effects::tests::ae\\_leader\\_eliminated\\_with\\_last\\_combat\\_unit\\_in\\_fire\\_combat](#)

## \$6.52 – Anglo-Egyptian Friendlys Brigade

**implemented**

manual page unknown

| File                                                                                | Symbol              | Code Snippet                                                                                                                                                                                                                                |
|-------------------------------------------------------------------------------------|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| omdurman-<br>rules/src/<br>unit.rs:317<br>GH:omdurman-<br>rules/src/<br>unit.rs:317 | <b>is_friendlys</b> | 315       /// "Friendlys" units obey several special rules<br>(\$5.21, \$5.23, \$6.52,<br>316       /// \$9.14 victory conditions).<br>317       pub fn is_friendlys(&self) -> bool {<br>318           matches!(<br>319               self, |
| omdurman-<br>types/src/<br>lib.rs:1024<br>GH:omdurman-<br>types/src/<br>lib.rs:1024 | <b>Friendlys</b>    | 1022       /// Native volunteer brigade -- the Shaggyeh<br>(\$6.52). Do not receive<br>1023       /// brigade integrity (\$5.54 enumerates only<br>British/Egyptian/Sudanese).<br>1024       Friendlys,<br>1025       }<br>1026             |

Proven by: [omdurman-rules::src::effects::fire::range\\_table\\_player\\_for\\_routes\\_friendlys\\_and\\_fok\\_to\\_the\\_dervish\\_table](#)

Covered by tests: [omdurman-rules::src::unit\\_profiles::friendlys\\_counters\\_score\\_by\\_bank\\_not\\_as\\_leaders](#)  
[omdurman-rules::src::effects::tests::friendlys\\_validate\\_and\\_resolve\\_on\\_dervish\\_table](#)

## \$6.53 – Royal Engineers demolition

**implemented**

manual page unknown

| File                                   | Symbol                | Code Snippet                                                                                                                 |
|----------------------------------------|-----------------------|------------------------------------------------------------------------------------------------------------------------------|
| omdurman-<br>rules/src/<br>unit.rs:288 | <b>RoyalEngineers</b> | 286       /// The Royal Engineers (\$6.53) -- a *specific*<br>unit, not a class, so we<br>287       /// model it explicitly. |

|                                                                                                             |                           |                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------------------------------------------------------------------------------------------------------------|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GH:omdurman-rules/src/unit.rs:288                                                                           |                           | <pre> 288       RoyalEngineers, 289   } 290   </pre>                                                                                                                                                                                                                                                                                                                                                            |
| omdurman-rules/src/unit.rs:528<br>GH:omdurman-rules/src/unit.rs:528                                         | <b>demolishing</b>        | <pre> 526       /// Set when the Royal Engineers are committed to a 527       /// (\$6.53) -- neither offensive fire nor melee 528       /// allowed that turn. 528       pub demolishing: bool, 529       /// Set when a gunboat has lost its engines to a 530       /// river mine (\$10.12, roll 530       /// 5-7): it may no longer move under power and 530       /// instead drifts two hexes per </pre> |
| omdurman-rules/src/effects/effect.rs:176<br>GH:omdurman-rules/src/effects/effect.rs:176                     | <b>Demolition</b>         | <pre> 174   175       /// Royal Engineers demolition (rulebook \$6.53). 176       Demolition { 177           unit_id: UnitId, 178           target: DemolitionTarget, </pre>                                                                                                                                                                                                                                    |
| omdurman-rules/src/effects/setup.rs:37<br>GH:omdurman-rules/src/effects/setup.rs:37                         | <b>apply_demolition</b>   | <pre> 35       /// resolution happens at end of turn via 36       /// ['apply_resolve_demolition'], which 36       /// checks the engineer is still adjacent and 37       /// undisrupted. 37       pub fn apply_demolition( 38           state: &amp;mut GameState, 39           unit_id: UnitId, </pre>                                                                                                       |
| omdurman-rules/src/combat.rs:150<br>GH:omdurman-rules/src/combat.rs:150                                     | <b>DemolitionTarget</b>   | <pre> 148       /// disrupted or driven off. 149       #[derive(Serialize, Deserialize, Clone, Copy, 150       PartialEq, Eq, Debug)] 150       pub enum DemolitionTarget { 151           Fort(UnitId), 152           WallHexside(HexsideRef), </pre>                                                                                                                                                           |
| omdurman-rules/src/effects/state/engineering.rs:39<br>GH:omdurman-rules/src/effects/state/engineering.rs:39 | <b>demolition_targets</b> | <pre> 37       /// the rules would accept. Empty when the unit 38       /// doesn't exist or has no 39       /// adjacent target. 39       pub fn demolition_targets(&amp;self, unit_id: UnitId) - 39       &gt; Vec&lt;DemolitionTarget&gt; { 40           let Ok(unit) = self.unit_or_err(unit_id) else { 41               return Vec::new(); </pre>                                                          |

Covered by tests: [omdurman-rules::src::effects::tests::demolition\\_requires\\_engineers\\_an\\_adjacent\\_target\\_and\\_a\\_pending\\_commit](#)  
[omdurman-rules::src::effects::tests::demolition\\_cancelled\\_when\\_engineer\\_disrupted](#)  
[omdurman-rules::src::effects::tests::demolition\\_cancelled\\_when\\_engineer\\_moved\\_away](#)  
[omdurman-rules::src::effects::tests::demolition\\_targets\\_finds\\_adjacent\\_fort\\_and\\_wall](#)

## §6.54 – Forts

**implemented**

manual page unknown

| File                                                                | Symbol           | Code Snippet                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------------------------------------------------------------------|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| omdurman-rules/src/unit.rs:561<br>GH:omdurman-rules/src/unit.rs:561 | <b>ZocReason</b> | <pre> 559       /// Used by the engine when answering "is this hex in 560       /// an enemy ZOC?". 560       #[derive(Serialize, Deserialize, Clone, Copy, 561       PartialEq, Eq, Debug)] 561       pub enum ZocReason { 562           /// Normal ZOC: any non-disrupted unit other than 563           /// an Anglo-Egyptian 563           /// leader (\$5.41) projects ZOC into each of its 563           /// six adjacent hexes. </pre> |
| omdurman-rules/src/unit.rs:568                                      | <b>Fort</b>      | <pre> 566       GunboatVsGunboat, 567       /// Forts project ZOC out of, but not into, an 567       /// empty fort (\$5.44, \$6.54). </pre>                                                                                                                                                                                                                                                                                                 |

|                                                                       |                                 |                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----------------------------------------------------------------------|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GH:omdurman-rules/src/unit.rs:568                                     |                                 | <pre> 568   Fort, 569       /// Walled-city ZOC: extends out through walls and           gates but not in, 570       /// across a breach in both directions (§5.44). </pre>                                                                                                                                                                                                                                        |
| omdurman-rules/src/unit.rs:516<br>GH:omdurman-rules/src/unit.rs:516   | <b>UnitState</b>                | <pre> 514       /// rather than one big enum. 515       #[derive(Serialize, Deserialize, Clone, Copy,           PartialEq, Eq, Hash, Debug, Default)] 516       pub struct UnitState { 517           /// Reference table: "Disrupted units: no ZOC; may               not move; may not fire 518           /// offensively or defensively; may not melee; are               turned face up at the </pre>           |
| omdurman-rules/src/combat.rs:65<br>GH:omdurman-rules/src/combat.rs:65 | <b>FireAttack</b>               | <pre> 63       /// modifiers (rulebook §6). 64       #[derive(Serialize, Deserialize, Clone, Debug,           PartialEq, Eq)] 65       pub struct FireAttack { 66           pub firing_player: Player, 67           pub phase: Phase, </pre>                                                                                                                                                                       |
| omdurman-rules/src/combat.rs:80<br>GH:omdurman-rules/src/combat.rs:80 | <b>FireAttack::net_modifier</b> | <pre> 78       impl FireAttack { 79           /// Sum of all fire modifiers applied to this               attack (rulebook §6.24). 80           pub fn net_modifier(&amp;self) -&gt; i16 { 81               // Saturating fold rather than `sum()`: the                   modifier list is unbounded 82               // and arrives over the network, so a plain sum                   can overflow `i16`. </pre> |

Covered by tests: `omdurman-rules::src::effects::tests::printed_fok_forts_are_british_forts`  
`omdurman-rules::src::board_data::fok_forts_are_not_walled_in`  
`omdurman-rules::src::effects::tests::retreat_before_melee_may_not_land_on_enemy_fort`

## §6.61 – Only artillery may fire at gunboats; 3+ to sink

implemented

manual page unknown

Only artillery may fire at gunboats. A result of 3 or more on the combat results table is required to sink a gunboat. Any other result is a miss.

| File                                                                                          | Symbol                        | Code Snippet                                                                                                                                                                                                                                                                              |
|-----------------------------------------------------------------------------------------------|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| omdurman-rules/src/unit.rs:167<br>GH:omdurman-rules/src/unit.rs:167                           | <b>WeaponClass</b>            | <pre> 165       Serialize, Deserialize, Clone, Copy, PartialEq, Eq,           PartialOrd, Ord, Hash, Debug, strum::Display, 166       )] 167       pub enum WeaponClass { 168           /// Dervish spears and swords -- no ranged fire at               all. 169           Melee, </pre> |
| omdurman-rules/src/effects/state/fire.rs:49<br>GH:omdurman-rules/src/effects/state/fire.rs:49 | <b>GameState::can_fire_at</b> | <pre> 47           /// modifier in the [`FireAttack`] and is               responsible for the LOS gate. 48           /// (Howitzer fire ignores LOS entirely -- §6.64.) 49           pub fn can_fire_at( 50               &amp;self, 51               firer: UnitId, </pre>              |

Covered by tests: `omdurman-rules::src::effects::tests::a_gunboat_sunk_by_artillery_scores_victory_points`  
`omdurman-rules::src::effects::tests::rifles_may_not_sink_a_gunboat`  
`omdurman-rules::src::effects::tests::artillery_sinks_gunboat_only_on_three_plus`

## §6.62 – Only artillery may fire at forts; 2+ to destroy

implemented

manual page unknown

Only artillery may fire at forts. A result of 2 or more on the combat results table is required to eliminate a fort. Any other result is a miss. If the fort contains any enemy units at the instant it is destroyed, one unit is eliminated with the fort.

| File                                                                                                          | Symbol                        | Code Snippet                                                                                                                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/unit.rs:167</a><br>GH:omdurman-rules/src/unit.rs:167                           | <b>WeaponClass</b>            | <pre> 165       Serialize, Deserialize, Clone, Copy, PartialEq, Eq, 166       PartialOrd, Ord, Hash, Debug, strum::Display, 167   )] 168   pub enum WeaponClass { 169       /// Dervish spears and swords -- no ranged fire at     all. 170       Melee,</pre> |
| <a href="#">omdurman-rules/src/effects/state/fire.rs:49</a><br>GH:omdurman-rules/src/effects/state/fire.rs:49 | <b>GameState::can_fire_at</b> | <pre> 47       /// modifier in the [`FireAttack`] and is     responsible for the LOS gate. 48       /// (Howitzer fire ignores LOS entirely -- §6.64.) 49       pub fn can_fire_at( 50           &amp;self, 51           firer: UnitId,</pre>                  |

Covered by tests: [omdurman-rules::src::effects::tests::a\\_destroyed\\_fort\\_takes\\_one\\_occupant\\_with\\_it\\_and\\_scores\\_it](#)  
[omdurman-rules::src::effects::tests::artillery\\_destroys\\_fort\\_on\\_two\\_or\\_better\\_only](#)  
[omdurman-rules::src::effects::tests::destroyed\\_fort\\_takes\\_one\\_occupant\\_with\\_it](#)

## §6.63 – Only artillery may breach wall hexsides; 2+ required

implemented

manual page unknown

Only artillery may fire to breach a wall hexside of Khartoum or the walled city of Omdurman. A result of 2 or more on the combat results table is required to breach a wall. Any other result is a miss. The effect of the breach is to negate the wall hexside for line of sight purposes. Place a “BREACH” marker in an adjacent hex so that the arrow points to the breached hexside. If any enemy units are adjacent to the wall hexside at the instant it is breached, one enemy unit is eliminated.

**\*\*6.64) Howitzer fire:\*\*** Five units in the game have howitzer fire capability. These are the five named British gunboats. They may fire their artillery factor as direct fire during the Direct Fire Subphase (see 4 and 6.41) and may then fire the same artillery factor as howitzer fire during the Maxim Second Fire and Howitzer Subphase (see 4 and 6.42). Exception: no howitzer fire is allowed during night game turns. To fire howitzer fire, select any target hex between 4 and 10 hexes from the firing gunboat (ignoring the Line of Sight Table) and roll the ten-sided die twice. The first die roll is the Combat Results Table die roll and the second roll is the impact hex die roll. Refer to the Howitzer Fire Scattergram on the marchsheet for the impact hex. The designated target hex is hit on a roll of 7-10. (see manual for full text)

See also: §6.41, §6.42, §6.64

| File                                                                                                    | Symbol                          | Code Snippet                                                                                                                                                                                                                                                                                                           |
|---------------------------------------------------------------------------------------------------------|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-types/src/lib.rs:198</a><br>GH:omdurman-types/src/lib.rs:198                       | <b>Breach</b>                   | <pre> 196       /// Breach in a wall (artillery/§6.63 or Royal     Engineers/§6.53). ZOC both 197       /// ways; LOS no longer blocked across the hexside. 198       Breach, 199       /// Khor -- gully/wadi. ZOCs do not extend across     (§5.44); advance after 200       /// combat may not cross (§6.82).</pre> |
| <a href="#">omdurman-rules/src/effects/effect.rs:282</a><br>GH:omdurman-rules/src/effects/effect.rs:282 | <b>ArtilleryBreachWall</b>      | <pre> 280       /// pre-rolled d10 used for the CRT lookup; range/     LOS are re-derived by the 281       /// engine from the firers and `target`. 282       ArtilleryBreachWall { 283           firers: Vec&lt;UnitId&gt;, 284           target: HexsideRef,</pre>                                                   |
| <a href="#">omdurman-rules/src/effects/fire.rs:840</a>                                                  | <b>apply_artillery_breach_v</b> | <pre> 838       /// artillery's CRT roll -- the rulebook specifies the     same "2+ required"</pre>                                                                                                                                                                                                                    |

|                                                                                                 |                  |                                                                                                                                                                                                                      |
|-------------------------------------------------------------------------------------------------|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GH:omdurman-rules/src/effects/fire.rs:840                                                       |                  | <pre> 839   /// threshold for both trigger styles. 840   pub fn apply_artillery_breach_wall( 841       state: &amp;mut GameState, 842       firers: &amp;[UnitId], </pre>                                            |
| omdurman-rules/src/effects/state/fire.rs:250<br>GH:omdurman-rules/src/effects/state/fire.rs:250 | can_fire_at_wall | <pre> 248       /// range band and resolving the CRT – this method only validates one 249       /// firer at a time. 250       pub fn can_fire_at_wall( 251           &amp;self, 252           firer: UnitId, </pre> |

Covered by tests: omdurman-rules::src::effects::tests::breach\_casualty\_stands\_at\_the\_wall  
omdurman-rules::src::unit\_profiles::breach\_marker\_cell\_returns\_none  
omdurman-rules::src::effects::tests::artillery\_breaches\_wall\_only\_on\_crt\_two\_or\_better  
omdurman-rules::src::effects::tests::wall\_breach\_eliminate\_one\_adjacent\_enemy\_unit  
omdurman-rules::src::effects::tests::breach\_is\_an\_opening\_for\_loS\_and\_effective\_kinds

## §6.64 – Howitzer fire

implemented

manual page unknown

| File                                                                                      | Symbol              | Code Snippet                                                                                                                                                                                                                                                                                                    |
|-------------------------------------------------------------------------------------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| omdurman-rules/src/howitzer_scatter.rs:8<br>GH:omdurman-rules/src/howitzer_scatter.rs:8   | ScatterHexDirection | <pre> 6   /// the diagram away from the firing player. 7   #[derive(serde::Serialize, serde::Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug)] 8   pub enum ScatterHexDirection { 9       UpperLeft, 10       UpperRight, </pre>                                                                           |
| omdurman-rules/src/howitzer_scatter.rs:38<br>GH:omdurman-rules/src/howitzer_scatter.rs:38 | howitzer_scatter    | <pre> 36   /// onto a hex-grid offset oriented away from the firer (see 37   /// `GameState::howitzer_impact_hex`). 38   pub fn howitzer_scatter(impact_roll: DieRoll) -&gt; ScatterHexDirection { 39   create::tables_data::SCATTERGRAM[(impact_roll.value() - 1) as usize] 40   } </pre>                      |
| omdurman-rules/src/unit.rs:119<br>GH:omdurman-rules/src/unit.rs:119                       | GunboatId           | <pre> 117   /// fire; "old" gunboats do not (rulebook §2.32). 118   #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)] 119   pub enum GunboatId { 120       /// One of the five new-type named gunboats with howitzer capability. 121       Named(NamedGunboat), </pre> |
| omdurman-rules/src/effects/effect.rs:89<br>GH:omdurman-rules/src/effects/effect.rs:89     | HowitzerFire        | <pre> 87       /// - CRT result applied to units at impact hex (not the original target). 88       /// - Firers marked as fired. 89       HowitzerFire { 90           attack: FireAttack, 91           combat_results_table_roll: DieRoll, </pre>                                                               |
| omdurman-rules/src/effects/fire.rs:18<br>GH:omdurman-rules/src/effects/fire.rs:18         | apply_howitzer_fire | <pre> 16   17   /// Validate and apply a howitzer fire attack (scatter path) (rulebook §6.64). 18   pub fn apply_howitzer_fire( 19       state: &amp;mut GameState, 20       attack: &amp;FireAttack, </pre>                                                                                                    |
| omdurman-rules/src/effects/state/fire.rs:49<br>GH:omdurman-                               | can_fire_at         | <pre> 47       /// modifier in the [`FireAttack`] and is responsible for the LOS gate. 48       /// (Howitzer fire ignores LOS entirely -- §6.64.) 49       pub fn can_fire_at( </pre>                                                                                                                          |

|                                        |            |                          |
|----------------------------------------|------------|--------------------------|
| rules/src/effects/<br>state/fire.rs:49 | 50  <br>51 | &self,<br>firer: UnitId, |
|----------------------------------------|------------|--------------------------|

Proven

by: [omdurman-rules::src::effects::opposite\\_is\\_an\\_involution](#) [omdurman-rules::src::effects::step\\_toward\\_lands\\_on\\_an\\_adjacent\\_hex](#)  
[omdurman-rules::src::howitzer\\_scatter::scatter\\_is\\_center\\_exactly\\_for\\_rolls\\_7\\_to\\_10](#)

Covered by tests: [omdurman-rules::src::howitzer\\_scatter::howitzer\\_on\\_target\\_7\\_to\\_10](#)  
[omdurman-rules::src::howitzer\\_scatter::howitzer\\_scatters\\_below\\_7](#)  
[omdurman-rules::src::howitzer\\_scatter::howitzer\\_each\\_miss\\_gets\\_its\\_printed\\_hex](#)  
[omdurman-rules::src::unit\\_profiles::named\\_and\\_old\\_gunboats\\_resolve](#)  
[omdurman-rules::src::effects::tests::named\\_gunboat\\_has\\_howitzer](#)  
[omdurman-rules::src::effects::tests::a\\_scattered\\_howitzer\\_shell\\_hits\\_a\\_friendly\\_occupied\\_hex](#)  
[omdurman-rules::src::effects::tests::a\\_rejected\\_howitzer\\_shot\\_leaves\\_no\\_impact\\_record](#)  
[omdurman-rules::src::effects::tests::named\\_gunboat\\_may\\_fire\\_howitzer\\_in\\_second\\_subphase](#)  
[omdurman-rules::src::effects::tests::named\\_gunboat\\_direct\\_fire\\_uses\\_artillery\\_weapon](#)  
[omdurman-rules::src::effects::tests::named\\_gunboat\\_no\\_howitzer\\_at\\_night](#)  
[omdurman-rules::src::effects::tests::dervish\\_gunboat\\_lacks\\_howitzer](#)

## §6.81 – Moving player may fire with all capable units

**implicit**

manual page unknown

During Offensive Fire phase, the moving player may fire with all of his units capable of firing, up to their maximum range, within the limitations imposed by the rules of combat.

## §6.82 – Advance after combat (offensive fire)

**implemented**

manual page unknown

If an enemy-occupied hex is vacated as a result of offensive fire, friendly units may advance after combat into the vacated hex. To be eligible to advance, the friendly units must have participated in the attack and must have been adjacent to the vacated hex. Note that artillery may not advance, nor may units advance across a wall hexside (except at a gate or breach). Units may never advance after combat across a khor.

| File                                                                                                      | Symbol                             | Code Snippet                                                                                                                                                                                                                                                                                                           |
|-----------------------------------------------------------------------------------------------------------|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/effects/effect.rs:163</a><br>GH:omdurman-rules/src/effects/effect.rs:163   | <b>AdvanceAfterCombat</b>          | <pre> 161       /// **Postconditions:** Unit position moved to           `to`; `vacated_by_combat` 162       /// entry consumed. 163       AdvanceAfterCombat { unit_id: UnitId, to:           HexCoord }, 164   165       // -- Unit state changes           ----- </pre>                                             |
| <a href="#">omdurman-rules/src/effects/dispatch.rs:45</a><br>GH:omdurman-rules/src/effects/dispatch.rs:45 | <b>apply_advance_after_combat</b>  | <pre> 43           } 44           GameEffect::AdvanceAfterCombat { unit_id, to }           =&gt; { 45               apply_advance_after_combat(state, *unit_id,                   *to) 46           } 47           GameEffect::RecoverUnit { unit_id } =&gt;               apply_recover_unit(state, *unit_id), </pre> |
| <a href="#">omdurman-types/src/lib.rs:248</a><br>GH:omdurman-types/src/lib.rs:248                         | <b>blocks_advance_after_combat</b> | <pre> 246   247       /// Whether advance-after-combat may *not* cross           this side (§6.82, §7.6). 248       pub fn blocks_advance_after_combat(self) -&gt; bool { 249           matches!( 250               self, </pre>                                                                                       |

|                                                                                                                   |                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------------------------------------------------------------------------------------------------------------------|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/effects/state/melee.rs:141</a><br>GH:omdurman-rules/src/effects/state/melee.rs:141 | <b>can_advance_after_comb</b> | <pre>139       /// player's unit, not artillery, adjacent to `to`,           and `to` now empty. 140       /// Wall/khor hexside restrictions are not enforced           (no hexside map data). 141       pub fn can_advance_after_combat(&amp;self, unit_id: UnitId, to: HexCoord) -&gt; Result&lt;(), RuleError&gt; { 142           let unit = self.unit_or_err(unit_id)?; 143           // §6.7: there is no advance after combat as a               result of defensive fire.</pre> |
|-------------------------------------------------------------------------------------------------------------------|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Proven by: [omdurman-rules::src::effects::advance\\_phase\\_is\\_atomic](#) [omdurman-types::src::lib::hexside\\_blocking\\_classifiers\\_are\\_exact](#)

Covered by tests: [omdurman-rules::src::effects::tests::can\\_advance\\_after\\_combat\\_rejects\\_wall\\_hexside](#)  
[omdurman-rules::src::effects::tests::can\\_advance\\_after\\_combat\\_rejects\\_khor\\_hexside](#)  
[omdurman-rules::src::effects::tests::advance\\_requires\\_combat\\_vacated\\_hex](#)  
[omdurman-rules::src::effects::tests::advance\\_requires\\_participation](#)  
[omdurman-rules::src::effects::tests::rejected\\_advance\\_phase\\_keeps\\_vacated\\_windows](#)

 7/8 implemented

## §7 – Melee Phase

### §7 – Melee Phase (chapter)

descriptive

manual page unknown

Melee Phase

### §7.1 – Melee strength printed on counter

implemented

manual page unknown

The melee strength of all units is printed on the counter. Note that gunboats have no melee strength. Gunboats may neither melee attack nor be melee attacked.

| File                                                                                    | Symbol                       | Code Snippet                                                                                                                                                                                                                                                                  |
|-----------------------------------------------------------------------------------------|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/scalars.rs:42</a><br>GH:omdurman-rules/src/scalars.rs:42 | <b>MeleeFactor</b>           | <pre>40   // Every possible value from the annotated counter       set is a named variant. 41   #[derive(Serialize, Deserialize, Clone, Copy,       PartialEq, Eq, Hash, Debug, strum::Display)] 42   pub enum MeleeFactor { 43       One = 1, 44       Three = 3,</pre>      |
| <a href="#">omdurman-rules/src/scalars.rs:53</a><br>GH:omdurman-rules/src/scalars.rs:53 | <b>MeleeFactor::sum</b>      | <pre>51   impl MeleeFactor { 52       // Sum multiple melee factors (rulebook §7.1). 53       pub fn sum&lt;'a&gt;(factors: impl IntoIterator&lt;Item =       &amp;'a MeleeFactor&gt;) -&gt; u16 { 54           factors.into_iter().map( f  f.value()).sum() 55       }</pre> |
| <a href="#">omdurman-types/src/lib.rs:965</a><br>GH:omdurman-types/src/lib.rs:965       | <b>may_be_melee_attacked</b> | <pre>963   964   // Gunboats neither attack nor are attacked in       melee (§7.1). 965   pub fn may_be_melee_attacked(self) -&gt; bool { 966       !matches!(self, UnitKind::Gunboat { .. }) 967   }</pre>                                                                   |

Proven by: [omdurman-types::src::lib::unit\\_kind\\_melee\\_capability\\_law\\_is\\_exact](#)

Covered by tests: [omdurman-rules::src::effects::tests::melee\\_rejects\\_duplicate\\_attackers\\_and\\_partial\\_defenders](#)  
[omdurman-rules::src::effects::tests::single\\_shot\\_melee\\_is\\_validated\\_like\\_a\\_declaration](#)  
[omdurman-rules::src::effects::tests::a\\_path\\_may\\_not\\_pass\\_through\\_an\\_enemy\\_hex](#)  
[omdurman-rules::src::tests::melee\\_factor\\_values\\_and\\_sum](#) [omdurman-rules::src::tests::unit\\_kind\\_melee\\_capability](#)

### §7.2 – Melee adjacent only, not across wall hexsides

implemented

manual page unknown

Melee simulates the hand-to-hand fighting of the period. Units may melee attack adjacent enemy units only. Units may not melee attack across a wall hexside, but may melee attack through a gate or breach hexside.

| File                                          | Symbol              | Code Snippet                                                                                                               |
|-----------------------------------------------|---------------------|----------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-types/src/lib.rs:237</a> | <b>blocks_melee</b> | <pre>235   // Effects Chart, Khor: "May not melee across").       Gates and breaches are 236   // passable to melee.</pre> |

|                                                                                                 |                  |                                                                                                                                                                                                                                                                                                                                   |
|-------------------------------------------------------------------------------------------------|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GH:omdurman-types/src/lib.rs:237                                                                |                  | <pre> 237       pub fn blocks_melee(self) -&gt; bool { 238           matches!( 239               self, </pre>                                                                                                                                                                                                                     |
| omdurman-rules/src/effects/state/melee.rs:13<br>GH:omdurman-rules/src/effects/state/melee.rs:13 | <b>can_melee</b> | <pre> 11           /// that may be melee-attacked (gunboats may not -- 12           /// thorn-hedge hexside blocks the attack (§7.2). 13       pub fn can_melee(&amp;self, attacker: UnitId, 14           defender_hex: HexCoord) -&gt; Result&lt;(), RuleError&gt; { 15           let unit = self.unit_or_err(attacker)?; </pre> |

Proven by: `omdurman-types::src::lib::hexside_blocking_classifiers_are_exact`

Covered by tests: `omdurman-rules::src::effects::tests::can_melee_gates_phase_adjacency_and_kind`  
`omdurman-rules::src::effects::tests::can_melee_rejects_wall_hexside`  
`omdurman-rules::src::effects::tests::can_melee_rejects_thorn_hedge_hexside`  
`omdurman-rules::src::effects::tests::can_melee_allows_gate_hexside`

## §7.3 – Simultaneous melee combat

**implemented**

manual page unknown

Melee combat is considered simultaneous, so that units eliminated by melee attacks still get a melee combat die roll.

| File                                                                                    | Symbol                    | Code Snippet                                                                                                                                                                                                                                                                                |
|-----------------------------------------------------------------------------------------|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| omdurman-rules/src/effects/effect.rs:112<br>GH:omdurman-rules/src/effects/effect.rs:112 | <b>MeleeCombat</b>        | <pre> 110           /// - Winner may advance into vacated hex (§7.6). 111           /// - Victory points awarded for eliminations. 112           MeleeCombat { 113               attack: MeleeAttack, 114               attacker_roll: DieRoll, </pre>                                      |
| omdurman-rules/src/effects/melee.rs:6<br>GH:omdurman-rules/src/effects/melee.rs:6       | <b>apply_melee_combat</b> | <pre> 4           /// Validated exactly like a declaration 5           ([`validate_melee_attack`]), then 6           /// resolved at once (no §7.5 reaction window). 7       pub fn apply_melee_combat( 8           state: &amp;mut GameState, 9           attack: &amp;MeleeAttack, </pre> |

Covered by tests: `omdurman-rules::src::effects::tests::melee_resolves_simultaneously`

## §7.4 – Who may melee attack / defend

**implemented**

manual page unknown

Only infantry, cavalry, camel units, and Dervish leaders may melee attack. All units (except gunboats — see 7.1) may melee defend.

See also: §7.1

| File                                                              | Symbol                  | Code Snippet                                                                                                                                                                                                                                                                                                                             |
|-------------------------------------------------------------------|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| omdurman-types/src/lib.rs:954<br>GH:omdurman-types/src/lib.rs:954 | <b>may_melee_attack</b> | <pre> 952           /// Rulebook §7.4 -- only infantry, cavalry, camel 953           /// and Dervish leaders may 954           /// melee *attack*. All others (except gunboats) 955           /// may melee *defend* (§7.1). 956       pub fn may_melee_attack(self) -&gt; bool { 957           matches!( 958               self, </pre> |

|                                                                                                                 |                     |                                                                                                                                                                                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-types/src/lib.rs:891</a><br>GH:omdurman-types/src/lib.rs:891                               | <b>UnitKind</b>     | <pre> 889     /// `Some(UnitKind::Marker)` or `None`. 890     #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)] 891     pub enum UnitKind { 892         /// Foot infantry (§2.3): fire / melee / movement. 893         Infantry { </pre>                                               |
| <a href="#">omdurman-types/src/lib.rs:731</a><br>GH:omdurman-types/src/lib.rs:731                               | <b>DervishTribe</b> | <pre> 729         strum::EnumIter, 730     }) 731     pub enum DervishTribe { 732         Baggara, 733         Jaalin, </pre>                                                                                                                                                                                                   |
| <a href="#">omdurman-rules/src/effects/state/melee.rs:13</a><br>GH:omdurman-rules/src/effects/state/melee.rs:13 | <b>can_melee</b>    | <pre> 11     /// that may be melee-attacked (gunboats may not -- §7.1), and no wall or 12     /// thorn-hedge hexside blocks the attack (§7.2). 13     pub fn can_melee(&amp;self, attacker: UnitId, defender_hex: HexCoord) -&gt; Result&lt;(), RuleError&gt; { 14         let unit = self.unit_or_err(attacker)?; 15   </pre> |

Proven by: [omdurman-types::src::lib::unit\\_kind\\_melee\\_capability\\_law\\_is\\_exact](#)

Covered by tests: [omdurman-rules::src::effects::tests::can\\_melee\\_gates\\_phase\\_adjacency\\_and\\_kind](#)

## §7.5 – Cavalry/camel retreat before melee

**implemented**

manual page unknown

Cavalry and camel units may retreat two hexes from an infantry melee attack. Note, however, that only one retreat per unit per turn is permitted. Thus, if their retreat places them adjacent to enemy units whose melee attacks have not yet been resolved, those enemy units may elect to attack the retreating unit(s).

| File                                                                                                      | Symbol                            | Code Snippet                                                                                                                                                                                                                                                                    |
|-----------------------------------------------------------------------------------------------------------|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/effects/effect.rs:149</a><br>GH:omdurman-rules/src/effects/effect.rs:149   | <b>RetreatBeforeMelee</b>         | <pre> 147     /// 148     /// **Postconditions:** Unit position moved to `to`. 149     RetreatBeforeMelee { unit_id: UnitId, to: HexCoord }, 150   151     /// An attacking unit advances into a hex vacated by combat (rulebook §6.82 </pre>                                   |
| <a href="#">omdurman-rules/src/effects/dispatch.rs:42</a><br>GH:omdurman-rules/src/effects/dispatch.rs:42 | <b>apply_retreat_before_melee</b> | <pre> 40     GameEffect::ResolveMelee =&gt; apply_resolve_melee(state), 41     GameEffect::RetreatBeforeMelee { unit_id, to } =&gt; { 42         apply_retreat_before_melee(state, *unit_id, *to) 43     } 44     GameEffect::AdvanceAfterCombat { unit_id, to } =&gt; { </pre> |
| <a href="#">omdurman-types/src/lib.rs:971</a><br>GH:omdurman-types/src/lib.rs:971                         | <b>may_retreat_before_melee</b>   | <pre> 969     /// Cavalry and camel units may retreat two hexes from an infantry melee 970     /// attack (§7.5). 971     pub fn may_retreat_before_melee(self) -&gt; bool { 972         matches!(self, UnitKind::Cavalry { .. }   UnitKind::Camel { .. }) 973     } </pre>     |
| <a href="#">omdurman-rules/src/scalars.rs:119</a><br>GH:omdurman-                                         | <b>HexDistance</b>                | <pre> 117     /// (rulebook §6.22, §7.5). 118     #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, PartialOrd, Ord, Hash, Debug)] 119     pub struct HexDistance(pub(crate) u16); </pre>                                                                            |

|                                                                                                                 |                                 |                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------------------------------------------------------------------------------------------------------------|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| rules/src/<br>scalars.rs:119                                                                                    |                                 | <pre> 120   121   impl HexDistance { </pre>                                                                                                                                                                                                                                                                                                                                                                              |
| <a href="#">omdurman-rules/src/effects/state/melee.rs:72</a><br>GH:omdurman-rules/src/effects/state/melee.rs:72 | <b>can_retreat_before_melee</b> | <pre> 70       /// two hexes away and empty. (Does not verify the 71       attacker is infantry -- 72       /// the caller offers the retreat only in response 73       to one.) 74       pub fn can_retreat_before_melee(&amp;self, unit_id: UnitId, to: HexCoord) -&gt; Result&lt;(), RuleError&gt; { 75           let unit = self.unit_or_err(unit_id)?; 76           if !matches!(self.phase, Phase::Melee) { </pre> |

Proven by: [omdurman-rules::src::effects::resolve\\_melee\\_is\\_atomic](#) [omdurman-rules::src::effects::advance\\_phase\\_is\\_atomic](#)  
[omdurman-types::src::lib::unit\\_kind\\_melee\\_capability\\_law\\_is\\_exact](#)

Covered by tests: [omdurman-rules::src::effects::tests::retreat\\_before\\_melee\\_only\\_cavalry\\_two\\_hexes](#)  
[omdurman-rules::src::effects::tests::retreat\\_opens\\_window\\_only\\_when\\_hex\\_emptyies](#)  
[omdurman-rules::src::effects::tests::rejected\\_resolve\\_melee\\_keeps\\_the\\_declaration](#)

## §7.6 – Advance after melee

**implemented**

manual page unknown

If a melee attack eliminates all of the defenders in an adjacent hex, the Dervish player **MUST** advance into the vacated hex. To be eligible to advance, the Dervish units must have been adjacent to the vacated hex and participated in the melee attack that eliminated the defenders. All surviving eligible Dervish units **MUST** advance, up to the stacking limit. The Anglo-Egyptian player may advance if desired. Note that only attacking units may advance.

| File                                                                                                              | Symbol                            | Code Snippet                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-------------------------------------------------------------------------------------------------------------------|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/effects/effect.rs:163</a><br>GH:omdurman-rules/src/effects/effect.rs:163           | <b>AdvanceAfterCombat</b>         | <pre> 161       /// **Postconditions:** Unit position moved to 162       `to`; `vacated_by_combat` 163       /// entry consumed. 164       AdvanceAfterCombat { unit_id: UnitId, to: HexCoord }, 165       // -- Unit state changes ----- </pre>                                                                                                                                                                                                                                          |
| <a href="#">omdurman-rules/src/effects/dispatch.rs:45</a><br>GH:omdurman-rules/src/effects/dispatch.rs:45         | <b>apply_advance_after_combat</b> | <pre> 43       } 44       GameEffect::AdvanceAfterCombat { unit_id, to } =&gt; { 45           apply_advance_after_combat(state, *unit_id, *to) 46       } 47       GameEffect::RecoverUnit { unit_id } =&gt; apply_recover_unit(state, *unit_id), </pre>                                                                                                                                                                                                                                  |
| <a href="#">omdurman-rules/src/effects/state/melee.rs:141</a><br>GH:omdurman-rules/src/effects/state/melee.rs:141 | <b>can_advance_after_combat</b>   | <pre> 139       /// player's unit, not artillery, adjacent to `to`, 140       and `to` now empty. 141       /// Wall/khor hexside restrictions are not enforced 142       (no hexside map data). 143       pub fn can_advance_after_combat(&amp;self, unit_id: UnitId, to: HexCoord) -&gt; Result&lt;(), RuleError&gt; { 144           let unit = self.unit_or_err(unit_id)?; 145           // §6.7: there is no advance after combat as a 146           result of defensive fire. </pre> |
| <a href="#">omdurman-rules/src/effects/melee.rs:6</a><br>GH:omdurman-rules/src/effects/melee.rs:6                 | <b>apply_melee_combat</b>         | <pre> 4       /// Validated exactly like a declaration 5       ([`validate_melee_attack`]), then 6       /// resolved at once (no §7.5 reaction window). 7       pub fn apply_melee_combat( 8           state: &amp;mut GameState, 9           attack: &amp;MeleeAttack, </pre>                                                                                                                                                                                                           |

Proven by: [omdurman-rules::src::effects::advance\\_phase\\_is\\_atomic](#)

Covered by tests: [omdurman-rules::src::effects::tests::advancing\\_consumes\\_the\\_units\\_eligibility](#)  
[omdurman-rules::src::effects::tests::dervish\\_advance\\_after\\_melee\\_is\\_mandatory](#)  
[omdurman-rules::src::effects::tests::rejected\\_advance\\_phase\\_keeps\\_vacated\\_windows](#)  
[omdurman-rules::src::effects::tests::melee\\_mandatory\\_advance\\_leaders\\_are\\_free\\_stacking](#)

## §7.7 – Melee modifiers

[implemented](#)

manual page unknown

To resolve melee, both the attacker and the defender roll on the Combat Results Table and apply the applicable melee modifier to their die roll. The Dervish player receives a +2 melee modifier, the Anglo-Egyptian player receives a +1 melee modifier. No terrain modifiers are applied to melee combat (Exception: Zariba hexsides in the historical scenario and the campaign game, if constructed — see 9.23). Melee losses must be taken from meleeing units first!

See also: §9.23

| File                                                                                                  | Symbol                                       | Code Snippet                                                                                                                                                                                                                                                                                                                              |
|-------------------------------------------------------------------------------------------------------|----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/combat.rs:108</a><br>GH:omdurman-rules/src/combat.rs:108               | <b>MeleeModifier</b>                         | <pre>106  <br/>107   #[derive(Serialize, Deserialize, Clone, Copy,<br/>108           PartialEq, Eq, Debug)]<br/>109   pub enum MeleeModifier {<br/>110       /// +2 to all Dervish melee rolls (§7.7).<br/>111       DervishStandard,</pre>                                                                                               |
| <a href="#">omdurman-rules/src/effects/melee.rs:405</a><br>GH:omdurman-rules/src/effects/melee.rs:405 | <b>mandatory_melee_modifier</b>              | <pre>403   /// `(attacker_modifiers, defender_modifiers)`; the<br/>404   engine applies exactly<br/>405   these at resolution and rejects a declared attack<br/>406   whose lists differ.<br/>407   pub fn mandatory_melee_modifiers(<br/>408       state: &amp;GameState,<br/>409       attack: &amp;MeleeAttack,</pre>                  |
| <a href="#">omdurman-rules/src/combat.rs:131</a><br>GH:omdurman-rules/src/combat.rs:131               | <b>MeleeAttack</b>                           | <pre>129   /// A melee attack: simultaneous, both sides roll on<br/>130   the Combat Results Table (§7.3, §7.7).<br/>131   #[derive(Serialize, Deserialize, Clone, Debug,<br/>132           PartialEq, Eq)]<br/>133   pub struct MeleeAttack {<br/>134       pub attacker_player: Player,<br/>135       pub attacker_hex: HexCoord,</pre> |
| <a href="#">omdurman-rules/src/combat.rs:112</a><br>GH:omdurman-rules/src/combat.rs:112               | <b>MeleeModifier::Anglo-EgyptianStandard</b> | <pre>110   DervishStandard,<br/>111       /// +1 to all Anglo-Egyptian melee rolls (§7.7).<br/>112       AngloEgyptianStandard,<br/>113       /// Inverted to -2 when Dervish units melee-attack<br/>114       across a trench into<br/>115       /// an entrenched defender (§9.232).</pre>                                              |
| <a href="#">omdurman-rules/src/combat.rs:115</a><br>GH:omdurman-rules/src/combat.rs:115               | <b>MeleeModifier::DervishVsDefender</b>      | <pre>113       /// Inverted to -2 when Dervish units melee-attack<br/>114       across a trench into<br/>115       /// an entrenched defender (§9.232).<br/>116       DervishVsTrenchedDefender,<br/>117   }</pre>                                                                                                                        |
| <a href="#">omdurman-rules/src/combat.rs:110</a><br>GH:omdurman-rules/src/combat.rs:110               | <b>MeleeModifier::DervishStandard</b>        | <pre>108   pub enum MeleeModifier {<br/>109       /// +2 to all Dervish melee rolls (§7.7).<br/>110       DervishStandard,<br/>111       /// +1 to all Anglo-Egyptian melee rolls (§7.7).<br/>112       AngloEgyptianStandard,</pre>                                                                                                      |

Proven by: [omdurman-rules::src::verification::melee\\_modifier\\_keeps\\_roll\\_legal](#)

Covered by tests: [omdurman-rules::src::effects::tests::melee\\_resolves\\_simultaneously](#)  
[omdurman-rules::src::effects::tests::melee\\_modifiers\\_are\\_engine\\_derived\\_and\\_mismatches\\_rejected](#)

2/3 implemented

## §8 – Night Game Turns

### §8 – Night Game Turns

descriptive

manual page unknown

#### Night Game Turns

### §8.1 – Night effects

implemented

manual page unknown

The effects of night game turns are: a) all Anglo-Egyptian movement allowances are halved (round down), b) there is no Anglo-Egyptian howitzer fire, and c) all fire ranges are halved for both sides (round down, but range 1 stays range 1). Range effects on fire combat are the same as during day game turns. For example, an Anglo-Egyptian infantry unit firing at night will be doubled at range 1, normal at range 2, and may not fire at range 3 or greater.

**\*\*8.2) Dervish Desertion Roll:\*\*** Once each campaign game, during the first night turn of the game, the Dervish player rolls one die to see how many of his units desert. The roll is made during the movement phase and the number of deserting Dervish units is equal to 1½ times the roll of one die. The Dervish player may choose which units desert by merely removing them from the mapsheet. The KHALIFA unit, gunboats, artillery units, and forts are the only Dervish units that may not be chosen. No victory points are awarded to the Anglo-Egyptian player for deserting Dervishes.

(see manual for full text)

See also: §8.2

| File                                                                                                | Symbol                             | Code Snippet                                                                                                                                                                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/scalars.rs:88</a><br>GH:omdurman-rules/src/scalars.rs:88             | <b>MovementAllowance::halve</b>    | <pre>86   impl MovementAllowance { 87       /// Night movement allowance = halved (round down)          (rulebook §8.1, §5.11). 88       pub fn halve(self) -&gt; Self { 89           let v = self.value() / 2; 90           MovementAllowance::try_from(v).expect("halved              value always a named variant")</pre> |
| <a href="#">omdurman-types/src/lib.rs:798</a><br>GH:omdurman-types/src/lib.rs:798                   | <b>DayNight</b>                    | <pre>796   /// (rulebook §8.1). 797   #[derive(Serialize, Deserialize, Clone, Copy,              PartialEq, Eq, Debug)] 798   pub enum DayNight { 799       Day, 800       Night,</pre>                                                                                                                                      |
| <a href="#">omdurman-rules/src/range_effects.rs:71</a><br>GH:omdurman-rules/src/range_effects.rs:71 | <b>night_max_range</b>             | <pre>69   70   /// The halved maximum range at night (§8.1): round      down, minimum 1. 71   pub fn night_max_range(weapon: WeaponClass, ae: bool) -      &gt; u8 { 72       let day = max_day_range(weapon, ae); 73       if day &lt;= 1 { 1 } else { day / 2 }</pre>                                                      |
| <a href="#">omdurman-rules/src/scalars.rs:201</a><br>GH:omdurman-rules/src/scalars.rs:201           | <b>effective_movement_at_night</b> | <pre>199   /// Apply night-turn movement halving for Anglo-      Egyptian units (§8.1): all 200   /// Anglo-Egyptian movement allowances are halved      (round down). 201   pub fn effective_movement_at_night( 202       allowance: MovementAllowance, 203       player: Player,</pre>                                     |
| <a href="#">omdurman-rules/src/effects/</a>                                                         | <b>NoHowitzerAtNight</b>           | <pre>110   // §6.64: no howitzer fire at night. 111   if kind == FireKind::Howitzer &amp;&amp; self.day_night</pre>                                                                                                                                                                                                          |

|                                                                      |  |                                                                                                             |
|----------------------------------------------------------------------|--|-------------------------------------------------------------------------------------------------------------|
| state/fire.rs:112<br>GH:omdurman-rules/src/effects/state/fire.rs:112 |  | <pre> == DayNight::Night { 112           return Err(RuleError::NoHowitzerAtNight); 113       } 114   </pre> |
|----------------------------------------------------------------------|--|-------------------------------------------------------------------------------------------------------------|

Proven by: `omdurman-rules::src::verification::movement_allowance_halve_never_panics`  
`omdurman-rules::src::verification::night_halving_is_ac_only_and_day_neutral`  
`omdurman-rules::src::range_effects::night_cap_is_halved_day_max_floored_at_one`  
`omdurman-rules::src::range_effects::night_range_effects_gates_the_day_band_by_the_cap`  
`omdurman-rules::src::effects::fire::night_capped_distance_is_some_exactly_within_the_night_max`

Covered by tests: `omdurman-rules::src::range_effects::night_max_ranges` `omdurman-rules::src::range_effects::night_max_ranges_remaining`  
`omdurman-rules::src::range_effects::ae_rifle_at_night_matches_rulebook_example`  
`omdurman-rules::src::range_effects::max_day_range_all_combos`  
`omdurman-rules::src::effects::tests::night_movement_overlay_allowance_halved`  
`omdurman-rules::src::effects::tests::named_gunboat_no_howitzer_at_night`

## §8.2 – Dervish Desertion Roll

`implemented`

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| File                                                                                    | Symbol                     | Code Snippet                                                                                                                                                                                                                                                                                                                                                   |
|-----------------------------------------------------------------------------------------|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| omdurman-rules/src/effects/effect.rs:191<br>GH:omdurman-rules/src/effects/effect.rs:191 | <b>DervishDesertion</b>    | <pre> 189       /// the effect. The Khalifa, gunboats, artillery, 190       and forts may not be 191       chosen. 192       DervishDesertion { 193           roll: DieRoll, 194           deserters: Vec&lt;UnitId&gt;, </pre>                                                                                                                                |
| omdurman-rules/src/unit.rs:366<br>GH:omdurman-rules/src/unit.rs:366                     | <b>is_desertion_exempt</b> | <pre> 364       /// Non-Dervish identities are trivially not 365       eligible to desert and so are 366       reported as exempt too. 367       pub fn is_desertion_exempt(&amp;self) -&gt; bool { 368           match self { 369               UnitIdentity::DervishLeader(DervishLeader::KhalifaAbdullah) </pre>                                            |
| omdurman-rules/src/turn_track.rs:63<br>GH:omdurman-rules/src/turn_track.rs:63           | <b>DervishDesertion</b>    | <pre> 61       None, 62       /// Dervish desertion roll (§8.2) -- occurs on the 63       first night turn. 64       DervishDesertion, 65       /// Dervish reinforcements are available. 66       DervishReinforcements, </pre>                                                                                                                               |
| omdurman-rules/src/turn_track.rs:63<br>GH:omdurman-rules/src/turn_track.rs:63           | <b>DervishDesertion</b>    | <pre> 61       None, 62       /// Dervish desertion roll (§8.2) -- occurs on the 63       first night turn. 64       DervishDesertion, 65       /// Dervish reinforcements are available. 66       DervishReinforcements, </pre>                                                                                                                               |
| omdurman-rules/src/turn_track.rs:60<br>GH:omdurman-rules/src/turn_track.rs:60           | <b>TurnEvent</b>           | <pre> 58       /// Special events that occur on specific turns 59       (rulebook §8.2, §9.112, §9.113). 60       #[derive(Serialize, Deserialize, Clone, Copy, 61               PartialEq, Eq, Debug)] 62       pub enum TurnEvent { 63           None, 64           /// Dervish desertion roll (§8.2) -- occurs on the 65           first night turn. </pre> |

Proven by: `omdurman-rules::src::effects::desertion_count_is_one_and_a_half_times_the_roll`  
`omdurman-rules::src::turn_track::desertion_event_is_unique_to_the_first_night_turn`

Covered by tests: `omdurman-rules::src::effects::tests::desertion_rejects_a_unit_listed_twice`  
`omdurman-rules::src::effects::tests::desertion_count_is_floor_one_and_a_half`

omdurman-rules::src::turn\_track::desertion\_on\_first\_night  
omdurman-rules::src::effects::tests::desertion\_roll\_required\_before\_first\_night\_movement\_ends  
omdurman-rules::src::effects::tests::desertion\_removes\_chosen\_count\_and\_respects\_exemptions

 23/33 implemented

## §9 – The Scenarios

### §9 – The Scenarios

descriptive

manual page unknown

The Scenarios

### §9.1 – The Campaign Game

descriptive

manual page unknown

The Campaign Game

### §9.2 – The Historical Scenario

descriptive

manual page unknown

The Historical Scenario

Players should note that the historical scenario is an exercise in futility for the Dervish player. It is, however, an interesting demonstration of the absolute imbecility of the Khalifa's generalship and vividly shows the superiority of entrenched firepower over traditional tribal arms in the colonial period.

### §9.3 – Bonus Game: Fall of Khartoum

descriptive

manual page unknown

Bonus Game: FALL OF KHARTOUM

### §9.11 – Set Up (Campaign)

descriptive

manual page unknown

Set Up

### §9.12 – Scenario Length (Campaign)

implemented

manual page unknown

Scenario Length

6:00 am, Sept. 1 through 8:00 am, Sept. 3, 22 Game Turns.

| File                                                                | Symbol          | Code Snippet                                                                                                                                               |
|---------------------------------------------------------------------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/turn_track.rs:11</a><br>GH:omdurman- | <b>GameTime</b> | <pre>9   /// starts at one of these twelve times. 10   #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 11   pub enum GameTime {</pre> |

|                                                                                               |                     |                                                                                                                                                                                                                                                                                                                                                                                |
|-----------------------------------------------------------------------------------------------|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| rules/src/<br>turn_track.rs:11                                                                |                     | 12   SixAM,<br>13   EightAM,                                                                                                                                                                                                                                                                                                                                                   |
| omdurman-<br>rules/src/<br>turn_track.rs:47<br>GH:omdurman-<br>rules/src/<br>turn_track.rs:47 | TurnEntry           | 45   /// A single entry on the Turn Record Track (rulebook<br>\$9.12, §9.22).<br>46   #[derive(Serialize, Deserialize, Clone, Copy,<br>PartialEq, Eq, Debug)]<br>47   pub struct TurnEntry {<br>48       /// 1-based turn number.<br>49       pub turn: u8,                                                                                                                    |
| omdurman-<br>rules/src/<br>turn_track.rs:95<br>GH:omdurman-<br>rules/src/<br>turn_track.rs:95 | CAMPAIGN_TURN_TRACK | 93   /// is turn 9, which carries the once-per-game Dervish<br>Desertion Roll (§8.2) --<br>94   /// the printed track prints "Dervish Desertion Roll /<br>NIGHT" on that cell.<br>95   pub const CAMPAIGN_TURN_TRACK: [TurnEntry; 22] = [<br>96       // Row 1, left->right: Sept 1, 6 am -> 8 pm, then<br>the first NIGHT.<br>97       entry(1, SixAM, Day, TurnEvent::None), |
| omdurman-<br>rules/src/turn_track.rs:242<br>GH:omdurman-<br>rules/src/<br>turn_track.rs:242   | TurnLabel           | 240   /// printed cell; `Blank` is for unused positions in<br>the 9x3 grid.<br>241   #[derive(Serialize, Deserialize, Debug, Clone, Copy,<br>PartialEq, Eq)]<br>242   pub enum TurnLabel {<br>243       Blank,<br>244       /// "SEPT. 1\n6:00 am" -- day header plus time<br>(turn 1).                                                                                        |

Proven by: omdurman-rules::src::turn\_track::scenario\_turn\_indexing\_is\_exact\_for\_every\_scenario  
omdurman-rules::src::turn\_track::turn\_labels\_agree\_with\_the\_rule\_bearing\_track

Covered  
by tests: omdurman-rules::src::turn\_track::campaign\_track\_22\_turns omdurman-rules::src::turn\_track::desertion\_on\_first\_night  
omdurman-rules::src::turn\_track::campaign\_track\_label\_and\_day\_night\_agree  
omdurman-rules::src::turn\_track::game\_time\_display\_all\_variants omdurman-rules::src::turn\_track::turn\_label\_display  
omdurman-rules::src::turn\_track::turn\_label\_out\_of\_range\_is\_none  
omdurman-rules::src::effects::tests::game\_over\_after\_campaign\_turns

§9.13 – Special Rules (Campaign)

descriptive  
manual page unknown

- Special Rules
- None.

§9.14 – Victory Conditions (Campaign)

implemented  
manual page unknown

Victory Conditions

The Mahdi’s Tomb in Omdurman was not only the tallest structure in the entire Sudan in 1898, it was also a Dervish holy shrine. Its loss or destruction would be a severe blow to the Mahdist cause. It is accordingly assigned 25 victory points which are awarded to the player who controls it at the conclusion of play. The Dervish player controls it at the start of play. As a tactical note, the Anglo-Egyptian player will find a decisive victory almost impossible unless he takes the Mahdi’s Tomb from the Dervish player. To take the Tomb hex, it must be occupied by one British leader plus any one non-“Friendlies” Anglo-Egyptian combat unit (both undisrupted) at the conclusion of play.

Additional victory points are awarded as follows:

**\*\*Dervish Player receives:\*\***

- 10 pts: each British leader eliminated

(see manual for full text)

See also: §5.21

| File                                                                                      | Symbol                                        | Code Snippet                                                                                                                                                                                                                                                                                                                              |
|-------------------------------------------------------------------------------------------|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/victory.rs:16</a><br>GH:omdurman-rules/src/victory.rs:16   | <b>VpSource</b>                               | <pre>14   /// the manual and the engine. 15   #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 16   pub enum VpSource { 17       // ----- Anglo-Egyptian player receives: 18       /// 25 pts -- the Mahdi's Tomb taken from the Dervish: held at the</pre>                                                           |
| <a href="#">omdurman-rules/src/victory.rs:48</a><br>GH:omdurman-rules/src/victory.rs:48   | <b>VpSource::points</b>                       | <pre>46   impl VpSource { 47       /// VP awarded to `who_scores()` (rulebook §9.14). 48       pub fn points(self) -&gt; VictoryPoints { 49           match self { 50               VpSource::MahdisTombTaken   VpSource::MahdisTombHeld =&gt; VictoryPoints::new(25),</pre>                                                              |
| <a href="#">omdurman-rules/src/victory.rs:63</a><br>GH:omdurman-rules/src/victory.rs:63   | <b>VpSource::who_scores</b>                   | <pre>61   62       /// Which player receives these victory points (rulebook §9.14). 63       pub fn who_scores(self) -&gt; Player { 64           match self { 65               VpSource::MahdisTombTaken</pre>                                                                                                                            |
| <a href="#">omdurman-rules/src/victory.rs:100</a><br>GH:omdurman-rules/src/victory.rs:100 | <b>VictoryLedger</b>                          | <pre>98   /// Cumulative victory ledger for one scenario (rulebook §9.14). 99   #[derive(Serialize, Deserialize, Clone, Debug, Default)] 100   pub struct VictoryLedger { 101       pub events: Vec&lt;VpEvent&gt;, 102   }</pre>                                                                                                         |
| <a href="#">omdurman-rules/src/victory.rs:106</a><br>GH:omdurman-rules/src/victory.rs:106 | <b>VpEvent</b>                                | <pre>104   /// A single victory-point scoring event (rulebook §9.14). 105   #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 106   pub struct VpEvent { 107       pub turn: GameTurnIndex, 108       pub source: VpSource,</pre>                                                                                      |
| <a href="#">omdurman-rules/src/victory.rs:113</a><br>GH:omdurman-rules/src/victory.rs:113 | <b>VictoryLedger::total_for</b>               | <pre>111   impl VictoryLedger { 112       /// Total victory points earned by a given player (rulebook §9.14). 113       pub fn total_for(&amp;self, player: Player) -&gt; VictoryPoints { 114           VictoryPoints( 115               self.events</pre>                                                                                |
| <a href="#">omdurman-rules/src/victory.rs:125</a><br>GH:omdurman-rules/src/victory.rs:125 | <b>VictoryLedger::superiority</b>             | <pre>123       /// Net superiority: positive = Anglo-Egyptian ahead, negative = Dervish ahead 124       /// (rulebook §9.14). 125       pub fn superiority(&amp;self) -&gt; VictoryPoints { 126           VictoryPoints( 127               self.total_for(Player::AngloEgyptian).value() - self.total_for(Player::Dervish).value(),</pre> |
| <a href="#">omdurman-rules/src/victory.rs:151</a><br>GH:omdurman-rules/src/victory.rs:151 | <b>CampaignVictoryLevel</b>                   | <pre>149   /// Campaign-game victory levels (§9.14). 150   #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 151   pub enum CampaignVictoryLevel { 152       Draw, 153       Marginal(Player),</pre>                                                                                                                   |
| <a href="#">omdurman-rules/src/victory.rs:160</a><br>GH:omdurman-rules/src/victory.rs:160 | <b>CampaignVictoryLevel::from_superiority</b> | <pre>158   impl CampaignVictoryLevel { 159       /// Assign a level from the net superiority (§9.14). 160       pub fn from_superiority(s: VictoryPoints) -&gt; Self { 161           let net = s.0; 162           // Positive -&gt; Anglo-Egyptian thresholds: 15/30/50</pre>                                                             |

|                                                                                                         |                          |                                                                                                                                                                                                                                                                                                                                                                            |
|---------------------------------------------------------------------------------------------------------|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/effects/victory.rs:75</a><br>GH:omdurman-rules/src/effects/victory.rs:75 | <b>score_elimination</b> | <pre>73   /// elimination under `cause`. The owner is derived 74   /// so unlike the historical signature there is no 75   /// caller-supplied player. 76   pub fn score_elimination(state: &amp;mut GameState, 77   unit_id: UnitId, cause: ElimCause) { 78       if let Some(unit) = state.find_unit(unit_id) { 79           let identity = unit.profile.identity;</pre> |
| <a href="#">omdurman-rules/src/scalars.rs:171</a><br>GH:omdurman-rules/src/scalars.rs:171               | <b>VictoryPoints</b>     | <pre>169   /// (rulebook §9.14). 170   #[derive(Serialize, Deserialize, Clone, Copy, 171   PartialEq, Eq, PartialOrd, Ord, Debug, Default)] 172   pub struct VictoryPoints(pub(crate) i32); 173   impl VictoryPoints {</pre>                                                                                                                                               |

Proven by: [omdurman-rules::src::verification::campaign\\_victory\\_levels\\_match\\_manual\\_superiority\\_table](#)  
[omdurman-rules::src::verification::vp\\_source\\_points\\_and\\_scorer\\_match\\_the\\_printed\\_schedule](#)  
[omdurman-rules::src::effects::victory::vp\\_source\\_for\\_routes\\_every\\_elimination\\_to\\_the\\_printed\\_source](#)  
[omdurman-rules::src::effects::victory::score\\_elimination\\_records\\_exactly\\_what\\_it\\_scores](#)

Covered by tests: [omdurman-rules::src::effects::tests::a\\_destroyed\\_fort\\_takes\\_one\\_occupant\\_with\\_it\\_and\\_scores\\_it](#)  
[omdurman-rules::src::effects::tests::a\\_mined\\_gunboat\\_scores\\_and\\_takes\\_its\\_friendlys\\_down](#)  
[omdurman-rules::src::effects::tests::a\\_gunboat\\_sunk\\_by\\_artillery\\_scores\\_victory\\_points](#)  
[omdurman-rules::src::effects::tests::friendlys\\_bank\\_scores\\_by\\_side](#)  
[omdurman-rules::src::effects::tests::mahdis\\_tomb\\_stays\\_dervish\\_without\\_a\\_leader](#)  
[omdurman-rules::src::effects::tests::mahdis\\_tomb\\_scores\\_for\\_anglo\\_egyptian\\_when\\_taken](#)  
[omdurman-rules::src::effects::tests::campaign\\_end\\_scores\\_an\\_untaken\\_tomb\\_for\\_the\\_dervish](#)  
[omdurman-rules::src::tests::vp\\_source\\_attributes](#)

## §9.21 – Set Up (Historical)

[descriptive](#)  
manual page unknown

Set Up

## §9.22 – Scenario Length (Historical)

[implemented](#)  
manual page unknown

Scenario Length  
6:00 am, September 2 through 12:00 noon, September 2. Four game turns.

| File                                                                                            | Symbol                 | Code Snippet                                                                                                                                                                                                         |
|-------------------------------------------------------------------------------------------------|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/turn_track.rs:132</a><br>GH:omdurman-rules/src/turn_track.rs:132 | <b>HISTORICAL_TURN</b> | <pre>130   131   /// Historical scenario track (§9.22 -- 4 turns, Sept 2 132   6:00 am -&gt; 12:00 pm). 133   pub const HISTORICAL_TURN_TRACK: [TurnEntry; 4] = [ 134       TurnEntry { 135           turn: 1,</pre> |

Proven by: [omdurman-rules::src::turn\\_track::scenario\\_turn\\_indexing\\_is\\_exact\\_for\\_every\\_scenario](#)

Covered by tests: [omdurman-rules::src::turn\\_track::historical\\_turn\\_all\\_four\\_turns](#)

## §9.23 – Special Rule: The Zariba

[descriptive](#)  
manual page unknown

Special Rule: “The Zariba”

\*\*9.231) Thorn hedge hexsides:\*\* -2 to die roll on all Dervish fire attacks; may not melee across in either direction; may not advance after combat across in either direction.

\*\*9.232) Trench hexsides:\*\* -4 to die roll on all Dervish fire attacks vs. entrenched units only; -2 (instead of +2) melee modifier to Dervish units melee attacking an entrenched unit; entrenched units may be fired “over” in both directions (i.e. they do not block line of sight); units are considered “entrenched” if they are directly adjacent to (and on the Nile River side of) a trench hexside.

(see manual for full text)

See also: §9.231, §9.232

§9.24 – Victory Conditions (Historical)

implemented

manual page unknown

Victory Conditions

Victory Levels are based solely on eliminating enemy units while conserving your own force as much as possible.

| Victory Level | Anglo-Egyptian Player (Dervish units eliminated) | Dervish Player (Anglo-Egyptian units eliminated)  
| |—|—|—| | 5 — DECISIVE | 100+ | 30+ | | 4 — STRATEGIC | 60–99 | 15–29 | | 3 — TACTICAL | 45–59 | 10–14 | | 2 — MARGINAL | 30–44 | 5–9 | | 1 — DRAW | 0–29 | 0–4 |

The lower value victory level is then subtracted from the higher level to determine a player’s net victory. For example, if the Anglo-Egyptian player eliminates 104 Dervish units (decisive victory) but loses 18 units doing it (Dervish Strategic), the Anglo-Egyptian player only nets out with a draw (decisive worth 5 minus strategic worth 4 = 1, draw).

(see manual for full text)

| File                                                                                      | Symbol                        | Code Snippet                                                                                                                                                                                                 |
|-------------------------------------------------------------------------------------------|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/victory.rs:190</a><br>GH:omdurman-rules/src/victory.rs:190 | <b>HistoricalVictoryLevel</b> | <pre>188   /// draw"). 189   #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, PartialOrd, Ord, Debug)] 190   pub enum HistoricalVictoryLevel { 191       Draw = 1, 192       Marginal = 2,</pre> |

Proven by: [omdurman-rules::src::verification::historical\\_victory\\_ladders\\_match\\_manual\\_bands](#)

Covered by tests: [omdurman-rules::src::tests::historical\\_victory\\_level\\_for\\_dervish](#)  
[omdurman-rules::src::tests::historical\\_victory\\_level\\_for\\_anglo\\_egyptian](#)

§9.31 – Bonus game map

implemented

manual page unknown

Only the small FALL OF KHARTOUM scenario map is used for this game.

| File                                                                | Symbol                      | Code Snippet                                                                                                        |
|---------------------------------------------------------------------|-----------------------------|---------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/board_data.rs:37</a><br>GH:omdurman- | <b>fall_of_khartoum_map</b> | <pre>35   36   /// The Fall-of-Khartoum board (§9.3). 37   pub fn fall_of_khartoum_map_data() -&gt; MapData {</pre> |

|                                |  |                                                                       |
|--------------------------------|--|-----------------------------------------------------------------------|
| rules/src/<br>board_data.rs:37 |  | <div>38   FOK.get_or_init(   parse(FOK_RON)).clone()<br/>39   }</div> |
|--------------------------------|--|-----------------------------------------------------------------------|

Covered by tests: `omdurman-app::src::tests::scenario_maps_to_board` `omdurman-app::src::tests::start_game_scenario_selects_board`

§9.32 – Set Up (Bonus)

`descriptive`  
manual page unknown

Set Up

§9.33 – Scenario Length (Bonus)

`implemented`  
manual page unknown

Scenario Length

Variable, see victory conditions (9.35). Rarely lasts five turns.

See also: §9.35

| File                                                                                            | Symbol                  | Code Snippet                                                                                                                                                                                                                                                                                                                          |
|-------------------------------------------------------------------------------------------------|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/turn_track.rs:172</a><br>GH:omdurman-rules/src/turn_track.rs:172 | <b>FALL_OF_KHARTOUM</b> | <div>170   /// (the rulebook fixes none); only `day_night` is<br/>171   rule-bearing (night halves<br/>171   /// Anglo-Egyptian movement and ranges and bars<br/>171   howitzer fire, §8.1).<br/>172   pub const FALL_OF_KHARTOUM_TURN_TRACK: [TurnEntry; 8] =<br/>172   [<br/>173       TurnEntry {<br/>174           turn: 1,</div> |

Proven by: `omdurman-rules::src::turn_track::scenario_turn_indexing_is_exact_for_every_scenario`  
`omdurman-rules::src::turn_track::fall_of_khartoum_night_window_is_turns_one_and_two`

Covered by tests: `omdurman-rules::src::turn_track::fall_of_khartoum_turn_one_is_night`  
`omdurman-rules::src::turn_track::fall_of_khartoum_turns_3_to_8_are_day`

§9.34 – Special Rules (Bonus)

`descriptive`  
manual page unknown

Special Rules

§9.35 – Victory Conditions (Bonus)

`implemented`  
manual page unknown

Victory Conditions

Victory is determined by how many turns it takes the Dervish player to eliminate the GORDON leader unit and how many Dervish units are eliminated:

- Dervish decisive: eliminate GORDON turn four or sooner.
- Dervish tactical: eliminate GORDON turn five.

- Dervish marginal: eliminate GORDON turn six.
- British marginal: GORDON survives end of turn six.
- British tactical: GORDON survives end of turn seven.

(see manual for full text)

| File                                                                                      | Symbol                          | Code Snippet                                                                                                                                                                                                                                                                                                                                                                                                           |
|-------------------------------------------------------------------------------------------|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/victory.rs:308</a><br>GH:omdurman-rules/src/victory.rs:308 | <b>FoKVictoryLevel::resolve</b> | <pre> 306       /// rulebook: GORDON dies turn 5 (tactical) with 24           Dervish losses (-2 307       /// levels) nets a British marginal. 308       pub fn resolve(gordon_died_turn: Option&lt;u8&gt;,           scenario_end_turn: u8, dervish_lost: i16) -&gt; Self { 309           let base = Self::base(gordon_died_turn,               scenario_end_turn); 310           let base_idx = Self::LADDER </pre> |

Proven by: [omdurman-rules::src::verification::fok\\_victory\\_ladder\\_penalties\\_shift\\_monotonically](#)

Covered by tests: [omdurman-rules::src::effects::tests::gordon\\_dying\\_in\\_melee\\_ends\\_fall\\_of\\_khartoum](#)  
[omdurman-rules::src::tests::fok\\_victory\\_level\\_worked\\_example](#) [omdurman-rules::src::tests::fok\\_victory\\_level\\_gordon\\_died\\_early](#)  
[omdurman-rules::src::tests::fok\\_victory\\_level\\_late\\_gordon\\_death](#) [omdurman-rules::src::tests::fok\\_victory\\_level\\_gordon\\_survived](#)

## §9.111 – Dervish set up (Campaign)

**implemented**

manual page unknown

Dervish player sets up first, moves second.

- Isa Zachneih infantry unit: anywhere on the east bank, in or south of El Debeba.
- KHALIFA ABDULLAH: in the walled city of Omdurman, in either palace hex.
- 3 artillery units, and all Taiasha units: anywhere in the walled city of Omdurman.
- 17 forts: anywhere on the mapsheet south of the Khor Shambat on the west bank, and/or south of all Halfaya hut hexes on the east bank and Nile River islands.
- 2 gunboats: any south edge Nile River hexes.

**\*\*9.112) Dervish reinforcements:\*\*** all reinforcements enter on the west edge of the mapsheet, south of the Khor Shambat. Each unit pays the terrain cost of the hex through which it enters, no matter how many units enter through

(see manual for full text)

See also: §9.112

| File                                                                                                              | Symbol                | Code Snippet                                                                                                                                                                                                                                                                                                                                           |
|-------------------------------------------------------------------------------------------------------------------|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/effects/state/setup.rs:105</a><br>GH:omdurman-rules/src/effects/state/setup.rs:105 | <b>setup_complete</b> | <pre> 103       /// currently shares the same "both sides deployed"           gate; when a scenario 104       /// needs a different minimum, branch on           `self.scenario` here. 105       pub fn setup_complete(&amp;self) -&gt; Result&lt;(),           RuleError&gt; { 106           let has =  player  { 107               self.units </pre> |

Covered by tests: [omdurman-rules::src::effects::tests::setup\\_is\\_sequential\\_first\\_side\\_then\\_second](#)  
[omdurman-app::src::scenario\\_setup::campaign\\_has\\_no\\_fixed\\_placements](#)  
[omdurman-rules::src::unit\\_profiles::hadendowa\\_first\\_cell\\_is\\_isa\\_zachneih](#)  
[omdurman-rules::src::effects::tests::campaign\\_deployment\\_is\\_boat\\_land\\_exclusive](#)  
[omdurman-rules::src::effects::tests::campaign\\_setup\\_rejects\\_non\\_initial\\_force](#)

## §9.112 – Dervish reinforcements (Campaign)

**implemented**

manual page unknown

| File                                                                                                      | Symbol                        | Code Snippet                                                                                                                                                                                                                                                                                                                                |
|-----------------------------------------------------------------------------------------------------------|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/effects/effect.rs:183</a><br>GH:omdurman-rules/src/effects/effect.rs:183   | <b>PlaceReinforcements</b>    | <pre> 181   // -- Reinforcement / placement ----- 182   /// Place reinforcements onto the map (rulebook \$9.112, \$9.113). 183       PlaceReinforcements(Vec&lt;UnitPlacement&gt;), 184   185   // -- Scenario-specific ----- </pre>                                                                                                        |
| <a href="#">omdurman-rules/src/effects/dispatch.rs:53</a><br>GH:omdurman-rules/src/effects/dispatch.rs:53 | <b>apply_place_reinforce</b>  | <pre> 51           GameEffect::Demolition { unit_id, target } =&gt; apply_demolition(state, *unit_id, *target), 52           GameEffect::PlaceReinforcements(placements) =&gt; { 53               apply_place_reinforcements(state, placements) 54           } 55           GameEffect::DervishDesertion { roll, deserters } =&gt; { </pre> |
| <a href="#">omdurman-rules/src/reinforcements.rs:69</a><br>GH:omdurman-rules/src/reinforcements.rs:69     | <b>dervish_campaign_sched</b> | <pre> 67   /// All reinforcements enter on the west edge, south of the Khor Shambat. 68   /// Each unit pays terrain cost of the hex it enters through. 69   pub fn dervish_campaign_schedule() -&gt; ReinforcementSchedule { 70       ReinforcementSchedule { 71           player: Player::Dervish, </pre>                                 |
| <a href="#">omdurman-types/src/lib.rs:531</a><br>GH:omdurman-types/src/lib.rs:531                         | <b>Location</b>               | <pre> 529   /// Named map landmarks (rulebook mapsheet, \$9.111, \$9.113, \$9.212 scenarios). 530   #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug, strum::Display)] 531   pub enum Location { 532       FortMakran, 533       NorthFort, </pre>                                                                        |
| <a href="#">omdurman-types/src/lib.rs:611</a><br>GH:omdurman-types/src/lib.rs:611                         | <b>SetupLetter</b>            | <pre> 609   /// Each letter marks a specific hex where a Dervish leader is placed. 610   #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug, strum::Display)] 611   pub enum SetupLetter { 612       Y, 613       K, </pre>                                                                                                 |
| <a href="#">omdurman-types/src/lib.rs:754</a><br>GH:omdurman-types/src/lib.rs:754                         | <b>Faction</b>                | <pre> 752   /// `Some(BrigadeId::friendlies)`'. 753   #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 754   pub enum Faction { 755       Dervish { 756           tribe: DervishTribe, </pre>                                                                                                                           |

Proven by: [omdurman-rules::src::reinforcements::wave\\_for\\_turn\\_answers\\_exactly\\_on\\_the\\_printed\\_schedule](#)

Covered [omdurman-app::src::tests::place\\_unit\\_translates\\_by\\_phase](#) by [omdurman-rules::src::reinforcements::dervish\\_schedule\\_has\\_three\\_waves](#) tests:  
[omdurman-rules::src::reinforcements::dervish\\_wave\\_one\\_has\\_baggaara\\_and\\_three\\_leaders](#)  
[omdurman-rules::src::reinforcements::dervish\\_wave\\_two\\_has\\_hadendowa](#)  
[omdurman-rules::src::reinforcements::dervish\\_wave\\_three\\_is\\_all\\_remaining](#)  
[omdurman-rules::src::reinforcements::wave\\_for\\_turn\\_returns\\_correct\\_wave](#)  
[omdurman-rules::src::effects::tests::campaign\\_reinforcements\\_gate\\_by\\_wave](#)  
[omdurman-rules::src::effects::tests::campaign\\_setup\\_rejects\\_non\\_initial\\_force](#)

## \$9.113 – Anglo-Egyptian reinforcements (Campaign)

**implemented**

manual page unknown

The Anglo-Egyptian player moves first. There are no Anglo-Egyptian units on the mapsheet at start. The GORDON unit is not used in this scenario.

- The leader units KITCHENER, GATACRE, and HUNTER may enter anytime during the first four game turns and do not count against the 12 units per turn limit. All three leaders must be in play by the end of turn four!
- All gunboats enter through any north edge Nile River hex, paying one movement point for the first hex entered. The “Friendlies” brigade enters through the Abu Alim hut hex on the east bank, paying eight movement points per unit. All other Anglo-Egyptian units enter through the west bank “ANGLO-EGYPTIAN ENTRANCE AREA”, each unit paying one movement point to enter the mapsheet.
- Turn 1) Any three gunboats; “Friendlies” brigade; Egyptian Cavalry; Horse Artillery; and two infantry brigades

(see manual for full text)

| File                                                                                                    | Symbol                        | Code Snippet                                                                                                                                                                                                                                                                                                                                                                       |
|---------------------------------------------------------------------------------------------------------|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/reinforcements.rs:126</a><br>GH:omdurman-rules/src/reinforcements.rs:126 | <b>anglo_egyptian_campaig</b> | <pre>124   /// - "Friendlies" enter via Abu Alim hut on the east       bank (8 MP per unit). 125   /// - All other AE units enter via the Anglo-Egyptian       Entrance Area (1 MP). 126   pub fn anglo_egyptian_campaign_schedule() -&gt;       ReinforcementSchedule { 127       let free_leaders = vec![ 128           CampaignLeader::British(BritishLeader::Kitchener),</pre> |

Proven by: [omdurman-rules::src::reinforcements::wave\\_for\\_turn\\_answers\\_exactly\\_on\\_the\\_printed\\_schedule](#)

Covered by tests: [omdurman-rules::src::reinforcements::anglo\\_egyptian\\_schedule\\_has\\_four\\_waves](#)  
[omdurman-rules::src::reinforcements::anglo\\_egyptian\\_leaders\\_available\\_each\\_wave](#)  
[omdurman-rules::src::reinforcements::anglo\\_egyptian\\_turn\\_four\\_is\\_all\\_remaining](#)  
[omdurman-rules::src::effects::tests::campaign\\_reinforcement\\_cap\\_and\\_double\\_entry](#)  
[omdurman-rules::src::effects::tests::campaign\\_gunboats\\_quota\\_three\\_per\\_turn](#)  
[omdurman-rules::src::effects::tests::reinforcement\\_rejected\\_onto\\_enemy\\_occupied\\_hex](#)  
[omdurman-rules::src::effects::tests::campaign\\_setup\\_rejects\\_non\\_initial\\_force](#)

## §9.211 – Anglo-Egyptian set up first, moves second (Historical)

**implemented**

manual page unknown

The Anglo-Egyptian player sets up first, and moves second:

- Not in play: GORDON leader unit, “Friendlies” brigade.
- Gunboats start in any Nile River hexes adjacent to the Zariba.
- Camel Corps, Egyptian Cavalry, and Horse Artillery start in the village of Kerreri hut hexes.
- All remaining Anglo-Egyptian units set up in the 13 hexes of the Zariba.

| File                                                                                                        | Symbol              | Code Snippet                                                                                                                                                                                                                                                      |
|-------------------------------------------------------------------------------------------------------------|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/effects/dispatch.rs:462</a><br>GH:omdurman-rules/src/effects/dispatch.rs:462 | <b>first_player</b> | <pre>460   461   /// The player who moves first in a scenario (\$4,       \$9.113, \$9.212, \$9.322). 462   pub fn first_player(scenario: Scenario) -&gt; Player { 463       match scenario { 464           Scenario::Campaign =&gt; Player::AngloEgyptian,</pre> |

Covered by tests: [omdurman-rules::src::effects::tests::setup\\_is\\_sequential\\_first\\_side\\_then\\_second](#)  
[omdurman-rules::src::effects::tests::historical\\_setup\\_rejects\\_not\\_in\\_play\\_units](#)

## §9.212 – Dervish set up (Historical) – deployment zones and leader-to-hex mapping

implemented

manual page unknown

The Dervish player sets up second, and moves first.

- Not in play: Isa Zachneih, gunboats, and forts.
- All Dervish units must be set up out of the line of sight of all Anglo-Egyptian units.
- Dervish leaders start on the lettered hexes:
  - A: Ali Wad Helu
  - D: Sheik El Din
  - Y: Yakub
  - K: Khalifa Abdullah
  - S: Sherif
  - O: Osman Digna

"A" is the hex of the Dervish leader Ali Wad Helu. "D" is the hex of the Dervish leader Sheik El Din. "Y" is the hex of the Dervish leader Yakub. "K" is the hex of the Dervish leader Khalifa Abdullah. "S" is the hex of the Dervish leader Sherif. "O" is the hex of the Dervish leader Osman Digna. (see manual for full text)

| File                                                                                                              | Symbol                             | Code Snippet                                                                                                                                                                                                                                                                                                                                                                                       |
|-------------------------------------------------------------------------------------------------------------------|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/effects/state/setup.rs:215</a><br>GH:omdurman-rules/src/effects/state/setup.rs:215 | <b>in_deployment_zone</b>          | <pre>213       /// plan / UI rather than this hex predicate.           Documented, not silently 214       /// dropped. 215       pub fn in_deployment_zone(&amp;self, player: Player,           hex: HexCoord, is_boat: bool) -&gt; bool { 216           // No board attached -&gt; permissive (unit tests,               unbound session). 217           if self.board.terrain.is_empty() {</pre> |
| <a href="#">omdurman-rules/src/unit.rs:78</a><br>GH:omdurman-rules/src/unit.rs:78                                 | <b>DervishLeader::setup_letter</b> | <pre>76       /// (§9.212): A→Ali Wad Helu, D→Sheik El Din,           Y→Yakub, K→Khalifa Abdullah, 77       /// S→Sherif, O→Osman Digna. Inverse of           [̀dervish_leader_for_setup_letter`]. 78       pub fn setup_letter(self) -&gt; SetupLetter { 79           match self { 80               DervishLeader::AliWadHelu =&gt;                   SetupLetter::A,</pre>                       |
| <a href="#">omdurman-rules/src/unit.rs:94</a><br>GH:omdurman-rules/src/unit.rs:94                                 | <b>dervish_leader_for_setup</b>    | <pre>92       /// inherent impl here, so the mapping is a free           function -- the bijective 93       /// inverse of [̀DervishLeader::setup_letter`]. 94       pub fn dervish_leader_for_setup_letter(letter:           SetupLetter) -&gt; DervishLeader { 95           match letter { 96               SetupLetter::A =&gt; DervishLeader::AliWadHelu,</pre>                                |

Covered by tests: [omdurman-rules::src::effects::tests::deploy\\_rejected\\_outside\\_zone](#)  
[omdurman-rules::src::unit\\_profiles::embedded\\_leaders\\_resolve\\_from\\_their\\_host\\_section](#)  
[omdurman-app::src::scenario\\_setup::historical\\_places\\_all\\_six\\_leaders\\_when\\_anchors\\_present](#)  
[omdurman-app::src::scenario\\_setup::missing\\_anchor\\_is\\_reported\\_not\\_dropped\\_silently](#)  
[omdurman-rules::src::effects::tests::setup\\_letter\\_dervish\\_leader\\_roundtrip](#)  
[omdurman-rules::src::effects::tests::setup\\_letter\\_to\\_dervish\\_leader\\_known\\_values](#)  
[omdurman-rules::src::effects::tests::historical\\_setup\\_rejects\\_not\\_in\\_play\\_units](#)

## §9.231 – Thorn hedge hexsides

implemented

manual page unknown

| File                                                            | Symbol                  | Code Snippet                                                                                                                                                                   |
|-----------------------------------------------------------------|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/combat.rs:28</a><br>GH:omdurman- | <b>ZaribaThornHedge</b> | <pre>26       Terrain(i16), 27       /// -2 thorn-hedge defensive modifier (§9.231). 28       ZaribaThornHedge, 29       /// -4 trench defensive modifier (§9.232). Only</pre> |

|                                                                                                       |                               |                                                                                                                                                                                                                                                                                                                                                |
|-------------------------------------------------------------------------------------------------------|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| rules/src/<br>combat.rs:28                                                                            |                               | applies vs. "entrenched"<br>30       /// units (those Nile-side of the trench hexside).                                                                                                                                                                                                                                                        |
| <a href="#">omdurman-types/src/lib.rs:206</a><br>GH:omdurman-types/src/lib.rs:206                     | <b>ZaribaThornHedge</b>       | 204       Crest,<br>205       /// Historical-scenario thorn-hedge segment of the<br>Zariba (§9.231).<br>206       ZaribaThornHedge,<br>207       /// Historical-scenario trench segment of the<br>Zariba (§9.232).<br>208       ZaribaTrench,                                                                                                  |
| <a href="#">omdurman-rules/src/effects/state.rs:318</a><br>GH:omdurman-rules/src/effects/state.rs:318 | <b>has_zariba_thorn_hedge</b> | 316       /// *effective* hexsides, so constructed (§5.3) and<br>authored hedges both<br>317       /// count.<br>318       pub fn has_zariba_thorn_hedge(&self, hex: HexCoord)<br>-> bool {<br>319           for n in hex.neighbors() {<br>320               if self.hexside_effective_is(hex, n,  k  k<br>== HexsideKind::ZaribaThornHedge) { |

Covered by tests: [omdurman-rules::src::effects::tests::zariba\\_fire\\_penalties\\_apply\\_to\\_dervish\\_fire\\_only](#)

## §9.232 – Trench hexsides

**implemented**

manual page unknown

| File                                                                                  | Symbol                        | Code Snippet                                                                                                                                                                                                                                                                                                             |
|---------------------------------------------------------------------------------------|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/combat.rs:31</a><br>GH:omdurman-rules/src/combat.rs:31 | <b>ZaribaTrenchEntrenched</b> | 29       /// -4 trench defensive modifier (§9.232). Only<br>applies vs. "entrenched"<br>30       /// units (those Nile-side of the trench hexside).<br>31       ZaribaTrenchEntrenched,<br>32       }<br>33                                                                                                              |
| <a href="#">omdurman-types/src/lib.rs:208</a><br>GH:omdurman-types/src/lib.rs:208     | <b>ZaribaTrench</b>           | 206       ZaribaThornHedge,<br>207       /// Historical-scenario trench segment of the<br>Zariba (§9.232).<br>208       ZaribaTrench,<br>209       /// One of the two end hexsides of a Zariba trench<br>segment that connect to<br>210       /// the Nile River (§9.233). Units may only enter/<br>leave the Zariba via |

Proven by: [omdurman-rules::src::verification::melee\\_modifier\\_keeps\\_roll\\_legal](#)

Covered by tests: [omdurman-rules::src::effects::tests::trench\\_entrenched\\_units\\_take\\_trench\\_modifiers](#)

## §9.233 – Zariba entry/exit costs

**implemented**

manual page unknown

Units may only enter and/or leave the Zariba via the two end hexsides that connect to the Nile River, paying +2 movement points to cross (Exception: advance after combat across an entrenched hexside).

| File                                                                              | Symbol                 | Code Snippet                                                                                                                                                                                                                       |
|-----------------------------------------------------------------------------------|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-types/src/lib.rs:265</a><br>GH:omdurman-types/src/lib.rs:265 | <b>blocks_movement</b> | 263       /// `omdurman-rules`). The trench *end* variants<br>are therefore intentionally<br>264       /// not blocking.<br>265       pub fn blocks_movement(self) -> bool {<br>266           matches!(<br>267               self, |

|                                                                                                         |                        |                                                                                                                                                                                                                                                                                                                                                                                                        |
|---------------------------------------------------------------------------------------------------------|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| omdurman-rules/src/effects/state.rs:331<br>GH:omdurman-rules/src/effects/state.rs:331                   | zariba_entry_surcharge | 329       /// +2 movement points to cross"). Trench ends are<br>authored FoK geography<br>330       /// (players cannot construct them), so this reads<br>the authored board.<br>331       pub fn zariba_entry_surcharge(&self, from:<br>HexCoord, to: HexCoord) -> i16 {<br>332           match self.board.hexside_between(from, to) {<br>333               Some(k) if k.is_zariba_trench_end() => 2, |
| omdurman-rules/src/effects/state/movement.rs:437<br>GH:omdurman-rules/src/effects/state/movement.rs:437 | movement_cost_for      | 435       ///<br>436       /// \$5.42: entering or leaving an enemy ZOC adds no<br>MP cost.<br>437       pub fn movement_cost_for(<br>438           &self,<br>439           unit: &UnitPlacement,                                                                                                                                                                                                      |
| omdurman-rules/src/effects/movement.rs:33<br>GH:omdurman-rules/src/effects/movement.rs:33               | apply_move_unit        | 31       /// engine from the board (\$5.11/\$5.24): the caller-<br>supplied `cost` is ignored<br>32       /// (it survives in the effect only for wire<br>compatibility).<br>33       pub fn apply_move_unit(<br>34           state: &mut GameState,<br>35           unit_id: UnitId,                                                                                                                  |

Proven by: omdurman-types::src::lib::zariba\_trench\_ends\_differ\_only\_in\_the\_entry\_rule

Covered by tests: omdurman-rules::src::effects::tests::zariba\_end\_hexside\_costs\_extra\_mp  
omdurman-rules::src::effects::tests::zariba\_thorn\_hedge\_blocks\_movement  
omdurman-app::src::picker::movement::trench\_end\_crossing\_costs\_two\_extra\_mp

## §9.321 – British set up (Bonus)

implemented

manual page unknown

The British player sets up first, moves second:

- General GORDON leader unit in the palace.
- Two old style (unnamed) gunboats in any Nile River hexes.
- Set up in any building or hut hexes of Khartoum, Forts Makran and/or Buri, and/or adjacent to any wall hex:
  - one Egyptian Battalion artillery unit
  - two British infantry units (represents Caucasian troops)
  - three Egyptian infantry units (represents Cairo “Bazouks”)
  - four Sudan infantry units (represents Sudanese blacks)
  - four “Friendlies” units (represents the Shaggyeh)

(see manual for full text)

| File                                                                                  | Symbol           | Code Snippet                                                                                                                                                                                                                                                                        |
|---------------------------------------------------------------------------------------|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| omdurman-rules/src/scenario_setup.rs:42<br>GH:omdurman-rules/src/scenario_setup.rs:42 | FALL_OF_KHARTOUM | 40       /// North Fort uses a campaign HadendowaForts counter<br>(one of the spare fort<br>41       /// sprites).<br>42       pub const FALL_OF_KHARTOUM_SETUP: &[FixedPlacement] =<br>&[<br>43           FixedPlacement {<br>44               section: SectionName::BritishBoats, |

Covered by tests: omdurman-rules::src::effects::tests::setup\_is\_sequential\_first\_side\_then\_second  
omdurman-rules::src::effects::tests::printed\_fok\_forts\_are\_british\_forts  
omdurman-rules::src::effects::tests::fall\_of\_khartoum\_has\_no\_maxim\_howitzzer\_subphase  
omdurman-rules::src::effects::tests::confirm\_ready\_rejected\_below\_scenario\_target  
omdurman-rules::src::effects::tests::remove\_deployed\_unit\_happy\_path

omdurman-app::src::scenario\_setup::fall\_of\_khartoum\_places\_gordon\_in\_the\_palace  
 omdurman-app::src::scenario\_setup::fall\_of\_khartoum\_reports\_missing\_palace  
 omdurman-app::src::scenario\_setup::fall\_of\_khartoum\_fort\_landmarks\_sit\_at\_the\_correct\_hexes  
 omdurman-rules::src::effects::tests::fok\_ae\_gunboat\_deploys\_only\_on\_nile  
 omdurman-rules::src::effects::tests::fok\_ae\_land\_unit\_rejected\_on\_nile  
 omdurman-rules::src::effects::tests::fok\_setup\_complete\_requires\_full\_oob  
 omdurman-rules::src::effects::tests::deploy\_via\_real\_sprite\_resolution\_matches\_engine  
 omdurman-rules::src::effects::tests::british\_boats\_named\_vs\_old\_gunboat\_detection  
 omdurman-rules::src::effects::tests::fok\_order\_of\_battle\_british      omdurman-rules::src::effects::tests::fok\_order\_of\_battle\_dervish  
 omdurman-rules::src::effects::tests::fok\_caps\_bind\_across\_counter\_variants

## §9.322 – Dervish enters turn one (Bonus)

implemented

manual page unknown

Dervish player moves first: enters turn one through any hexes on the south or east edge of the map.

- 32 Mulazmin units (represents combined forces of Wad El Nejumi, Abu Girgeh, and Sheik El Obeid)
- 2 Hadendowa; 6 Kehena; 5 Degheim (represents Mahdi's combined west bank forces)
- 3 Dervish artillery units.

| File                                                                                  | Symbol              | Code Snippet                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------------------------------------------------------------------------------|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| omdurman-rules/src/unit_profiles.rs:392<br>GH:omdurman-rules/src/unit_profiles.rs:392 | ali_wad_helu        | <pre> 390   /// §9.2/§9.3) unplaceable -- the FoK order-of-battle       table 391   /// (`fok_cap_group`) is keyed by identity. 392   pub fn ali_wad_helu(col: u32, row: u32) -&gt;       Option&lt;Classification&gt; { 393       match (col, row) { 394           (0, 0) =&gt;       dervish_leader(DervishLeader::AliWadHelu),           </pre>                                                     |
| omdurman-types/src/lib.rs:855<br>GH:omdurman-types/src/lib.rs:855                     | sections_for_picker | <pre> 853       /// counter of those two forces and leave the       §9.322 entry force -- and 854       /// with it the setup Ready gate (§9.2/§9.3) --       unplaceable. 855       pub fn sections_for_picker(self) -&gt; Option&lt;&amp;'static       [SectionName]&gt; { 856           match self { 857               Scenario::Campaign   Scenario::Historical       =&gt; None,           </pre> |

Covered by tests: omdurman-rules::src::effects::tests::deployment\_requires\_the\_canonical\_counter  
 omdurman-rules::src::effects::tests::fok\_reinforcements\_need\_the\_owners\_movement\_phase\_and\_unique\_ids  
 omdurman-rules::src::unit\_profiles::ali\_wad\_helu\_block\_resolves\_leader\_and\_degelim\_tribes  
 omdurman-rules::tests::fok\_setup\_flow::degheim\_counters\_resolve\_to\_the\_degheim\_tribe  
 omdurman-rules::src::effects::tests::fok\_setup\_complete\_requires\_full\_oob  
 omdurman-rules::src::effects::tests::fok\_dervish\_land\_unit\_rejected\_on\_nile  
 omdurman-types::src::lib::fok\_picker\_allowlist\_has\_dervish\_entry\_force\_blocks  
 omdurman-rules::src::effects::tests::fok\_order\_of\_battle\_dervish  
 omdurman-rules::src::effects::tests::fok\_dervish\_east\_edge\_on\_diamond\_board

## §9.341 – Turn 1 is always a night turn (Bonus)

implemented

manual page unknown

Turn 1 is always a night turn (see 8.1).

See also: §8.1

| File                                                                            | Symbol                  | Code Snippet                                                                                                                                                                                                                                                                                            |
|---------------------------------------------------------------------------------|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| omdurman-rules/src/turn_track.rs:172<br>GH:omdurman-rules/src/turn_track.rs:172 | <b>FALL_OF_KHARTOUM</b> | <pre> 170   /// (the rulebook fixes none); only `day_night` is       rule-bearing (night halves 171   /// Anglo-Egyptian movement and ranges and bars       howitzer fire, §8.1). 172   pub const FALL_OF_KHARTOUM_TURN_TRACK: [TurnEntry; 8] =       [ 173       TurnEntry {           turn: 1, </pre> |

Proven by: `omdurman-rules::src::turn_track::scenario_turn_indexing_is_exact_for_every_scenario`  
`omdurman-rules::src::turn_track::fall_of_khartoum_night_window_is_turns_one_and_two`

Covered by tests: `omdurman-rules::src::turn_track::fall_of_khartoum_turn_one_is_night`

## §9.342 – All hexes are playable, including half hexes (Bonus)

implemented

manual page unknown

All hexes are playable, including hexes showing up half or less.

| File                                                                          | Symbol                      | Code Snippet                                                                                                                                                                     |
|-------------------------------------------------------------------------------|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| omdurman-rules/src/board_data.rs:37<br>GH:omdurman-rules/src/board_data.rs:37 | <b>fall_of_khartoum_map</b> | <pre> 35   36   /// The Fall-of-Khartoum board (§9.3). 37   pub fn fall_of_khartoum_map_data() -&gt; MapData { 38       FOK.get_or_init(   parse(FOK_RON)).clone() 39   } </pre> |

Covered by tests: `omdurman-rules::src::effects::tests::fall_of_khartoum_board_excludes_no_hexes`

## §9.343 – Both players use the Dervish Range Effects Table (Bonus)

implemented

manual page unknown

Both players must use the Dervish Range Effects Table.

| File                                                                              | Symbol                | Code Snippet                                                                                                                                                                                            |
|-----------------------------------------------------------------------------------|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| omdurman-rules/src/effects/fire.rs:78<br>GH:omdurman-rules/src/effects/fire.rs:78 | <b>range_band_for</b> | <pre> 76   /// in FALL OF KHARTOUM *both* players use the Dervish       Range Effects Table 77   /// (§9.343). 78   pub fn range_band_for( 79       scenario: Scenario, 80       player: Player, </pre> |

Proven by: `omdurman-rules::src::effects::fire::range_band_for_routes_to_the_right_faction_table`  
`omdurman-rules::src::effects::fire::range_table_player_for_routes_friendlyes_and_fok_to_the_dervish_table`

Covered by tests: `omdurman-rules::src::effects::tests::fok_both_players_use_dervish_range_table`

## §9.344 – Dervish controls the North Fort (Bonus)

implemented

manual page unknown

The Dervish player controls the “North Fort” and may fire its guns.

| File                                                                                  | Symbol                    | Code Snippet                                                                                                                                                                                                                                                                                                                                       |
|---------------------------------------------------------------------------------------|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| omdurman-rules/src/scenario_setup.rs:42<br>GH:omdurman-rules/src/scenario_setup.rs:42 | <b>FALL_OF_KHARTOUM</b>   | <pre> 40   /// North Fort uses a campaign HadendowaForts counter       (one of the spare fort 41   /// sprites). 42   pub const FALL_OF_KHARTOUM_SETUP: &amp;[FixedPlacement] =       &amp;[ 43       FixedPlacement { 44           section: SectionName::BritishBoats,</pre>                                                                      |
| omdurman-rules/src/effects/state.rs:425<br>GH:omdurman-rules/src/effects/state.rs:425 | <b>hex_has_enemy_fort</b> | <pre> 423       /// positions (§9.321), while the North Fort is the       Dervish one (§9.344, 424       /// represented by its fort counter). 425       pub fn hex_has_enemy_fort(&amp;self, hex: HexCoord,       mover: Player) -&gt; bool { 426           self.printed_fort_owner(hex) 427           .is_some_and( owner  owner != mover)</pre> |

Covered by tests: **omdurman-app::src::scenario\_setup::fall\_of\_khartoum\_places\_gordon\_in\_the\_palace**  
**omdurman-app::src::scenario\_setup::fall\_of\_khartoum\_fort\_landmarks\_sit\_at\_the\_correct\_hexes**  
**omdurman-app::src::scenario\_setup::placement\_done\_gate\_matches\_by\_identity\_not\_allocated\_id**  
**omdurman-rules::src::effects::tests::fok\_order\_of\_battle\_dervish**  
**omdurman-types::src::lib::fok\_picker\_allowlist\_has\_dervish\_entry\_fork\_blocks**

## §9.345 – Gunboat White Nile <-> Blue Nile crossing (Bonus)

**implemented**

manual page unknown

The British gunboats may move from the White Nile to the Blue Nile and vice-versa at an off-board movement cost of six “upstream” movement points.

| File                                                                                  | Symbol                          | Code Snippet                                                                                                                                                                                                                                                                                                                                      |
|---------------------------------------------------------------------------------------|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| omdurman-rules/src/effects/state.rs:404<br>GH:omdurman-rules/src/effects/state.rs:404 | <b>is_nile_mouth_crossing</b>   | <pre> 402       /// must be named on the board, else this is       `false` and the move falls 403       /// through to the ordinary contiguous-Nile rules. 404       pub fn is_nile_mouth_crossing(&amp;self, from:       HexCoord, to: HexCoord) -&gt; bool { 405           let white = self 406           .board</pre>                          |
| omdurman-types/src/lib.rs:547<br>GH:omdurman-types/src/lib.rs:547                     | <b>Location::WhiteNileMouth</b> | <pre> 545       /// The off-board mouth of the White Nile branch       (FALL OF KHARTOUM §9.345) -- 546       /// a British gunboat may cross to the Blue Nile       mouth for 6 upstream MP. 547       WhiteNileMouth, 548       /// The off-board mouth of the Blue Nile branch       (FALL OF KHARTOUM §9.345). 549       BlueNileMouth,</pre> |

Covered by tests: **omdurman-rules::src::effects::tests::fok\_gunboat\_crosses\_between\_nile\_mouths**

## §9.346 – GORDON immobile, eliminated only at the Palace (Bonus)

**implemented**

manual page unknown

The GORDON leader unit starts in the palace and may not move during the scenario. He may only be eliminated by a Dervish unit passing through or occupying the palace hex (as normal movement or as advance after combat).

| File | Symbol | Code Snippet |
|------|--------|--------------|
|------|--------|--------------|

|                                                                                                             |                                |                                                                                                                                                                                                                                                                                               |
|-------------------------------------------------------------------------------------------------------------|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/effects/dispatch.rs:603</a><br>GH:omdurman-rules/src/effects/dispatch.rs:603 | <b>check_gordon_palace</b>     | <pre>601   /// (overrun in passing, §6.51). A no-op outside FoK or       while he lives 602   /// unthreatened. 603   pub fn check_gordon_palace(state: &amp;mut GameState) { 604       if state.scenario != Scenario::FallOfKhartoum { 605           return;</pre>                           |
| <a href="#">omdurman-rules/src/unit.rs:332</a><br>GH:omdurman-rules/src/unit.rs:332                         | <b>UnitIdentity::is_gordon</b> | <pre>330       /// Whether this is the GORDON leader unit (§9.32,       §9.346) -- the immobile 331       /// palace defender whose elimination ends FALL OF       KHARTOUM (§9.35). 332       pub fn is_gordon(&amp;self) -&gt; bool { 333           matches!( 334               self,</pre> |

Covered by tests:

- omdurman-rules::src::effects::tests::breach\_casualty\_stands\_at\_the\_wall
- omdurman-rules::src::unit\_profiles::gordon\_is\_an\_immobile\_british\_leader
- omdurman-rules::src::effects::tests::gordon\_survives\_means\_no\_elimination
- omdurman-rules::src::effects::tests::gordon\_dying\_in\_melee\_ends\_fall\_of\_khartoum
- omdurman-rules::src::effects::tests::fire\_never\_kills\_gordon
- omdurman-app::src::scenario\_setup::fall\_of\_khartoum\_places\_gordon\_in\_the\_palace

 7/10 implemented

## §10 – Optional Rules

### §10 – Optional Rules

descriptive

manual page unknown

Optional Rules (Campaign game only)

It is suggested that the most intriguing employment of the following two options is to permit the Dervish player to have either one or the other, but the Anglo-Egyptian player doesn't know which one until he stumbles onto it. Players are advised that the employment of both optionals in the same game is not recommended.

### §10.1 – River Mines

descriptive

manual page unknown

River Mines

The Khalifa twice tried (unsuccessfully) to submerge a powerful mine in the Nile to sink or damage British gunboats. This option assumes that both attempts were successful.

### §10.2 – River Chain

descriptive

manual page unknown

River Chain

The Khalifa also tried (also unsuccessfully) to string a heavy chain across the Nile to stop or slow down the British gunboats. This option assumes the chain was emplaced.

### §10.11 – Secretly record mine hexes

implemented

manual page unknown

Prior to the commencement of play the Dervish player secretly records two Nile River hexes to be mined (the mines may not both be placed in the same hex). These hexes must be south of the E–W hexrow in which the Khor Shambat empties into the Nile.

| File                                                                                                  | Symbol                  | Code Snippet                                                                                                                                                                                                                                                                                                                            |
|-------------------------------------------------------------------------------------------------------|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/effects/setup.rs:350</a><br>GH:omdurman-rules/src/effects/setup.rs:350 | <b>apply_place_mine</b> | <pre>348   /// Lay a river mine during setup (§10.11). Validated       by 349   /// [<code>GameState::can_place_mine`</code>]. 350   pub fn apply_place_mine(state: &amp;mut GameState, hex:       HexCoord) -&gt; Result&lt;(), RuleError&gt; { 351       state.can_place_mine(hex)?; 352       state.mines.push(MinePlacement {</pre> |
| <a href="#">omdurman-rules/src/effects/state.rs:212</a><br>GH:omdurman-rules/src/effects/state.rs:212 | <b>GameState::mines</b> | <pre>210   211           friendlies_transport: None, 212           optional_rules: Vec::new(), 213           mines: Vec::new(), 214           chain: None,               board: Arc::new(BoardInfo::default()),</pre>                                                                                                                   |
| <a href="#">omdurman-rules/src/</a>                                                                   | <b>OptionalRule</b>     | <pre>56   /// two should be in play (rulebook §10). 57   #[derive(Serialize, Deserialize, Clone, Copy,</pre>                                                                                                                                                                                                                            |

|                                                                |                                                                                                          |
|----------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| <a href="#">turn.rs:58</a><br>GH:omdurman-rules/src/turn.rs:58 | <pre>PartialEq, Eq, Debug)] 58   pub enum OptionalRule { 59       RiverMines, 60       RiverChain,</pre> |
|----------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|

Covered by tests: [omdurman-rules::src::effects::tests::mine\\_and\\_chain\\_limits\\_enforced\\_in\\_setup](#)  
[omdurman-rules::src::effects::tests::mines\\_and\\_chain\\_require\\_their\\_optional\\_rule](#)

### §10.12 – Mine resolution

implemented

manual page unknown

When a British gunboat enters a mined hex, the Dervish player must order it to stop as it has struck a mine. The Dervish player then resolves the effect of the mine’s blast by rolling the ten-sided die:

- 1–4: No effect
- 5–7: Gunboat damaged, lost use of its engines and must drift two hexes per turn (with the current) for the rest of the game. No effect on guns or Maxims unless they drift out of range.
- 8–10: Gunboat sunk!

| File                                                                                                    | Symbol                  | Code Snippet                                                                                                                                                                                                      |
|---------------------------------------------------------------------------------------------------------|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/effects/effect.rs:201</a><br>GH:omdurman-rules/src/effects/effect.rs:201 | <b>RiverMine</b>        | <pre>199   // -- Optional rules ----- 200   /// River mine resolution (rulebook §10.12). 201   RiverMine { 202       gunboat_id: UnitId, 203       hex: HexCoord,</pre>                                           |
| <a href="#">omdurman-rules/src/effects/river.rs:128</a><br>GH:omdurman-rules/src/effects/river.rs:128   | <b>apply_river_mine</b> | <pre>126   127   /// Apply a river-mine resolution (rulebook §10.12). 128   pub fn apply_river_mine( 129       state: &amp;mut GameState, 130       gunboat_id: UnitId,</pre>                                     |
| <a href="#">omdurman-rules/src/transport.rs:63</a><br>GH:omdurman-rules/src/transport.rs:63             | <b>MineResult</b>       | <pre>61   /// British gunboat enters a mined hex. 62   #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 63   pub enum MineResult { 64       /// Roll 1-4: no effect. 65       NoEffect,</pre> |

Proven by: [omdurman-rules::src::verification::mine\\_result\\_bands\\_match\\_the\\_printed\\_rule](#)  
[omdurman-rules::src::effects::river::river\\_mine\\_triggers\\_even\\_on\\_a\\_harmless\\_roll](#)  
[omdurman-rules::src::effects::river::river\\_mine\\_engine\\_loss\\_armes\\_the\\_drift\\_flag](#)  
[omdurman-rules::src::effects::river::river\\_mine\\_sinking\\_removes\\_the\\_gunboat](#)

Covered by tests: [omdurman-rules::src::effects::tests::a\\_mined\\_gunboat\\_scores\\_and\\_takes\\_its\\_friendlies\\_down](#)  
[omdurman-rules::src::effects::tests::mine\\_fires\\_once\\_and\\_sparees\\_dervish](#)

### §10.13 – Mines consumed after both rolled for

implemented

manual page unknown

After both mines have been rolled for, no more are available.

| File | Symbol | Code Snippet |
|------|--------|--------------|
|------|--------|--------------|

|                                                                                                       |                         |                                                                                                                                                                                 |
|-------------------------------------------------------------------------------------------------------|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/effects/river.rs:128</a><br>GH:omdurman-rules/src/effects/river.rs:128 | <b>apply_river_mine</b> | <pre> 126   127   /// Apply a river-mine resolution (rulebook §10.12). 128   pub fn apply_river_mine( 129       state: &amp;mut GameState, 130       gunboat_id: UnitId, </pre> |
|-------------------------------------------------------------------------------------------------------|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Proven by: [omdurman-rules::src::effects::river::river\\_mine\\_triggers\\_even\\_on\\_a\\_harmless\\_roll](#)  
[omdurman-rules::src::effects::river::a\\_triggered\\_mine\\_never\\_fires\\_again](#)

Covered by tests: [omdurman-rules::src::effects::tests::mine\\_fires\\_once\\_and\\_spares\\_dervish](#)

## §10.14 – Dervish gunboats pass safely

**implemented**

manual page unknown

The Dervish player’s gunboats may pass through the mined hexes with no ill effect (he knows where the mines are).

| File                                                                                                  | Symbol                  | Code Snippet                                                                                                                                                                    |
|-------------------------------------------------------------------------------------------------------|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/effects/river.rs:128</a><br>GH:omdurman-rules/src/effects/river.rs:128 | <b>apply_river_mine</b> | <pre> 126   127   /// Apply a river-mine resolution (rulebook §10.12). 128   pub fn apply_river_mine( 129       state: &amp;mut GameState, 130       gunboat_id: UnitId, </pre> |

Proven by: [omdurman-rules::src::effects::river::dervish\\_gunboats\\_pass\\_mined\\_hexes\\_unharmed](#)

Covered by tests: [omdurman-rules::src::effects::tests::mine\\_fires\\_once\\_and\\_spares\\_dervish](#)

## §10.21 – Secretly record chain hexes

**implemented**

manual page unknown

Prior to the commencement of play the Dervish player secretly records a line of river hexes (not exceeding four hexes long) across which the chain is strung. The hexes must be south of the E–W hexrow in which the Khor Shambat empties into the Nile.

| File                                                                                                              | Symbol                   | Code Snippet                                                                                                                                                                                                                                                                                                                                                                                          |
|-------------------------------------------------------------------------------------------------------------------|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/effects/state/setup.rs:526</a><br>GH:omdurman-rules/src/effects/state/setup.rs:526 | <b>can_place_chain</b>   | <pre> 524       /// Read-only check of a river-chain placement in 525       setup (§10.21): Setup phase 526       /// and at most [<code>MAX_CHAIN_HEXES</code>] hexes. 527       pub fn can_place_chain(&amp;self, hexes: &amp;[HexCoord]) - 528       &gt; Result&lt;(), RuleError&gt; { 529           self.require_setup_phase()?; 530           self.require_setup_turn(Player::Dervish)?; </pre> |
| <a href="#">omdurman-rules/src/effects/state/setup.rs:361</a><br>GH:omdurman-rules/src/effects/state/setup.rs:361 | <b>apply_place_chain</b> | <pre> 359       /// Lay (or replace) the river chain during setup 360       (§10.21). Validated by 361       /// [<code>can_place_chain</code>]. 362       pub fn apply_place_chain(state: &amp;mut GameState, hexes: 363       &amp;[HexCoord]) -&gt; Result&lt;(), RuleError&gt; { 364           state.can_place_chain(hexes)?; 365           state.chain = Some(ChainPlacement { </pre>            |
| <a href="#">omdurman-rules/src/effects/state/setup.rs:91</a><br>GH:omdurman-                                      | <b>MAX_CHAIN_HEXES</b>   | <pre> 89   90   /// Maximum contiguous Nile hexes the river chain may 91   span (§10.21). 92   pub const MAX_CHAIN_HEXES: usize = 4; </pre>                                                                                                                                                                                                                                                           |

|                                         |  |                                            |
|-----------------------------------------|--|--------------------------------------------|
| rules/src/effects/<br>state/setup.rs:91 |  | 92  <br>93   <code>impl GameState {</code> |
|-----------------------------------------|--|--------------------------------------------|

Covered by tests: `omdurman-rules::src::effects::tests::mine_and_chain_limits_enforced_in_setup`  
`omdurman-rules::src::effects::tests::mines_and_chain_require_their_optional_rule`

§10.22 – Gunboat stops on chained hex

implemented

manual page unknown

When a British gunboat enters a “chained” river hex it must stop and may move no further that turn.

| File                                                                                                                    | Symbol                        | Code Snippet                                                                                                                                                                                                                                                     |
|-------------------------------------------------------------------------------------------------------------------------|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/effects/state/movement.rs:468</a><br>GH:omdurman-rules/src/effects/state/movement.rs:468 | <code>can_move_gunboat</code> | <pre>466   // The engine computes the cost (one MP per hex, a       flat 6 for the §9.345 467   // mouth crossing); a larger caller `cost` must       also fit. 468       pub fn can_move_gunboat(               &amp;self, 469           unit_id: UnitId,</pre> |

Covered by tests: `omdurman-rules::src::effects::tests::chain_stops_gunboat_until_sunk`

§10.23 – Sinking the chain

implemented

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No gunboats (British or Dervish) may cross the chain until it has been sunk by the British player. He may sink the chain by a) having an infantry or cavalry unit spend one complete turn on either riverbank adjacent to a “chained” river hex, or b) firing at the chain with artillery and achieving a 3 or more on the Combat Results Table.

| File                                                                                                  | Symbol                        | Code Snippet                                                                                                                                                                                                                                                                                                                                              |
|-------------------------------------------------------------------------------------------------------|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/effects/river.rs:173</a><br>GH:omdurman-rules/src/effects/river.rs:173 | <code>apply_sink_chain</code> | <pre>171   // Sink the river chain (rulebook §10.23). Marks the       placed chain cleared so it 172   // no longer stops gunboats (§10.22). 173       pub fn apply_sink_chain(state: &amp;mut GameState) -&gt;       Result&lt;(), RuleError&gt; { 174           match state.chain.as_mut() { 175               Some(chain) if !chain.sunk =&gt; {</pre> |

Covered by tests: `omdurman-rules::src::effects::tests::chain_stops_gunboat_until_sunk`

0/1 implemented

## §11 – Historical Notes

### §11 – Historical Notes

descriptive

manual page unknown

#### Historical Notes

In 1881 Mohammed Ahmed Ibn Al-Sayid Abdullah, the son of an obscure carpenter in the hinterlands of the Sudan, proclaimed himself the “Mahdi” — the Messiah of the Islamic faith. His timing was propitious indeed. Since the early 1820’s a corrupt Egypt, with the Sultan of Turkey’s blessing, had incessantly raped the Sudan, taking ivory and flooding the slave markets with some half million captured Sudanese blacks. By 1880, nearly 40,000 Egyptian troops occupied outposts scattered throughout the Sudan, enforcing Egypt’s hold on this lucrative ivory and slave trade and squeezing the native population dry through vicious and corrupt tax officials. All was controlled from Khartoum via the office of Governor General of the Sudan. The title had been held by a succession of individuals, including General Charles Gordon, whose appointment was an attempt to reinstate some rudimentary justice in the Sudan after France and Britain assumed joint political control of a bankrupt Egypt.

By 1881, however, Gordon’s term had expired and a new Governor General, again corrupt and incompetent,  
(see manual for full text)

See also: §9.1, §9.2, §9.3

0/1 implemented

## **Credits**

### **§Credits – Credits**

**descriptive**

manual page unknown

 1/1 implemented

## Combat Results Table (shared reference)

### §CRT – Combat Results Table (shared by §6.22 fire and §7.7 melee)

implemented

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| File                                                                                                              | Symbol                      | Code Snippet                                                                                                                                                                                                                                                                                                                                 |
|-------------------------------------------------------------------------------------------------------------------|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/combat_results_table.rs:10</a><br>GH:omdurman-rules/src/combat_results_table.rs:10 | <b>FireFactorRow</b>        | <pre>8       serde::Serialize, serde::Deserialize, Clone, Copy, 9       PartialEq, Eq, PartialOrd, Ord, Hash, Debug, 10   } 11   pub enum FireFactorRow { 12       /// 1-5 factors 13       Row01to05,</pre>                                                                                                                                 |
| <a href="#">omdurman-rules/src/combat_results_table.rs:46</a><br>GH:omdurman-rules/src/combat_results_table.rs:46 | <b>from_total</b>           | <pre>44       /// Determine which row a given total fire factor 45       falls into (rulebook §6.22). 46       pub fn from_total(total: u16) -&gt; Self { 47           match total { 48               0..=5 =&gt; FireFactorRow::Row01to05,</pre>                                                                                            |
| <a href="#">omdurman-rules/src/combat_results_table.rs:88</a><br>GH:omdurman-rules/src/combat_results_table.rs:88 | <b>combat_results_table</b> | <pre>86       /// D = `Disrupt` (1/2 of target units, round up) 87       /// 1...5 = `Eliminate(n)` (that many units removed) 88       pub fn combat_results_table(row: FireFactorRow, roll: 89           DieRoll) -&gt; CombatResult { 90           crate::tables_data::CRT[row.index()][roll.value() 91               - 1] as usize]</pre> |
| <a href="#">omdurman-rules/src/combat.rs:97</a><br>GH:omdurman-rules/src/combat.rs:97                             | <b>CombatResult</b>         | <pre>95       /// * `--` -- no effect 96       #[derive(Serialize, Deserialize, Clone, Copy, 97           PartialEq, Eq, Debug)] 98       pub enum CombatResult { 99           NoEffect, 100           Disrupt,</pre>                                                                                                                        |

Proven by: [omdurman-rules::src::verification::disrupt\\_half\\_is\\_rounded\\_up](#)  
[omdurman-rules::src::combat\\_results\\_table::crt\\_eliminate\\_count\\_stays\\_within\\_printed\\_bounds](#)  
[omdurman-rules::src::combat\\_results\\_table::crt\\_is\\_monotone\\_in\\_the\\_die\\_roll\\_for\\_every\\_row](#)

Covered by tests: [omdurman-rules::src::combat\\_results\\_table::ae\\_combat\\_results\\_table\\_lowest\\_is\\_no\\_effect](#)  
[omdurman-rules::src::combat\\_results\\_table::ae\\_combat\\_results\\_table\\_highest\\_is\\_eliminate\\_5](#)  
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[omdurman-rules::src::combat\\_results\\_table::crt\\_lowest\\_row\\_is\\_worst\\_highest\\_row\\_is\\_best](#)  
[omdurman-rules::src::combat\\_results\\_table::crt\\_eliminate\\_never\\_exceeds\\_5](#)  
[omdurman-rules::src::combat\\_results\\_table::crt\\_every\\_cell\\_matches\\_the\\_table](#)

0/1 implemented

## **| Reference – Charts and Tables**

### **§Reference – Charts and Tables**

**out-of-scope**

manual page unknown

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